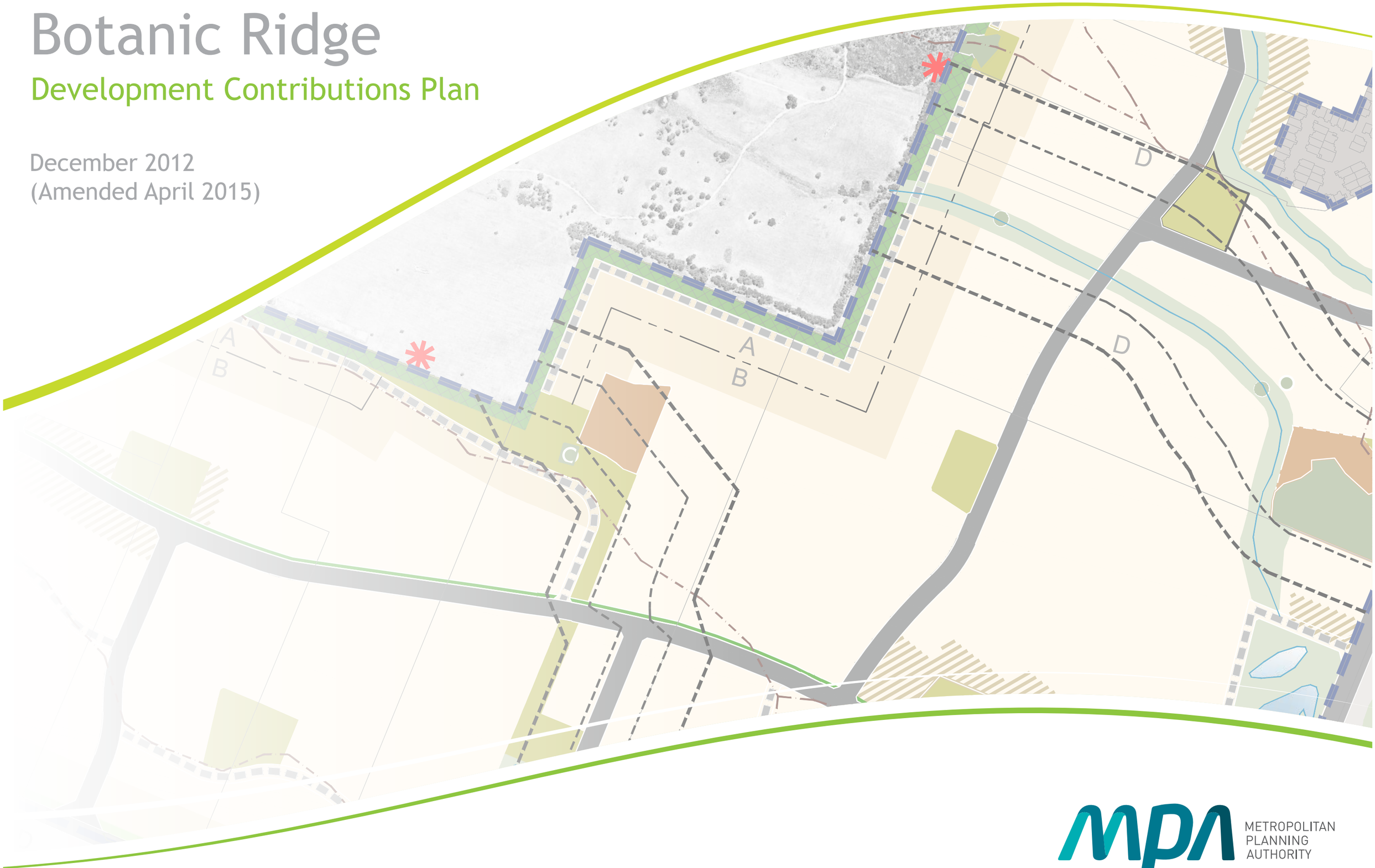
Botanic Ridge

Development Contributions Plan

December 2012
(Amended April 2015)



METROPOLITAN
PLANNING
AUTHORITY

Version	Date	Incorporated into the planning scheme by amendment	Description of changes
1	December 2012	Casey C133	N/A
2	April 2015	Casey C197	Update to future urban structure to reflect revised bushfire management provisions in the Botanic Ridge Precinct Structure Plan.

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INTRODUCTION

The Botanic Ridge Development Contributions Plan has been prepared by the Growth Areas Authority with the assistance of the Casey City Council, Government agencies, service authorities and major stakeholders.

The Development Contributions Plan has been developed to support the provision of works, services, and facilities within the Botanic Ridge DCP area as shown on Plan 1 of this DCP. The regional context of the DCP area is also shown on Plan 3 of this DCP.

The Development Contributions Plan outlines projects required to ensure that future residents, visitors and workers within the Botanic Ridge area can be provided with timely access to services and transport necessary to support a quality, affordable lifestyle.

The DCP details requirements that must be met by future land use and development and provides developers, investors and local communities with certainty about development contribution requirements and how they will be administered.

This Development Contributions Plan establishes a framework for development proponents to contribute a fair proportion towards the cost of works, services and facilities for the new community at Botanic Ridge. It ensures the cost of providing new infrastructure and services is shared equitably between various developers and the wider community. Fairness requires costs to be apportioned according to the projected share of use and the required works, services and facilities items.

Development Contributions Plan Structure

The Development Contributions Plan document comprises five main sections

1.0 STRATEGIC BASIS

Section 1 clearly explains the strategic basis for the Development Contributions Plan, which includes information about the Botanic Ridge Precinct Structure Plan and justification for the various infrastructure projects included in the Development Contributions Plan.

2.0 CALCULATION OF CONTRIBUTIONS

Section 2 sets out how the development contributions are calculated and costs apportioned

3.0 ADMINISTRATION AND IMPLEMENTATION

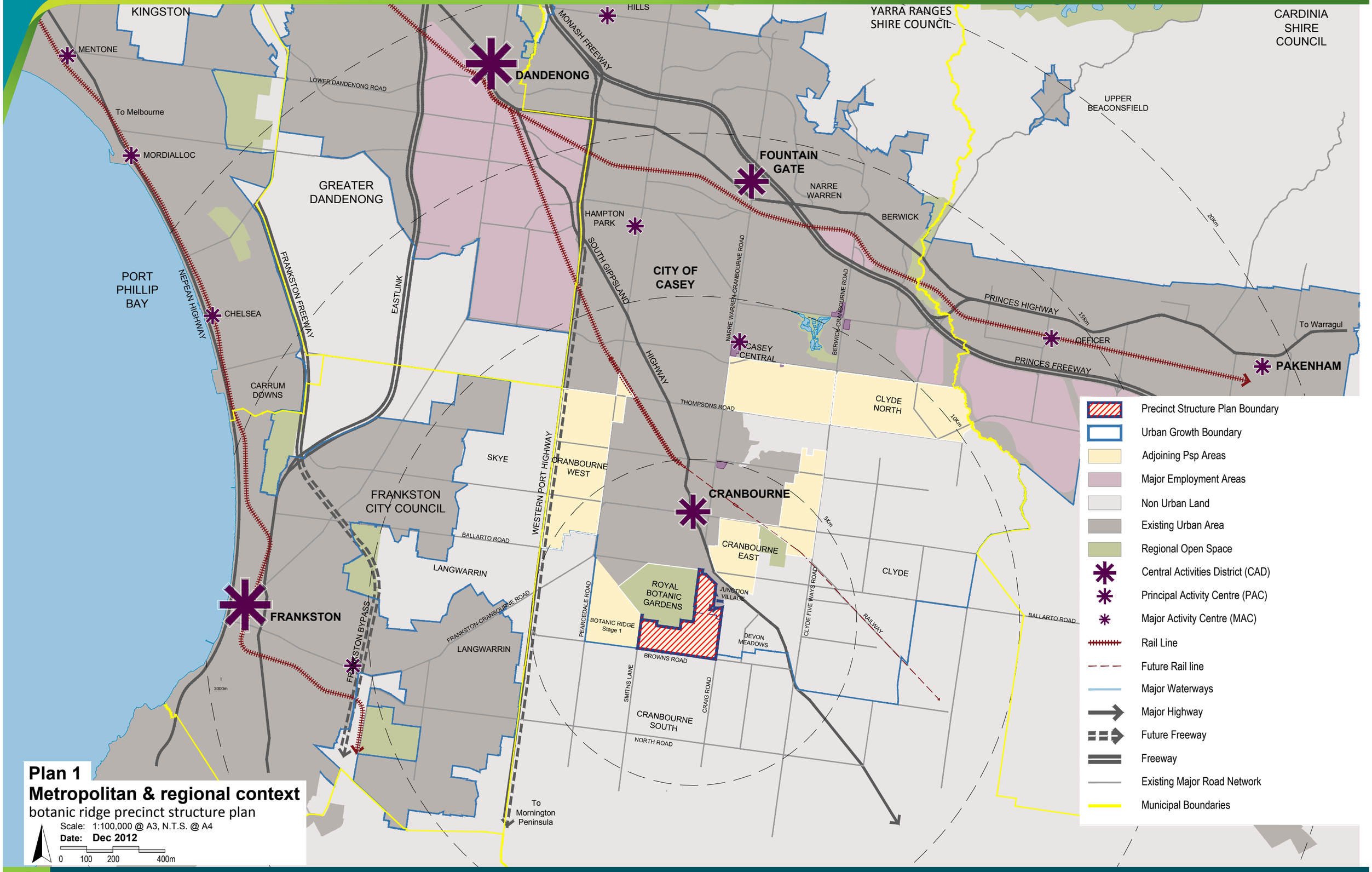
Section 3 focuses on the administrative and implementation of the Development Contributions Plan

4.0 OTHER INFORMATION

Section 4 provides other information focusing on the administration and implementation of the Development Contributions Plan.

5.0 PROJECT COSTS

Section 5 provides further information on projects, including cost estimate breakdowns.



Plan 1
Metropolitan & regional context
botanic ridge precinct structure plan

Scale: 1:100,000 @ A3, N.T.S. @ A4
Date: Dec 2012
0 100 200 400m

1.0 STRATEGIC BASIS

The strategic basis for this Development Contributions Plan is established by the State and Local Planning Policy Framework of the Casey Planning Scheme. Key documents are the Growth Corridor Plans (June 2012), the Municipal Strategic Statement and the Botanic Ridge Precinct Structure Plan (and supporting documents), which set out a broad, long term vision for the sustainable development of the Development Contributions Plan area.

The Growth Area Framework Plans, have been incorporated into the Victoria Planning Provisions and illustrate the planned extent of residential, employment and other development as well as the location of larger activity centres for each growth area. They also include key elements of infrastructure and services including the regional open space network, the location of public transport networks, freeways and arterial roads.

The Botanic Ridge Precinct Structure Plan has been developed following a comprehensive planning process and provides an appropriate level of detail to guide the preparation of the Development Contributions Plan.

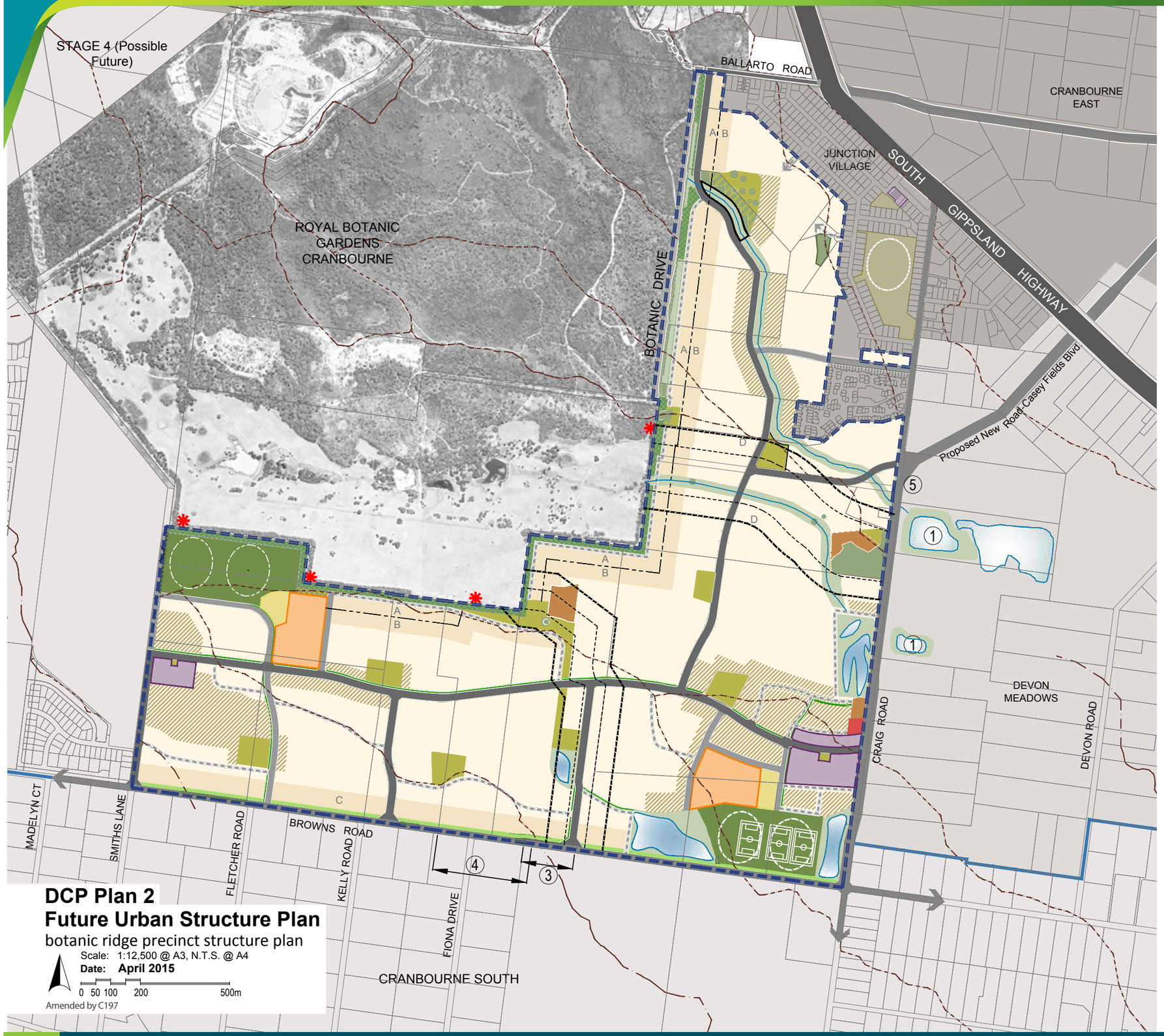
1.1 PLANNING AND ENVIRONMENT ACT 1987

This Development Contributions Plan has been prepared in accordance with Part 3B of the Planning and Environment Act 1987 (the Act) and has been developed in line with the State and Local Planning Policy Frameworks of the Casey Planning Scheme as well as relevant Victorian Government policy and Guidelines.

The Development Contributions Plan provides a 'development infrastructure levy' and a 'community infrastructure levy' as enabled by Part 3B of the Planning and Environment Act, as some items are classified as community infrastructure under the Act.

This Development Contributions Plan forms part of the Casey Planning Scheme pursuant to Section 46I of the Act, and is an incorporated document under Clause 81 of the Casey Planning Scheme.

The Development Contributions Plan is included in Schedule 14 of the Development Contributions Plan Overlay in the Casey Planning Scheme.



DCP Plan 2
Future Urban Structure Plan
botanic ridge precinct structure plan
Scale: 1:12,500 @ A3, N.T.S. @ A4
Date: April 2015
0 50 100 200 500m
Amended by C197

- Precinct Structure Plan Area
- Urban Growth Boundary
- Village Housing Zone
- Interface Housing Zone (A, B & C)
- Neighbourhood Housing Zone
- Town Centre
- Potential Mixed use
- Council Community Facility
- Government Education Facility (Primary)
- Active Open Space/Playing Fields
- Passive Open Space
- Conservation area set aside subject to VCAT enforcement order (in addition to NVPP requirements)
- Defendable space - RBGC Perimeter ∗
- Conservation Open Space (NVPP Requirement) ∗
- Waterway modification works minimised ∗
- Waterway / Drainage / Waterbodies / Wetlands ∗
- Tree Reserve along Browns Roads ∗
- Heritage site / Open Space (Aboriginal) ∗
- Scattered Trees ∗
- Non Urban Land / Existing Urban Area
- Other Undeveloped Land Within UGB
- Major Arterial Road (South Gipps Highway)
- Arterial Road - Craig Rd. / Browns Rd. (2 - 4 lane)
- Connector Street (27m)
- Connector Street (with Landscape and trail 32m)
- Access Street - Level 2 (22m)
- Access Street - Level 2 (with Landscape and trail 28m)
- Key Access Street - Level 1
- Key Access Street - Level 1 (with Landscape and trail 24m)
- Ridge Line
- RBGC Potential Pedestrian Access Point subject to RBGC approval.
- ① Wetland retarding basin located outside Precinct subject to Melbourne Water PAO
- ② Eastern 5m of Smiths Lane road reserve used for Active Open Space
- ∗ Encumbered - No POS Credit
- ③ Flexibility for Connector Road Location. Refer Plan 7 & Appx. D
- ④ Alternative for interim access to initial stages of development via a local street. Refer Plan 7 & Appx. D
- ⑤ Potential for intersection to shift South to align with existing Road Reserve. Refer Section 3.7
- Land subject to EPBC Act (Commonwealth) Determination - Refer Section 3.9 & Appx. D.

1.2 PRECINCT STRUCTURE PLAN

The Botanic Ridge Precinct Structure Plan is located to the south of Cranbourne in the outer south-east region of Melbourne.

The Botanic Ridge Precinct Structure Plan area is expected to:

- Provide for a population of approximately 9,650 people, accommodated in approximately 3,450 households; and
- Generate around 1000 jobs in the Precinct Structure Plan area.

The Precinct Structure Plan establishes the future urban structure of the new community which includes a range of networks including transport, open space and active recreation, social infrastructure, activity centres, and residential neighbourhoods (refer to Plan 2).

The need for the infrastructure has been determined according to the anticipated development scenario for Botanic Ridge as described in the Botanic Ridge Precinct Structure Plan. The Development Contributions Plan emanates from the Precinct Structure Plan, as the Precinct Structure Plan provides the rationale and justification for infrastructure items that have been included within the Development Contributions Plan.

Accordingly, the Development Contributions Plan is an implementation based planning tool which identifies the infrastructure items required by the new community and apportions the cost of this infrastructure in an equitable manner across the plan area.

STAGE 4
(Possible Future)

ROYAL BOTANIC
GARDENS
CRANBOURNE

STAGE 1 (Under
Development)

BALLARTO ROAD

CRANBOURNE
EAST

BOTANIC DRIVE

JUNCTION
VILLAGE

SOUTH GIPPSLAND HIGHWAY

Existing playing
fields

STAGE 2
(North)

PROPOSED CASEY BLVD

STAGE 3
(South)

SMITHS LANE

FLETCHER ROAD

KELLY ROAD

FIONA DRIVE

BROWNS ROAD

CRAIG ROAD

DEVON ROAD

DEVON
MEADOWS

CRANBOURNE
SOUTH

DCP Plan 3
Catchment area
botanic ridge precinct structure plan

Scale: 1:12,500 @ A3, N.T.S. @ A4
Date: Dec 2012
0 50 100 200 500m

- DCP (Main Catchment Area)
- Urban Growth Boundary
- Stage 2 (North)
- Stage 3 (South)
- Regional Open Space
- Non Urban Land
- Existing Urban Area
- Existing Roads

1.3 THE AREA TO WHICH THE DEVELOPMENT CONTRIBUTIONS PLAN APPLIES

In accordance with Section 46K(1)(a) of the Planning and Environment Act 1987, the Botanic Ridge Development Contributions Plan applies to land shown on Plan 3. The area is also indicated in the relevant Development Contributions Plan Overlay in the Casey Planning Scheme (Schedule 14).

The DCP area includes approximately 318 hectares of land as shown in Plan 3.

The Development Contributions Plan forms a single catchment area as shown in Plan 3, for all projects, consistent with the Botanic Ridge Precinct Structure Plan.

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1.4 INFRASTRUCTURE PROJECT JUSTIFICATION

1.4.1 INTRODUCTION

The need for infrastructure has been determined according to the anticipated development scenario for Botanic Ridge as described in the Botanic Ridge Precinct Structure Plan and its supporting documents.

Items can be included in a Development Contributions Plan provided there is a need for the works, services and facilities generated by the proposed development of land in the area. Proposed development does not have to trigger the need for new items in its own right. The development is charged in line with its projected share of use. An item can be included in a Development Contributions Plan regardless of whether it is within or outside the Development Contributions Plan area.

In selecting items, consideration has been given to ensure they are not already wholly funded through another contribution mechanism, such as the Precinct Infrastructure Plan set out in the Botanic Ridge Precinct Structure Plan, an existing local development contributions plan, an agreement under Section 173 of the Act, or by condition on an existing planning permit.

All items included in this Development Contributions Planhave been assessed to ensure they have a clear relationship or nexus to proposed development in the Botanic Ridge Precinct Structure Plan area. The cost apportionment methodology adopted in this Development Contributions Plan relies on the nexus principle. A new development is deemed to have a nexus with an item if it is likely to make use of that item.

The items that have been included in the Development Contributions Plan all have the following characteristics:

- They are essential to the health, safety and well-being of the community;
- They will be used by a broad cross-section of the community;
- They reflect the vision and strategic aspirations as expressed in the Botanic Ridge Precinct Structure Plan; and,
- They are not recurrent items

1.4.2 ITEMS NOT INCLUDED IN THE DEVELOPMENT CONTRIBUTIONS PLAN

The following items are not included in the Development Contributions Plan, as they are not considered to be higher order items, but must be provided by developers as a matter of course:

- Internal access level streets and connector streets, and associated traffic management measures*.
- Flood mitigation works;
- Local drainage systems;
- Main drainage works;
- Intersections connecting the development to the existing road network, except where specified as Development Contribution Plan projects;
- Water, sewerage, underground power, gas, and telecommunication services;
- Local pathways and connections to the regional and/or district pathway network;
- Basic leveling, water tapping and landscaping of open space;
- Passive open space reserves and any agreed associated works required by the Precinct Structure Plan
- Council's plan checking and supervision costs; and,
- Bus stops.

* Exceptions to this rule apply to delivery of a constrained section of access road (RD07 & RD08) connecting to Craig Road in the stage 2 (north) of the precinct and contingency for the possible connector road and trail crossings of Southern Brown Bandicoot (SBB) corridors should these be required (CP01, CP02, CP03, CP04a and 4b).

DCP Plan 4
Transport projects
botanic ridge precinct structure plan

Scale: 1:12,500 @ A3, N.T.S. @ A4
Date: Dec 2012

0 50 100 200 500m

STAGE 4 (Possible Future)
STAGE 1 (Under Development)
ROYAL BOTANIC GARDENS CRANBOURNE
CRANBOURNE EAST
CRANBOURNE SOUTH
DEVON MEADOWS
JUNCTION VILLAGE
BALLARTO ROAD
BOTANIC DRIVE
SOUTH GIPPSLAND HIGHWAY
Proposed New Road-Casey Fields Blvd
CRAIG ROAD
DEVON ROAD
BROWNS ROAD
MADELYN CT
SMITHS LANE
FLETCHER ROAD
KELLY ROAD
FIONA DRIVE

RD01 RD02 RD03 RD04 RD05 RD06 RD07 RD08
IN01 IN02 IN03 IN04 IN05 IN06 IN07
C1-A C1-B C1-C C2 C3 C4-A C4-B C5-A C5-B C5-C C6 C7
AL1-K AL1-T AL2 AL2-A AL2-B AL2-T
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26

- Precinct Structure Plan Area
- Urban Growth Boundary
- Land
- Construction
- Town Centre Main Street (Connector Level)
- Major Arterial Road (South Gipps Highway)
- Arterial Road (2 - 4 lanes)
- Connector Street
- Access Street - Level 2
- Access Street - Key Access Level 1 Streets
- Other Development / Open Space
- Potential Mixed Use
- Regional Open Space
- Non Urban Land
- Existing Urban Area
- Intersection Roundabout large
- Intersection Roundabout
- Traffic Lights TBC

1.5 INFRASTRUCTURE PROJECTS

1.5.1 TRANSPORT

The Transport related projects in the Development Contributions Plan are based on the transport network depicted in Plan 4 which is derived from the Transport Assessment carried out for the Botanic Ridge PSP (GTA, November 2011). The transport projects comprise two categories:

- Road construction and land requirements; and,
- Construction of major controlled intersections.

The transport plan was prepared taking into account the requirements and objectives of the Botanic Ridge Precinct Structure Plan.

The transport plan was prepared taking into account the requirements and objectives of the Botanic Ridge Precinct Structure Plan.

The road and intersection projects to be funded by the Development Contributions Plan are shown on Plan 4 and Table 1.

Table 1: TRANSPORT PROJECTS

DCP PROJECT NUMBER	PROJECT TITLE	PROJECT DESCRIPTION
RD01	Land - Casey fields Boulevard extension from South Gippsland Highway to Craig Road - includes land for intersection.	Land to achieve a 20m wide interim ultimate road reserve and appropriate intersection flaring. Note, project cost incorporates a base value for the road reserve land with additional allowances for potential 'consequential losses' due to land severance etc.
RD02	Land - Craig Road Widening - (within precinct).	Land required to achieve a ultimate 34m road reserve.
RD03	Construction - Casey Fields Boulevard - Interim construction from South Gippsland Highway to intersection with Craig Road.	Construction of interim 2 lane, 10m wide carriageway as per cross section - all new works 737 linear metres length, not including intersections.
RD04	Construction - Craig Road - interim upgrade to urban standard from Browns Road to Retirement Village.	Upgrade of existing 6m wide 2 lane to interim 2 lane, 10m wide carriageway as per cross section - 1189 linear metres length not including intersections.
RD05	Construction - Browns Road - upgrade to rural arterial standard from Smiths Lane to Craig Road.	Upgrade as per Botanic Ridge Stage 1 upgrades - shoulder upgrades and kerb/channel - 2140 linear metres not including intersections.
RD06	Construction - Ballarto Road west of South Gippsland Hwy - upgrade to urban connector to facilitate public transport and trail connections.	Upgrade and construction of modified connector within constrained 21-22m existing road reserve. Includes crushed rock shared path. 202 linear metres.
RD07	Land - Constrained access level 2 street adjacent north boundary of retirement village.	Land to facilitate road delivery. Existing land is already constrained by a carriageway easement in favour of 100 Botanic Drive. Note, project cost incorporates a base value for the road reserve land with additional allowances due to acquisition process and easement rights.
RD08	Construction - Constrained access level 2 street adjacent north boundary of retirement village.	Construction of 18.5 - 22.6m wide and 275m long connector road within constrained reservation.
IN01	Construction - Casey Fields Bvd / Sth Gippsland Hwy Intersection.	New intersection connection to south side of South Gippsland Hwy to integrate with roundabout intersection being delivered by the Cranbourne East DCP.
IN02	Construction - South Gippsland Highway / Ballarto Road Intersection.	New intersection connection to south side of South Gippsland Hwy to integrate with roundabout intersection being delivered by the Cranbourne East DCP. Ballarto Road, west of South Gippsland Highway will have Connector Road status.
IN03	Construction - Craig Road / Casey Fields Boulevard Intersection.	Construction of interim roundabout to connecting Craig Road to Casey Fields Boulevard.
IN04	Construction - Craig Road / Connector C1 Intersection.	Construct controlled intersection with traffic lights compatible with interim Craig Road upgrade treatment.
IN05	Land - Browns Road deceleration lane at Connector C4.	Flaring / widening to accommodate future deceleration lane.
IN06	Construction - Browns Road / Connector C4 Intersection.	Construction of a roundabout to facilitate full turning movements for upgraded 2 lane Browns Road.
IN07	Construction - Browns Road / Connector C3 Intersection.	Construction of a roundabout to facilitate full turning movements for upgraded 2 lane Browns Road.

STAGE 4
(Possible Future)

ROYAL BOTANIC
GARDENS
CRANBOURNE

BALLARTO ROAD

CRANBOURNE
EAST

BOTANIC DRIVE

JUNCTION
VILLAGE

Existing playing
fields

SOUTH GIPPSLAND HIGHWAY

BALLARTO ROAD

PROPOSED CASEY BLVD

STAGE 2
(North)

AR01, AR02, AR03, AR04 & AR05
Active Recreation - West
2 Football / Cricket Ovals & Pavilion, Cricket Nets,
2 Netball Courts, 4 Tennis Courts, Pavillion,
Indoor Sports Facility, Car Parking

STAGE 1 (Under
Development)

C1 & C2

STAGE 3
(South)

DEVON ROAD

DEVON
MEADOWS

CRAIG ROAD

SMITHS LANE

FLETCHER ROAD

KELLY ROAD

FIONA DRIVE

BROWNS ROAD

AR06, AR07 & AR08
Active Recreation - East
2 Football / Cricket Ovals, 3
Soccer Fields, Cricket Nets,
Pavillion, Car Parking

CRANBOURNE
SOUTH

- Precinct Structure Plan Area
- Urban Growth Boundary
- Active Recreation Land & Construction
- Community Facilities Land & Construction
- Other Development / Open Spaces
- Potential Mixed Use
- Non Urban Land
- Existing Urban Area
- Major Arterial Road (South Gipps Highway)
- Arterial Road (2 - 4 lanes)
- Connector Street
- Access Street - Level 2
- Access Street - Key Access Level 1 Streets

DCP Plan 5
Community facilities & recreation projects
botanic ridge precinct structure plan

Scale: 1:12,500 @ A3, N.T.S. @ A4
Date: Dec 2012
0 50 100 200 500m

1.5.2 COMMUNITY FACILITIES

The Community Infrastructure Assessment undertaken by ASR Research and assessment by Casey City Council determined the requirement for a range of community facilities which are identified on Plan 5.

The community facility projects funded by the Development Contributions Plan include:

Table 2: COMMUNITY FACILITIES

DCP PROJECT NUMBER	PROJECT TITLE	PROJECT DESCRIPTION
C1	Land - Community Centre 1.	Land for north western community centre.
C2	Construction - Community Centre 1.	Construction of integrated multi-purpose Community Centre to include x4 Early Years kindergarten rooms and associated external play areas, consulting rooms for maternal and child health (including 2 offices and waiting area), flexible meeting space for play groups etc.

1.5.3 RECREATION

The analysis undertaken by ASR Research and Casey City Council determined a number of facilities required to be built on the various active open space areas to meet the needs of the future community, refer Plan 5.

The recreation projects funded by the Development Contributions Plan include:

Table 3: RECREATION PROJECTS

DCP PROJECT NUMBER	PROJECT TITLE	PROJECT DESCRIPTION
AR01	Land - Active Recreation West.	Land for active recreation inclusive of sportsfields, tennis, netball pavilions and carparking.
AR02	Construction - Active Recreation West - Sportsfields.	Construction of x2 full size AFL / cricket ovals, cricket nets, x4 tennis courts, x2 netball courts, car parking and associated landscaping. Oval construction will require balance cut and fill.
AR03	Construction - Active Recreation West - Main Pavilion.	Construction of pavilion to service cricket, football and netball.
AR04	Construction - Active Recreation West - Tennis Pavilion.	Construction of pavilion to service tennis.
AR05	Construction - Active Recreation West - Indoor Sports Facility.	Lower order and flexible x2 basketball court sized indoor facility. Delivery shared with primary school.
AR06	Land - Active Recreation East.	Land for active recreation inclusive of sportsfields, pavilion and carparking.
AR07	Construction - Active Recreation East - Sportsfields.	Construction of x2 full size cricket oval combined with x3 soccer fields, cricket nets and car parking.
AR08	Construction - Active Recreation East - Pavilion.	Construction of pavilion to service cricket and soccer.

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1.5.4 PROJECT TIMING

General

Each item in the Development Contributions Plan has an assumed indicative provision trigger specified in Table 7. The timing of the provision of the items in this Development Contributions Plan is consistent with information available at the time that the Development Contributions Plan was prepared. The Development Agency will monitor and assess the required timing for individual items and may seek an amendment to the Casey Planning Scheme to adjust indicative provision triggers as part of its 5 year review (or earlier if justified) of the Scheme.

While indicative provision triggers are provided these do not preclude the early provision of certain infrastructure to be constructed/ provided by development proponents as works or land in-kind, if agreed to by the Collecting Agency.

Casey Fields Boulevard

Casey Fields Boulevard extension and associated intersection treatments(DCP Projects RD01, RD03, IN01 and IN03) must be delivered prior to the issue of a statement of compliance for 1,600th lot, unless otherwise agreed in writing by VicRoads.

1.5.5 DISTINCTION BETWEEN COMMUNITY AND DEVELOPMENT INFRASTRUCTURE

In accordance with the Planning and Environment Act 1987 and the Ministerial Direction on Development Contributions, the Development Contributions Plan makes a distinction between "development" and "community" infrastructure. The timing of payment of contributions is linked to the type of infrastructure in question.

For community infrastructure, contributions are to be made by the home-builder at the time of building approval. Contributions relating to community infrastructure will be paid for at a "per dwelling" rate.

The Community infrastructure Levy payable must be calculated in accordance with section 46L of the Planning and Environment Act, including any variation to the maximum amount made by order of the Governor in Council. The current rate per dwelling is $900.

The following infrastructure projects are deemed to be community infrastructure:

Table 4: COMMUNITY INFRASTRUCTURE PROJECTS

DCP PROJECT NUMBER	PROJECT TITLE	PROJECT DESCRIPTION
AR03	Construction - Active Recreation West - Main Pavilion.	Construction of pavilion to service cricket, football and netball.
AR04	Construction - Active Recreation West - Tennis Pavilion.	Construction of pavilion to service tennis.
AR05	Construction - Active Recreation West - Indoor Sports Facility.	Lower order and flexible x2 basketball court sized indoor facility. Delivery shared with primary school.
AR08	Construction - Active Recreation East - Pavilion.	Construction of pavilion to service cricket and soccer.

All other infrastructure projects are considered to be in the development infrastructure category. Contributions relating to development infrastructure are to be paid prior to the commencement of development and use of the land, or in the case of subdivision, prior to the issue of the statement of compliance (for the whole or relevant stage of the subdivision).

2.0 CALCULATION OF CONTRIBUTIONS

Part 1 sets out the strategic basis for this Development Contributions Plan and identifies infrastructure items to be included in the Development Contributions Plan. Part 2 focuses on the calculation of contributions and apportionment of costs.

The general cost apportionment method includes the following steps:

- Calculation of the net developable area and demand units (refer Tables 5 and 6);
- Calculation of projects costs (refer Table 7);
- Allowance for external use, and cost apportionment (refer Table 7);
- Calculation of service catchments (refer Table 7);
- Identification of development types required to pay the levy (refer Table 7);
- A charge per net developable hectare for all land included in the Development Contributions Plan (Table 7); and
- Summary of costs payable by development type and precinct for each infrastructure category (refer Table 8a and Table 8b).

2.1 CALCULATION OF NET DEVELOPABLE AREA AND DEMAND UNITS

2.1.1 INTRODUCTION

The following section sets out how Net Developable Area (NDA) is calculated, provides a detailed land budget for every property within the Botanic Ridge Precinct Structure Plan and outlines the residential and employment projection anticipated for the Precinct Structure Plan area.

2.1.2 NET DEVELOPABLE AREA

In this Development Contributions Plan, all development infrastructure contributions are payable on the net developable land on any given development site.

Residential development land including town centres will contribute to all items forming part of the Development Contributions Plan.

For the purposes of this Development Contributions Plan, Net Developable Area is defined as the total amount of land within the precinct that is made available for development of housing and employment, including lots and local and connector streets. It is the total precinct area minus community facilities, government schools, existing educational facilities, open space, encumbered land*, and arterial roads. Additional small local pocket parks and / or pedestrian links defined at the subdivision stage are included in Net Developable Area (NDA).

The NDA of each property within the DCP area is set out in Tables 5 and 6 and provides the basis for levy apportionment under this Plan.

Tables 5 and 6 should be read in conjunction with Plan 6.

* Note – encumbered land includes land set aside for drainage, flora and fauna conservation, aboriginal cultural heritage and defendable space (fire buffers).

2.1.3 LAND BUDGET AND DEMAND UNITS

Net developable hectares (NDHA) is the demand unit for this Development Contributions Plan.

The NDA is the basis upon which the development contribution levies are payable.

2.1.4 DEVELOPMENT AND POPULATION PROJECTIONS

The Botanic Ridge Precinct Structure Plan covers a total area of 317.7 hectares.

The NDA for the Botanic Ridge Precinct Structure Plan is calculated in Table 5.

It should be noted that the areas designated for the Local Town Centres have been included as part of the NDA and NDA within these centres attracts the payment of levies.

2.2 CALCULATION OF CONTRIBUTION CHARGES

2.2.1 CALCULATION OF COSTS

Each project has been assigned a land and/or construction cost. These costs are listed in Table 7. The costs are expressed in November 2012 dollars and will be indexed annually in accordance with the indexation method specified in Section 3.1.6. A summary of the total project costs by infrastructure category (including the Development Infrastructure rate per NDHa) is provided in Tables 8a and 8b.

VALUATION OF LAND

The cost of each land project was determined by a registered valuer appointed by the Growth Areas Authority based on a compensation based valuation to determine the current market value of the land required in accordance with the Precinct Structure Plan and Development Contributions Plan.

CALCULATION OF CONSTRUCTION COSTS

All road construction costs have been determined by a certified engineer appointed by the Growth Areas Authority.

All sports fields and community building construction costs have been determined by a certified engineer appointed by the Growth Areas Authority, and familiar with City of Casey requirements and standards.

Detailed project cost sheets are included in Section 5 of this DCP.

2.2.2 EXTERNAL USE

The strategic planning analysis underpinning this DCP has informed the allowance for other use external to the Main Catchment Area (MCA) for specific projects – that is use that does not emanate from the Botanic Ridge Precinct Structure Plan Development Contributions Plan area alone.

Table 7 quantifies any external demand (as a percentage) for each infrastructure project. Where this is the case, a percentage discount has been made to the dollar amount that will be recovered. The relevant projects are:

- C1 - Community Centre 1 – Land – refer Plan 5 for highlighted land area; and
- C2 - Community Centre 1 – Construction
- AR05 – Active Recreation West - Indoor Sports Facility

2.2.3 COST APPORTIONMENT

This Development Contributions Plan apportions a charge to new development according to its projected share of use of identified infrastructure items. Since development contributions charges are levied 'up-front', a measure of actual use by individual development sites is not possible. Therefore costs must be shared in accordance with the estimated share of use.

This Development Contributions Plan cannot and does not require payment from existing or approved development. However, the share of use that existing development receives from these items is taken into account when calculating the contributions expected from new development. This means new development only pays its fair share of the estimated cost of new infrastructure and services (and does not pay for the use by existing development).

This Development Contributions Plan calculates what each new development should pay towards provision of the identified infrastructure item. This is the total cost of the items (after deducting other funding sources and making allowance for any external demand) divided by total (existing and proposed) demand units within its catchment, then aggregated for all items used by a new development.

If a new development is not in the catchment for a particular item, it does not pay towards the cost of that item. The balance of the cost of the items not recovered under this Development Contributions Plan will be funded from alternative sources.

To support this approach, a Main Catchment Area has been determined for each item.

2.2.4 MAIN CATCHMENT AREAS

The Main Catchment Area (MCA) is the geographic area from which a given item of infrastructure will draw most of its use. The Development Contributions Plan has a single MCA to which the use of local infrastructure has been allocated.

The spatial extent of the MCA and charge areas is identified as the DCP area as shown on Plan 3.

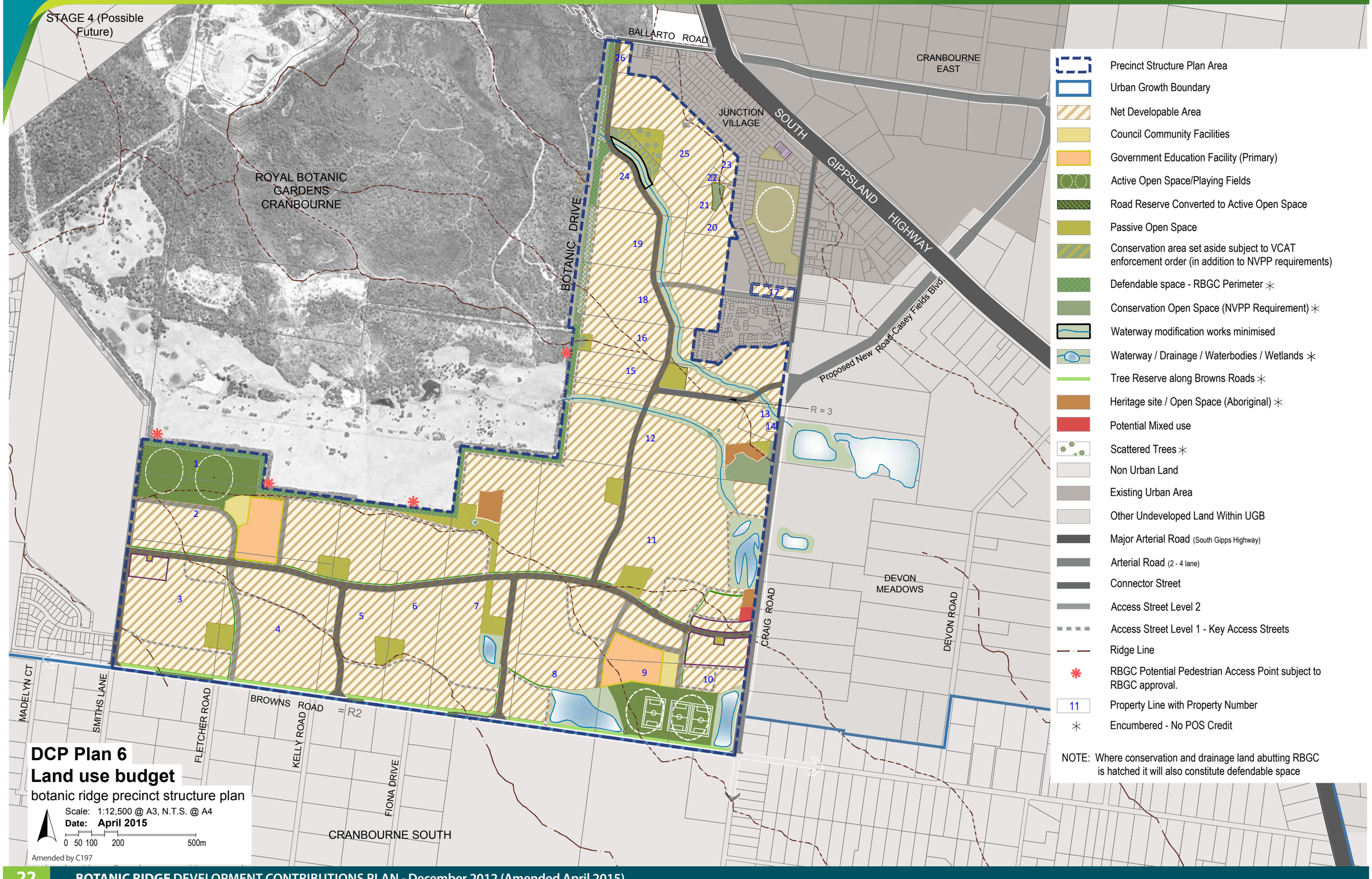
It is important to note that the number of demand units (net developable hectares) in the MCA is based on the land budgets in Tables 5 and 6.

The per hectare contributions will not be amended to respond to minor changes to land budgets that may result from the subdivision process. In other words, the Development Contributions Plan is permanently linked to the Detailed Land Budget in Table 6.

For the purposes of the Development Contributions Plan, the number of developable hectares will only change if the Collecting Agency agrees to a variation to the Precinct and Detailed Land Budget and associated tables. Table 6 should be used to determine the number of developable hectares (for Development Contributions Plan purposes) on individual parcels.

Table 5: SUMMARY LAND USE BUDGET

	NORTH (STAGE 2)			SOUTH (STAGE 3)			TOTAL		
	HECTARES	% OF TOTAL PRECINCT	% OF NDA	HECTARES	% OF TOTAL PRECINCT	% OF NDA	HECTARES	% OF TOTAL PRECINCT	% OF NDA
TOTAL PRECINCT AREA	**71.21**	**100.0%**		**246.36**	**100.0%**		**317.57**	**100.0%**	
TRANSPORT									
Arterial Roads / widening - Casey Fields Bvd / Craig Rd & Browns Rd	0.00	0.00%	0.00%	2.65	1.08%	1.44%	2.65	0.84%	1.10%
Road Reserve Not Available for Development	2.47	3.47%	4.36%	0.96	0.39%	0.52%	3.43	1.08%	1.42%
Tree reserve (Browns Road)	0.00	0.00%	0.00%	2.93	1.19%	1.59%	2.93	0.92%	1.22%
SUB-TOTAL	**2.47**	**3.47%**	**4.36%**	**6.55**	**2.66%**	**3.55%**	**9.01**	**2.84%**	**3.74%**
COMMUNITY FACILITIES									
Community Services Facilities	0.00	0.00%	0.00%	1.61	0.65%	0.87%	1.61	0.51%	0.67%
SUB-TOTAL	**0.00**	**0.00%**	**0.00%**	**1.61**	**0.65%**	**0.87%**	**1.61**	**0.51%**	**0.67%**
GOVERNMENT EDUCATION									
Government Schools	0.00	0.00%	0.00%	7.00	2.84%	3.79%	7.00	2.21%	2.90%
SUB-TOTAL	**0.00**	**0.00%**	**0.00%**	**7.00**	**2.84%**	**3.79%**	**7.00**	**2.21%**	**2.90%**
OPEN SPACE									
ENCUMBERED LAND AVAILABLE FOR RECREATION *									
Waterway / Drainage Line / Wetland / retarding	5.61	7.87%	9.90%	13.29	5.39%	7.20%	18.89	5.95%	7.83%
Heritage - Aboriginal	0.00	0.00%	0.00%	2.22	0.90%	1.20%	2.22	0.70%	0.92%
RBGC defendable space 'outer zone'	1.78	2.50%	3.14%	4.50	1.83%	2.44%	6.28	1.98%	2.60%
Conservation	2.81	3.95%	4.97%	2.16	0.87%	1.17%	4.97	1.56%	2.06%
SUB-TOTAL ENCUMBERED LAND FOR RECREATION	**10.20**	**14.32%**	**18.01%**	**22.16**	**9.00%**	**12.01%**	**32.36**	**10.19%**	**13.41%**
UNENCUMBERED LAND AVAILABLE FOR RECREATION									
Active Open Space	0.00	0.0%	0.00%	16.20	6.6%	8.78%	16.20	5%	6.72%
Passive Open Space	1.90	2.7%	3.36%	8.26	3.4%	4.48%	10.16	3.2%	4.21%
SUB-TOTAL UNENCUMBERED LAND FOR RECREATION	**1.90**	**2.67%**	**3.36%**	**24.47**	**9.93%**	**13.26%**	**26.37**	**8.30%**	**10.93%**
SUB-TOTAL - ALL OPEN SPACE	**12.10**	**17.0%**	**21.36%**	**46.63**	**18.9%**	**25.26%**	**58.73**	**18.5%**	**24.35%**
NET DEVELOPABLE AREA (NDA) HA	**56.64**	**79.54%**		**184.58**	**74.92%**		**241.22**	**75.96%**	



DCP Plan 6
Land use budget

botanic ridge precinct structure plan

Scale: 1:12,500 @ A3, N.T.S. @ A4
Date: **April 2015**

0 50 100 200 500m

Amended by C197

- Precinct Structure Plan Area
- Urban Growth Boundary
- Net Developable Area
- Council Community Facilities
- Government Education Facility (Primary)
- Active Open Space/Playing Fields
- Road Reserve Converted to Active Open Space
- Passive Open Space
- Conservation area set aside subject to VCAT enforcement order (in addition to NVPP requirements)
- Defendable space - RBGC Perimeter ⁎
- Conservation Open Space (NVPP Requirement) ⁎
- Waterway modification works minimised
- Waterway / Drainage / Waterbodies / Wetlands ⁎
- Tree Reserve along Browns Roads ⁎
- Heritage site / Open Space (Aboriginal) ⁎
- Potential Mixed use
- Scattered Trees ⁎
- Non Urban Land
- Existing Urban Area
- Other Undeveloped Land Within UGB
- Major Arterial Road (South Gipps Highway)
- Arterial Road (2 - 4 lane)
- Connector Street
- Access Street Level 2
- Access Street Level 1 - Key Access Streets
- Ridge Line
- RBGC Potential Pedestrian Access Point subject to RBGC approval.
- 11 Property Line with Property Number
- ⁎ Encumbered - No POS Credit

NOTE: Where conservation and drainage land abutting RBGC is hatched it will also constitute defendable space

Table 6: PROPERTY SPECIFIC LAND USE BUDGET

		TRANSPORT		COMMUNITY		ENCUMBERED LAND FOR RECREATION					UNENCUMBERED LAND FOR RECREATION		OTHER	TOTAL NET DEVELOPABLE AREA (HECTARES)	KEY PERCENTAGES				52.01 OPEN SPACE DEL TARGET %	DIFFERENCE % NDA	DIFFERENCE - EQUIVALENT LAND AREA Ha
PROPERTY NUMBER	TOTAL AREA (HECTARES)	ARTERIAL ROAD - CASEY FILEDS BVD / BROWNS RD	ROAD RESERVE NOT AVAILABLE FOR DEVELOPMENT	COMMUNITY FACILITIES	GOVERNMENT SCHOOLS	POWER EASEMENT	WATERWAY / DRAINAGE LINE / WETLAND / RETARDING	HERITAGE (Aboriginal)	RBGC DEFENDABLE SPACE 'OUTER ZONE'	CONSERVATION	ACTIVE OPEN SPACE	PASSIVE OPEN SPACE	TREE RESERVE (Browns Road)		NET DEVPT AREA % OF PRECINCT	ACITVE OPEN SPACE% NDA	PASSIVE OPEN SPACE % NDA	TOTAL PASSIVE & ACTIVE OPEN SPACE %			
SOUTH																					
Property 1	6.01								1.15		4.86			0.00	0.00%	n.a.	n.a.	n.a.	4.21%	0.00%	0.00
Property 2	20.06			0.80	3.11				0.66		3.77	0.38		11.34	56.54%	33.25%	3.39%	36.64%	4.21%	-0.83%	-0.09
Property 3	18.59											0.47	0.50	17.61	94.76%	0.00%	2.67%	2.67%	4.21%	-1.54%	-0.27
Property 4	19.95				0.40							0.81	0.53	18.22	91.32%	0.00%	4.45%	4.45%	4.21%	0.24%	0.04
Property 5	17.16								0.36			1.01	0.27	15.53	90.51%	0.00%	6.48%	6.48%	4.21%	2.26%	0.35
Property 6	15.45								0.50			0.92	0.25	13.78	89.20%	0.00%	6.66%	6.66%	4.21%	2.44%	0.34
Property 7	32.37	0.05					0.99	1.06	1.15	0.05		2.58	0.34	26.15	80.80%	0.00%	9.85%	9.85%	4.21%	5.64%	1.47
Property 8	9.09						2.69						0.41	5.98	65.79%	0.00%	0.00%	0.00%	4.21%	-4.21%	-0.25
Property 9	8.43			0.56	1.97		1.31			0.02	4.24		0.34	0.00	0.00%	n.a.	0.00%	n.a.	4.21%	-4.21%	0.00
Property 10	8.28	0.26					1.96				3.34		0.29	2.44	29.41%	136.98%	0.00%	136.98%	4.21%	-4.21%	-0.10
Property 11	64.58			0.25	1.53		3.85	0.36	0.15			1.89		56.55	87.57%	0.00%	3.34%	3.34%	4.21%	-0.87%	-0.49
Property 12	22.90						2.49	0.80	0.53	1.89		0.21		16.98	74.14%	0.00%	1.22%	1.22%	4.21%	-2.99%	-0.51
SUB-TOTAL	**242.86**	**0.31**	**0.00**	**1.61**	**7.00**	**0.00**	**13.29**	**2.22**	**4.50**	**1.96**	**16.20**	**8.26**	**2.93**	**184.58**	**76.00%**	**8.78%**	**4.48%**	**13.26%**	**4.21%**	**9.04%**	**0.49**
ROAD RESERVES																					
1 (Smiths La)	0.96		0.96						0.00		0.00			0.00	0.00%	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
2 (Browns Rd)	2.54	2.35								0.20				0.00	0.00%	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
SUB-TOTAL	**3.50**	**2.35**	**0.96**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.20**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00%**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
TOTAL SOUTH	**246.36**	**2.65**	**0.96**	**1.61**	**7.00**	**0.00**	**13.29**	**2.22**	**4.50**	**2.16**	**16.20**	**8.26**	**2.93**	**184.58**	**74.92%**	**8.78%**	**4.48%**	**13.26%**			
NORTH																					
Property 13	1.43						0.20							1.23	86.06%	0.00%	0.00%	0.00%	4.21%	-4.21%	-0.05
Property 14	0.14													0.14	100.00%	0.00%	0.00%	0.00%	4.21%	-4.21%	-0.01
Property 15	10.78						1.05		0.31			0.84		8.57	79.57%	0.00%	9.85%	9.85%	4.21%	5.63%	0.48
Property 16	6.68						0.64		0.22	0.28		0.20		5.35	80.09%	0.00%	3.72%	3.72%	4.21%	-0.50%	-0.03
Property 17	0.68													0.68	100.00%	0.00%	0.00%	0.00%	4.21%	-4.21%	-0.03
Property 18	13.85						1.05		0.06	0.14				12.60	91.01%	0.00%	0.00%	0.00%	4.21%	-4.21%	-0.53
Property 19	8.12						1.06		0.05	0.14				6.87	84.57%	0.00%	0.00%	0.00%	4.21%	-4.21%	-0.29
Property 20	1.26													1.26	100.00%	0.00%	0.00%	0.00%	4.21%	-4.21%	-0.05
Property 21	1.97									0.37				1.59	81.03%	0.00%	0.00%	0.00%	4.21%	-4.21%	-0.07
Property 22	0.41													0.41	100.00%	0.00%	0.00%	0.00%	4.21%	-4.21%	-0.02
Property 23	0.72													0.72	100.00%	0.00%	0.00%	0.00%	4.21%	-4.21%	-0.03
Property 24	8.50						1.36		0.67	1.09		0.00		5.38	63.26%	0.00%	0.00%	0.00%	4.21%	-4.21%	-0.23
Property 25	11.73						0.16		0.22	0.19		0.86		10.30	87.80%	0.00%	8.33%	8.33%	4.21%	4.12%	0.42
Property 26	0.97								0.21					0.76	78.71%	0.00%	0.00%	0.00%	4.21%	-4.21%	-0.03
SUB-TOTAL	**67.23**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**5.51**	**0.00**	**1.74**	**2.21**	**0.00**	**1.90**	**0.00**	**55.87**	**83.10%**	**0.00%**	**3.40%**	**3.40%**	**4.21%**	**-0.81%**	**-0.45**
ROAD RESERVES																					
3	1.68		0.77				0.09		0.04					0.77	46.21%	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
4 (Botanic Dr)	2.31		1.70							0.61				0.00	0.00%	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
SUB-TOTAL	**3.98**	**0.00**	**2.47**	**0.00**	**0.00**	**0.00**	**0.09**	**0.00**	**0.04**	**0.61**	**0.00**	**0.00**	**0.00**	**0.77**	**19.44%**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**	**n.a.**
TOTAL NORTH	**71.21**	**0.00**	**2.47**	**0.00**	**0.00**	**0.00**	**5.61**	**0.00**	**1.78**	**2.81**	**0.00**	**1.90**	**0.00**	**56.64**	**79.54%**	**0.00%**	**3.36%**	**3.36%**			
TOTAL PRECINCT	**317.57**	**2.65**	**3.43**	**1.61**	**7.00**	**0.00**	**18.89**	**2.22**	**6.28**	**4.97**	**16.20**	**10.16**	**2.93**	**241.22**	**75.96%**	**6.72%**	**4.21%**	**10.93%**			

Table 7: INFRASTRUCTURE PROJECT TIMING AND CALCULATION OF COSTS (all costs in 2012 dollars)

DCP PROJECT NO.	PROJECT	INFRASTRUCTURE CATEGORY	WORKS DESCRIPTION	INDICATIVE PROVISION TRIGGER	LAND AREA HA	PROJECT COST: LAND	PROJECT COST: CONSTRUCTION	TOTAL ESTIMATED PROJECT COST: LAND & CONSTRUCTION	PROPORTION OF COST ATTRIBUTABLE TO THE MCA %	TOTAL COST ATTRIBUTABLE TO MCA	DEVELOPMENT TYPES MAKING CONTRIBUTION	LAND COST PER NDHA	CONSTRUCTION COST PER NDHA	TOTAL COST PER NDHA	COMMUNITY INFRASTRUCTURE LEVY (PER DWELLING)
ROAD PROJECTS															
RD01	Land - Casey fields Boulevard extension from South Gippsland Highway to Craig Road - includes land for intersection.	Development	Land to achieve a 20m wide interim ultimate road reserve and appropriate intersection flaring. Note, project cost incorporates a base value for the road reserve land with additional allowances for potential 'consequential losses' due to land severance etc.	prior to certificate of compliance for 1,600th lot.	0.90	$550,000	$0	$550,000	100%	$550,000	Residential and town centres	$2,280	$0	$2,280	$0
RD02	Land - Craig Road Widening - (within precinct).	Development	Land required to achieve a ultimate 34m road reserve.	By completion of the DCP	0.26	$144,100	$0	$144,100	100%	$144,100	Residential and town centres	$597	$0	$597	$0
RD03	Construction - Casey Fields Boulevard - Interim construction from South Gippsland Highway to intersection with Craig Road.	Development	Construction of interim 2 lane, 10m wide carriageway as per cross section - all new works 737 linear metres length, not including intersections.	prior to certificate of compliance for 1,600th lot.		$0	$2,142,112	$2,142,112	100%	$2,142,112	Residential and town centres	$0	$8,880	$8,880	$0
RD04	Construction - Craig Road - interim upgrade to urban standard from Browns Road to Retirement Village.	Development	Upgrade of existing 6m wide 2 lane to interim 2 lane, 10m wide carriageway as per cross section - 1189 linear metres length not including intersections.	By completion of the DCP		$0	$4,332,670	$4,332,670	100%	$4,332,670	Residential and town centres	$0	$17,962	$17,962	$0
RD05	Construction - Browns Road - upgrade to rural arterial standard from Smiths Lane to Craig Road.	Development	Upgrade as per Botanic Ridge Stage 1 upgrades - shoulder upgrades and kerb/channel - 2140 linear metres not including intersections.	By completion of the DCP		$0	$2,737,083	$2,737,083	100%	$2,737,083	Residential and town centres	$0	$11,347	$11,347	$0
RD06	Construction - Ballarto Road west of South Gippsland Hwy - upgrade to urban connector to facilitate public transport and trail connections.	Development	Upgrade and construction of modified connector within constrained 21-22m existing road reserve. Includes crushed rock shared path. 202 linear metres.	By completion of the DCP		$0	$644,398	$644,398	100%	$644,398	Residential and town centres	$0	$2,671	$2,671	$0
RD07	Land - Constrained access level 2 street adjacent north boundary of retirement village.	Development	Land to facilitate road delivery. Existing land is already constrained by a carriageway easement in favour of 100 Botanic Drive. Note, project cost incorporates a base value for the road reserve land with additional allowances due to acquisition process and easement rights.	Where practicable during early stages of development.	0.51	$305,000	$0	$305,000	100%	$305,000	Residential and town centres	$1,264	$0	$1,264	$0
RD08	Construction - Constrained access level 2 street adjacent north boundary of retirement village.	Development	Construction of 18.5 - 22.6m wide and 275m long connector road within constrained reservation.	Where practicable during early stages of development, otherwise as demand requires.		$0	$812,005	$812,005	100%	$812,005	Residential and town centres	$0	$3,366	$3,366	$0
INTERSECTION PROJECTS															
IN01	Construction - Casey Fields Bvd / Sth Gippsland Hwy Intersection.	Development	New intersection connection to south side of South Gippsland Hwy to integrate with roundabout intersection being delivered by the Cranbourne East DCP.	prior to certificate of compliance for 1,600th lot.		$0	$217,555	$217,555	100%	$217,555	Residential and town centres	$0	$902	$902	$0
IN02	Construction - South Gippsland Highway / Ballarto Road Intersection.	Development	New intersection connection to south side of South Gippsland Hwy to integrate with roundabout intersection being delivered by the Cranbourne East DCP. Ballarto Road, west of South Gippsland Highway will have Connector Road status.	By completion of the DCP		$0	$528,575	$528,575	100%	$528,575	Residential and town centres	$0	$2,191	$2,191	$0
IN03	Construction - Craig Road / Casey Fields Boulevard Intersection.	Development	Construction of interim roundabout to connecting Craig Road to Casey Fields Boulevard.	prior to certificate of compliance for 1,600th lot.		$0	$628,990	$628,990	100%	$628,990	Residential and town centres	$0	$2,608	$2,608	$0
IN04	Construction - Craig Road / Connector C1 Intersection.	Development	Construct controlled intersection with traffic lights compatible with interim Craig Road upgrade treatment.	At the time of eastern town centre development or otherwise as demand requires		$0	$2,294,866	$2,294,866	100%	$2,294,866	Residential and town centres	$0	$9,514	$9,514	$0
IN05	Land - Browns Road deceleration lane at Connector C4.	Development	Flaring / widening to accommodate future deceleration lane.	At the time of subdivision.	0.03	$16,500	$0	$16,500	100%	$16,500	Residential and town centres	$68	$0	$68	$0
IN06	Construction - Browns Road / Connector C4 Intersection.	Development	Construction of a roundabout to facilitate full turning movements for upgraded 2 lane Browns Road.	At the time of subdivision.		$0	$720,894	$720,894	100%	$720,894	Residential and town centres	$0	$2,989	$2,989	$0
IN07	Construction - Browns Road / Connector C3 Intersection.	Development	Construction of a roundabout to facilitate full turning movements for upgraded 2 lane Browns Road.	At the time of subdivision.		$0	$720,894	$720,894	100%	$720,894	Residential and town centres	$0	$2,989	$2,989	$0

Table 7: INFRASTRUCTURE PROJECT TIMING AND CALCULATION OF COSTS (all costs in 2012 dollars) *continued*

DCP PROJECT NO.	PROJECT	INFRASTRUCTURE CATEGORY	WORKS DESCRIPTION	INDICATIVE PROVISION TRIGGER	LAND AREA HA	PROJECT COST: LAND	PROJECT COST: CONSTRUCTION	TOTAL ESTIMATED PROJECT COST: LAND & CONSTRUCTION	PROPORTION OF COST ATTRIBUTABLE TO THE MCA %	TOTAL COST ATTRIBUTABLE TO MCA	DEVELOPMENT TYPES MAKING CONTRIBUTION	LAND COST PER NDHA	CONSTRUCTION COST PER NDHA	TOTAL COST PER NDHA	COMMUNITY INFRASTRUCTURE LEVY (PER DWELLING)
COMMUNITY CENTRES															
C1	Land - Community Centre 1.	Development	Land for north western community centre.	When population growth triggers requirement for additional facilities.	0.80	$440,550	$0	$440,550	71%	$312,791	Residential and town centres	$1,297	$0	$1,297	$0
C2	Construction - Community Centre 1.	Development	Construction of integrated multi-purpose Community Centre to include x4 Early Years kindergarten rooms and associated external play areas, consulting rooms for maternal and child health (including 2 offices and waiting area), flexible meeting space for play groups etc.	When population growth triggers requirement for additional facilities.		$0	$5,795,696	$5,795,696	71%	$4,114,944	Residential and town centres	$0	$17,059	$17,059	$0
ACTIVE RECREATION															
AR01	Land - Active Recreation West.	Development	Land for active recreation inclusive of sportsfields, tennis, netball pavilions and carparking.	When population growth triggers requirement for additional facilities.	8.63	$4,746,500	$0	$4,746,500	100%	$4,746,500	Residential and town centres	$19,677	$0	$19,677	$0
AR02	Construction - Active Recreation West - Sportsfields.	Development	Construction of x2 full size AFL / cricket ovals, cricket nets, x4 tennis courts, x2 netball courts, car parking and associated landscaping. Oval construction will require balance cut and fill.	When population growth triggers requirement for additional facilities.		$0	$5,050,052	$5,050,052	100%	$5,050,052	Residential and town centres	$0	$20,936	$20,936	$0
AR03	Construction - Active Recreation West - Main Pavilion.	Community	Construction of pavilion to service cricket, football and netball.	When population growth triggers requirement for additional facilities.		$0	$2,567,582	$2,567,582	100%	$2,567,582	Residential and town centres	$0	$10,644	$10,644	Funded via CIL
AR04	Construction - Active Recreation West - Tennis Pavilion.	Community	Construction of pavilion to service tennis.	When population growth triggers requirement for additional facilities.		$0	$738,241	$738,241	100%	$738,241	Residential and town centres	$0	$3,060	$3,060	Funded via CIL
AR05	Construction - Active Recreation West - Indoor Sports Facility.	Community	Lower order and flexible x2 basketball court sized indoor facility. Delivery shared with primary school.	When population growth triggers requirement for additional facilities.		$0	$2,802,917	$2,802,917	50%	$1,401,459	Residential and town centres	$0	$5,810	$5,810	Funded via CIL
AR06	Land - Active Recreation East.	Development	Land for active recreation inclusive of sportsfields, pavilion and carparking.	When population growth triggers requirement for additional facilities.	7.57	$4,165,700	$0	$4,165,700	100%	$4,165,700	Residential and town centres	$17,270	$0	$17,270	$0
AR07	Construction - Active Recreation East - Sportsfields.	Development	Construction of x2 full size cricket oval combined with x3 soccer fields, cricket nets and car parking.	When population growth triggers requirement for additional facilities.		$0	$5,473,464	$5,473,464	100%	$5,473,464	Residential and town centres	$0	$22,691	$22,691	$0
AR08	Construction - Active Recreation East - Pavilion.	Community	Construction of pavilion to service cricket and soccer.	When population growth triggers requirement for additional facilities.		$0	$2,220,348	$2,220,348	100%	$2,220,348	Residential and town centres	$0	$9,205	$9,205	Funded via CIL
CONTINGENCY															
CP01 to CP08	Development Infrastructure Contingency Pending Commonwealth Decision On The Southern Brown Bandicoot Sub Regional Strategy	Development	Includes allowance for net additonal land / construction project costs and increase in rate due to reduction in developable area.	At time of subdivision							Residential and town centres			**$16,080**	**$0**

Apportionment Notes:

C1 & C2 - Based on 29% of facility demand being generated by stage 1 of the precinct, which is an already permitted development not part of the current PSP or DCP.

AR05 - Based on a single court facility being a typical primary school provision, therefore funding 50% of a 2 court facility.

Table 8a: SCHEDULE OF COSTS (all costs in 2012 dollars)

DCP PROJECT NO.	PROJECT	INFRASTRUCTURE CATEGORY	WORKS DESCRIPTION	TOTAL ESTIMATED PROJECT COST: LAND & CONSTRUCTION	TOTAL FUNDED VIA BOTANIC RIDGE DCP	COMMUNITY INFRASTRUCTURE LEVY
ROAD PROJECTS						
RD01	Land - Casey fields Boulevard extension from South Gippsland Highway to Craig Road - includes land for intersection.	Development	Land to achieve a 20m wide interim ultimate road reserve and appropriate intersection flaring. Note, project cost incorporates a base value for the road reserve land with additional allowances for potential 'consequential losses' due to land severance etc.	$550,000	$550,000	$0
RD02	Land - Craig Road Widening - (within precinct).	Development	Land required to achieve a ultimate 34m road reserve.	$144,100	$144,100	$0
RD03	Construction - Casey Fields Boulevard - Interim construction from South Gippsland Highway to intersection with Craig Road.	Development	Construction of interim 2 lane, 10m wide carriageway as per cross section - all new works 737 linear metres length, not including intersections.	$2,142,112	$2,142,112	$0
RD04	Construction - Craig Road - interim upgrade to urban standard from Browns Road to Retirement Village.	Development	Upgrade of existing 6m wide 2 lane to interim 2 lane, 10m wide carriageway as per cross section - 1189 linear metres length not including intersections.	$4,332,670	$4,332,670	$0
RD05	Construction - Browns Road - upgrade to rural arterial standard from Smiths Lane to Craig Road.	Development	Upgrade as per Botanic Ridge Stage 1 upgrades - shoulder upgrades and kerb/channel - 2140 linear metres not including intersections.	$2,737,083	$2,737,083	$0
RD06	Construction - Ballarto Road west of South Gippsland Hwy - upgrade to urban connector to facilitate public transport and trail connections.	Development	Upgrade and construction of modified connector within constrained 21-22m existing road reserve. Includes crushed rock shared path. 202 linear metres.	$644,398	$644,398	$0
RD07	Land - Constrained access level 2 street adjacent north boundary of retirement village.	Development	Land to facilitate road delivery. Existing land is already constrained by a carriageway easement in favour of 100 Botanic Drive. Note, project cost incorporates a base value for the road reserve land with additional allowances due to acquisition process and easement rights.	$305,000	$305,000	$0
RD08	Construction - Constrained access level 2 street adjacent north boundary of retirement village.	Development	Construction of 18.5 - 22.6m wide and 275m long connector road within constrained reservation.	$812,005	$812,005	$0
SUB-TOTAL ROADS				**$11,667,368**	**$11,667,368**	**$0**
INTERSECTION PROJECTS						
IN01	Construction - Casey Fields Bvd / Sth Gippsland Hwy Intersection.	Development	New intersection connection to south side of South Gippsland Hwy to integrate with roundabout intersection being delivered by the Cranbourne East DCP.	$217,555	$217,555	$0
IN02	Construction - South Gippsland Highway / Ballarto Road Intersection.	Development	New intersection connection to south side of South Gippsland Hwy to integrate with roundabout intersection being delivered by the Cranbourne East DCP. Ballarto Road, west of South Gippsland Highway will have Connector Road status.	$528,575	$528,575	$0
IN03	Construction - Craig Road / Casey Fields Boulevard Intersection.	Development	Construction of interim roundabout to connecting Craig Road to Casey Fields Boulevard.	$628,990	$628,990	$0
IN04	Construction - Craig Road / Connector C1 Intersection.	Development	Construct controlled intersection with traffic lights compatible with interim Craig Road upgrade treatment.	$2,294,866	$2,294,866	$0
IN05	Land - Browns Road deceleration lane at Connector C4.	Development	Flaring / widening to accommodate future deceleration lane.	$16,500	$16,500	$0
IN06	Construction - Browns Road / Connector C4 Intersection.	Development	Construction of a roundabout to facilitate full turning movements for upgraded 2 lane Browns Road.	$720,894	$720,894	$0
IN07	Construction - Browns Road / Connector C3 Intersection.	Development	Construction of a roundabout to facilitate full turning movements for upgraded 2 lane Browns Road.	$720,894	$720,894	$0
SUB-TOTAL INTERSECTIONS				**$5,128,274**	**$5,128,274**	**$0**

Table 8a: SCHEDULE OF COSTS (all costs in 2012 dollars) *continued*

DCP PROJECT NO.	PROJECT	INFRASTRUCTURE CATEGORY	WORKS DESCRIPTION	TOTAL ESTIMATED PROJECT COST: LAND & CONSTRUCTION	TOTAL FUNDED VIA BOTANIC RIDGE DCP	COMMUNITY INFRASTRUCTURE LEVY
COMMUNITY CENTRES						
C1	Land - Community Centre 1.	Development	Land for north western community centre.	$440,550	$312,791	$0
C2	Construction - Community Centre 1.	Development	Construction of integrated multi-purpose Community Centre to include x4 Early Years kindergarten rooms and associated external play areas, consulting rooms for maternal and child health (including 2 offices and waiting area), flexible meeting space for play groups etc.	$5,795,696	$4,114,944	$0
SUB-TOTAL COMMUNITY CENTRES				**$6,236,246**	**$4,427,735**	**$0**
RECREATION						
AR01	Land - Active Recreation West.	Development	Land for active recreation inclusive of sportsfields, tennis, netball pavilions and carparking.	$4,746,500	$4,746,500	$0
AR02	Construction - Active Recreation West - Sportsfields.	Development	Construction of x2 full size AFL / cricket ovals, cricket nets, x4 tennis courts, x2 netball courts, car parking and associated landscaping. Oval construction will require balance cut and fill.	$5,050,052	$5,050,052	$0
AR03	Construction - Active Recreation West - Main Pavilion.	Community	Construction of pavilion to service cricket, football and netball.	$2,567,582	Funded via CIL	Funded via CIL
AR04	Construction - Active Recreation West - Tennis Pavilion.	Community	Construction of pavilion to service tennis.	$738,241	Funded via CIL	Funded via CIL
AR05	Construction - Active Recreation West - Indoor Sports Facility.	Community	Lower order and flexible x2 basketball court sized indoor facility. Delivery shared with primary school.	$2,802,917	Funded via CIL	Funded via CIL
AR06	Land - Active Recreation East.	Development	Land for active recreation inclusive of sportsfields, pavilion and carparking.	$4,165,700	$4,165,700	$0
AR07	Construction - Active Recreation East - Sportsfields.	Development	Construction of x2 full size cricket oval combined with x3 soccer fields, cricket nets and car parking.	$5,473,464	$5,473,464	$0
AR08	Construction - Active Recreation East - Pavilion.	Community	Construction of pavilion to service cricket and soccer.	$2,220,348	Funded via CIL	Funded via CIL
SUB-TOTAL RECREATION				**$27,764,804**	**$19,435,716**	**$0**
TOTALS				**$50,796,692**	**$40,659,093**	**$0**
TOTAL DEVELOPMENT INFRASTRUCTURE LEVY PER NDHA (see Note 1 following page)					**$168,558**	
TOTAL COMMUNITY INFRASTRUCTURE LEVY PER DWELLING						**$900**
TOTAL COMMUNITY INFRASTRUCTURE LEVY ESTIMATED TOTAL RAISED VIA BOTANIC RIDGE PSP						**$3,087,459**

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2.2.5 APPLICATION OF SCHEDULE OF COSTS

1. If the Commonwealth Government Minister administering the EPBC Act makes a determination that does not require the setting aside of land within the Botanic Ridge precinct for SBB habitat corridors, the rate of $168,558 per NDHa (shown in the previous table 8a) will apply to this DCP.

2. However if the Commonwealth Government Minister administering the EPBC Act makes a determination that does require the setting aside of land within the Botanic Ridge precinct for SBB habitat corridors the rate of $184,639 per NDHa (shown in the following table 8b) will apply.

3. Prior to the Commonwealth Government Minister administering the EPBC Act making a determination regarding the setting aside of land for SBB habitat, the higher rate of $184,639 per NDHa applies as per table 8b.

The higher rate includes a contingency for:

- additional projects required due to the SBB Habitat Corridors
- alterations to existing projects and;
- an adjustment due to the change (reduction) in Net Developable Area

For further detail on the contingency projects and detailed cost estimates refer to Plan 7, Table 9 and cost sheets in Appendix A to this DCP.

Table 8b: SCHEDULE OF COSTS INCLUDING CONTINGENCY PROJECTS

DCP PROJECT NO.	PROJECT	INFRASTRUCTURE CATEGORY	WORKS DESCRIPTION	TOTAL ESTIMATED PROJECT COST: LAND & CONSTRUCTION	TOTAL FUNDED VIA BOTANIC RIDGE DCP	COMMUNITY INFRASTRUCTURE LEVY
CONTINGENCY						
CP01 to CP08	Development Infrastructure Contingency Projects Pending Commonwealth determination regarding the setting aside of land for SBB habitat within the Botanic Ridge precinct.	Development	Includes allowance for net additonal land / construction project costs and increase in rate due to reduction in developable area.	$3,648,648	$3,648,648	$0
SUB-TOTAL CONTINGENCY				**$3,648,648**	**$3,648,648**	**$0**
SUB TOTAL CONTINGENCY PROJECTS DEVELOPMENT INFRASTRUCTURE LEVY PER NDHA					**$16,080**	
TOTALS DEVELOPMENT INFRASTRUCTURE LEVY PER NDHA (INCLUSIVE OF SBB CORRIDOR CONTINGENCY PROJECTS) (see note 2 and 3 above)					**$184,639**	
TOTAL COMMUNITY INFRASTRUCTURE LEVY PER DWELLING						**$900**
COMMUNITY INFRASTRUCTURE LEVY ESTIMATED TOTAL RAISED VIA BOTANIC RIDGE PSP						**$2,867,473**

3.0 ADMINISTRATION AND IMPLEMENTATION

3.1 ADMINISTRATION OF THE DEVELOPMENT CONTRIBUTIONS PLAN

This section clearly sets out how the Development Contributions Plan will be administered and includes the timing of payment, provision of works and land in kind and how the Development Contributions Plan fund will be managed in terms of reporting, indexation and review periods.

The Botanic Ridge Precinct Structure Plan Development Contributions Plan Infrastructure Levy applies to subdivision and/or development of land.

The Botanic Ridge Precinct Structure Plan Community Infrastructure Levy applies to the construction of dwellings and must be paid prior to the issue of a Building Permit.

3.1.1 PAYMENT OF CONTRIBUTION LEVIES AND TIMING

DEVELOPMENT INFRASTRUCTURE

For subdivision of land

- An infrastructure levy must be paid to the Collecting Agency for the land after certification of the relevant plan of subdivision but not more than 21 days prior to the issue of a Statement of Compliance with respect to that plan.
- Where the subdivision is to be developed in stages the infrastructure levy for the stage to be developed may only be paid to the Collecting Agency within 21 days prior to the issue of a Statement of Compliance for that stage provided that a Schedule of Development Contributions is submitted with each stage of the plan of subdivision. This Schedule must show the amount of the development contributions payable for each stage and the value of the contributions for prior stages to the satisfaction of the Collecting Agency.
- If the Collecting Agency agrees to works or provision of land in lieu of the payment of the infrastructure levy, the land owner must enter into an agreement under Section 173 of the Planning and Environment Act with respect to the proposed works or provision of land in lieu to specify implementation requirements.

For development of land where no subdivision is proposed

Provided an infrastructure levy has not already been paid on the subject land, an infrastructure levy must be paid to the Collecting Agency in accordance with the provision of the approved Development Contributions Plan for each demand unit (net developable hectare) proposed to be developed prior to the commencement of any development (for example, development includes buildings, car parks, access ways, landscaping and ancillary components). The Collecting Agency may require that contributions be made at either the planning or building permit stage for Development Infrastructure.

If the Collecting Agency agrees to works or provision of land in lieu of the payment of the infrastructure levy, the land owner must enter into an agreement or other suitable arrangement under Section 173 of the Planning and Environment Act in relation to the proposed works or land in lieu.

Where no planning permit is required

The following requirements apply where no planning permit is required.

The land may only be used and developed subject to the following requirements being met.

- Unless some other arrangement has been agreed to by Collecting Agency in a Section 173 Agreement, prior to the commencement of any development, an infrastructure levy must be paid to the Collecting Agency in accordance with the provisions of this approved Development Contributions Plan for the land.
- If the Collecting Agency agrees to works or provision of land in lieu of the payment of the infrastructure levy, the land owner must enter into an agreement under Section 173 of the Planning and Environment Act in respect of the proposed works or provision of land in lieu.

Community Infrastructure

Contributions relating to community infrastructure are to be made by the home-builder prior to the issue of a Building Permit. However, development proponents are encouraged to pay the levy prior to the issue of a statement of compliance to reduce the administrative burden of collection from individual home builders.

Levies for 'residential buildings' will be calculated at the rate for a single dwelling. In all other forms of accommodation, the dwelling is the individual unit (such as each dwelling in a residential village, retirement village, serviced apartment and so on). 'Corrective institutions' are exempt.

A community infrastructure levy is not payable for a dwelling on a lot which was created prior to the date that this Development Contributions Plan was first incorporated into the Casey Planning Scheme.

Non Government Schools

Where land is subdivided or developed for the purpose of a non government school and the use of that land is subsequently for a purpose other than a non government school, the owner of that land must pay to the Collecting Agency development contributions in accordance with the provisions of this Development Contributions Plan. The development infrastructure levy and where applicable, the community infrastructure levy must be paid within 28 days of the date of the commencement of the construction of any buildings or works for that alternative use.

3.1.2 WORKS IN KIND

Works may be constructed in-lieu of a cash contribution on some projects. This is only possible where the arrangement has been approved in writing by the Collecting Agency.

As outlined in Section 3.1.1, payment of development contributions is to be made in cash.

Alternatively, infrastructure works and land may be provided by developers, with a credit provided against their development contribution, subject to the written agreement of the Collecting Agency. In determining whether to agree to the provision of works in lieu of cash the Collecting Agency will have regard to the following:

- Only works or land identified in the Development Contributions Plan can be provided in lieu of cash;
- Works must be provided to a standard that generally accords with the Development Contributions Plan unless agreed between the Collecting Agency and the developer;
- Detailed design must be approved by the Collecting Agency and generally accord with the standards outlined in the Development Contributions Plan unless agreed by the Collecting Agency and the developer;
- The construction of works must be completed to the satisfaction of the Collecting Agency; and,
- There is no additional financial impact on the Development Contributions Plan.

Where the Collecting Agency agrees that works are to be provided by a development proponent in lieu of cash contributions:

- The credit for the works provided shall equal the value identified in the Development Contributions Plan, taking into account the impact of indexation;
- The value of works provided in accordance with the principles outlined above, will be offset against the development contributions liable to be paid by the development proponent; and;
- The development proponent will not be required to make cash payments for contributions until the agreed value of any credits for the provision of agreed works–in-kind are exhausted/balanced.

3.1.3 CREDIT FOR OVER PROVISION

General Provisions

Where the Collection Agency agrees that a development proponent can physically provide an infrastructure item (either works and/or land) the situation may arise where the developer makes a contribution with a value that exceeds that required by the Development Contributions Plan for the individual project.

In such a case the developer may be entitled to credits against other projects in the Development Contributions Plan to the extent that they 'over contributed'. Alternatively, a developer may seek an agreement with the Collecting Agency to provide for a cash reimbursement where a significant over contribution has been made on a particular project.

The details of credits and reimbursements will need to be negotiated with, and agreed to by the Collecting Agency.

Contingency Project Provisions

Note – refer Tables 8a and 8b and Appendix A for detail on contingency projects.

If the Commonwealth Government Minister administering the EPBC Act approves a Sub-Regional Species Strategy for the Southern Brown Bandicoot that does not require the setting aside of land within the Botanic Ridge precinct for SBB habitat, the lower per NDHa DCP rate as shown in table 8a applies.

Development proponents who have paid the required higher per NDHa DCP rate (as shown in table 8b) in the interim, will be considered to be to have over contributed. DCP credits or reimbursements for such over contributions, will be administered as per the general provisions outlined above.

3.1.4 FUNDS ADMINISTRATION

The administration of the contributions made under the Development Contributions Plan will be transparent and development contributions charges will be held until required for provision of the item. Details of funds received and expenditures will be held by the Collecting Agency in accordance with the provisions of the Local Government Act 1993 and the Planning and Environment Act 1987.

The administration of contributions made under the Development Contributions Plan will be transparent and demonstrate:

- The amount and timing of funds collected;
- The source of the funds collected;
- The amount and timing of expenditure on specific projects;
- The project on which the expenditure was made;
- The account balances for individual projects; and
- Clearly show any pooling of funds to deliver specific projects where applicable.

The Collecting Agency will provide for regular monitoring, reporting and review of the monies received and expended in accordance with this Development Contributions Plan.

The Collecting Agency will establish interest bearing accounts and all monies held in these accounts will be used solely for the provision of infrastructure as itemised in this Development Contributions Plan, as required under Section 46QB of the Planning and Environment Act 1987.

Should the Collecting Agency resolve not to proceed with any of the infrastructure projects listed in this Development Contributions Plan, the funds collected for these items will be used for the provision of additional works, services or facilities where approved by the Minister responsible for the Planning and Environment Act, or will be refunded to developers and/or owners of land subject to these infrastructure charges.

3.1.5 OPEN SPACE PROVISION

This DCP does not provide funding for passive local open space. Local passive open space is dealt with under Clause 52.01 of the Casey Planning Scheme.

Unencumbered local active open space must be provided as required in accordance with the Botanic Ridge Development Contributions Plan. A Direct Provision Model for the delivery of unencumbered local active open space is encouraged in lieu of cash where the Botanic Ridge PSP and DCP include proposals for active open space/playing fields.

3.1.6 CONSTRUCTION AND LAND VALUE COSTS AND INDEXATION

Capital costs of all infrastructure items except for land are in 1st November 2012 dollars and will be indexed by the Collecting Agency annually for inflation in the following way.

In relation to the costs associated with the infrastructure items other than land, the cost must be adjusted according to the following method:

- The capital cost for each infrastructure item will be adjusted by applying the Building Price Index, as published in the latest edition of Rawlinson's Australian Construction handbook on 1st January and 1st July each year;
- In relation to the cost of land, the land value must be adjusted by adopting a revised value determined according to the following method:
- The land value will be adjusted on 1 July each year following site specific land valuations undertaken by a registered valuer. Within 14 days of the adjustments being made, the Collecting Agency will publish the amended contributions on the Collecting Agency's website.

3.1.7 DEVELOPMENT CONTRIBUTIONS PLAN REVIEW PERIOD

This Development Contributions Plan adopts a long-term outlook for development. It takes into account future development in Botanic Ridge. A 'full development' horizon of land within the current Urban Growth Boundary to the year 2030 has been adopted for this Development Contributions Plan.

This Development Contributions Plan commences on the date when it was first incorporated into the Casey Planning Scheme. This Development Contributions Plan will end when development within the Development Contributions Plan area is complete or when the Development Contributions Plan is removed from the Planning Scheme.

The Development Contributions Plan is expected to be revised and updated every 5 years (or more often if required). This will require an amendment to the Casey Planning Scheme to replace this document with an alternative, revised document. This review is anticipated to include:

- Update any aspect of the plan which is required.
- Review of projects required, as well as their costs and scope (as relevant) and indicative provision trigger.

- Review of estimated net developable area (this will also be required if the Precinct Structure Plan is subject to a substantive amendment).
- Review of land values for land to be acquired through the plan.

3.1.8 COLLECTING AGENCY (AGENCY RESPONSIBLE FOR COLLECTING INFRASTRUCTURE LEVY)

Casey City Council is the Collecting Agency pursuant to Section 46K(1)(fa) of the Planning and Environment Act (1987) which means that it is the public authority to whom all levies are payable. As the Collecting Agency, Casey City Council is also responsible for the administration of this Development Contributions Plan and also its enforcement pursuant to Section 46QC of the Planning and Environment Act (1987).

3.1.9 DEVELOPMENT AGENCY (AGENCY RESPONSIBLE FOR WORKS)

Casey City Council is the Development Agency and is responsible for the provision of the works funded. In the future the Development Agency for several road and intersection infrastructure projects may change from Casey City Council to VicRoads. However, any such transfer of responsibility would be dependent on written agreement from VicRoads.

3.2 IMPLEMENTATION STRATEGY

This section provides further details regarding how the Collecting Agency intends to implement the Development Contributions Plan. In particular, this section clearly identifies the rationale for the implementation strategy and details the various measures that have been adopted to reduce the risk posed by the Development Contributions Plan to all parties.

3.2.1 RATIONALE FOR THE IMPLEMENTATION STRATEGY

This Implementation Strategy has been incorporated into the Development Contributions Plan to provide certainty to both the Collecting Agency and development proponents. The implementation strategy recognises the complexities associated with infrastructure provision and funding and seeks to minimise risk to the Collecting Agency, Development Agency, development proponent and future community.

The implementation strategy has been formulated by:

- Assessing the risk posed by the Precinct Structure Plan layout (identifying high risk items);
- Having regard to the development context;
- Assessing the need for finance requirements – upfront financing and pooling of funds;
- Agreeing the land value and indexing it appropriately (where possible);
- Seeking direct delivery of infrastructure and land by development proponents where appropriate;
- Identifying preferred implementation mechanism to achieve the above outcomes and reducing the risk associate with the Development Contributions Plan to ensure that it will be delivered as intended; and,
- Provision of adequate resources to administer the Development Contributions Plan.

3.2.2 IMPLEMENTATION MECHANISM

As set out in Section 46P of the Planning and Environment Act (1987), the Collecting Agency may accept (with the consent of the Development Agency where the Collecting Agency is not also the Development Agency) the provision of land, works, services or facilities by the applicant in part or full satisfaction of the amount of levy payable.

This can be agreed with the Collecting Agency before or after the application for the permit is made or before or after the development is carried out.

To coordinate the provision of infrastructure, the Urban Growth Zone Schedule 4 for the Botanic Ridge PSP requires an application for subdivision to be accompanied by an infrastructure plan to the satisfaction of the Responsible Authority. The Public Infrastructure and Land Plan needs to show the location, type, staging and timing of infrastructure on the land as identified in the Botanic Ridge PSP or reasonably required as a result of the subdivision of the land and address the following:

- Stormwater drainage works;
- Road works internal or external to the land consistent with any relevant traffic report or assessment;
- The reserving or encumbrance of land for infrastructure, including for public open space and community facilities;
- Any infrastructure works which an applicant proposes to provide in lieu of development contributions in accordance with the Botanic Ridge DCP;
- The effects of the provision of infrastructure on the land or any other land; and,
- Any other relevant matter related to the provision of infrastructure reasonably required by the Responsible Authority.

Through the approval of this plan the Council (acting as the Collecting Agency) would consider if and what infrastructure should be provided as in-kind works under the Botanic Ridge DCP in accordance with Section 46P of the Act. The approved Infrastructure Plan must include a list of DCP infrastructure which the Collecting Agency has agreed in writing to allow to be provided as works in lieu.

Once approved, the Public Infrastructure and Land Plan must be implemented to the satisfaction of the Responsible Authority. The implementation of the Public Infrastructure Plan may include the requirement to enter into a Section 173 Agreement.

4.0 OTHER INFORMATION

4.1 ACRONYMS

'the Act' Planning and Environment Act 1987

AHD Australian Height Datum

AFL Australian Football League ovals

CAD Central Activities District

CIL Community Infrastructure Levy

DCP Development Contributions Plan

DEECD Department of Education & Early Childhood Development

DIL Development Infrastructure Levy

DPCD Department of Planning & Community Development

DoT Department of Transport

DSE Department of Sustainability & Environment

GAA Growth Areas Authority

GDA Gross Developable Area

Ha Hectare

MCA Main Catchment Area

MCH Maternal & Child Health

MSS Municipal Strategic Statement

NAC Neighbourhood Activity Centre

NDA Net Developable Area

NDHa Net Developable Hectare

PAC Principle Activity Centre

PPTN Principle Public Transport Network

PSP Precinct Structure Plan

P-6 State School Prep to Year 6

P-12 State School Prep to Year 12

RBGC Royal Botanic Gardens Cranbourne

m^2 Sqm Square Metres

UGB Urban Growth Boundary

UGZ Urban Growth Zone

4.2 GLOSSARY

ACTIVE OPEN SPACE

Land set aside for the specific purpose of formal/organised club based sports.

ACTIVITY CENTRE

See 'Town Centre'.

ARTERIAL ROAD

A higher order road providing for moderate to high volumes at relatively high speeds typically used for inter-suburban journeys and linking to freeways, and identified under the Road Management Act 2004. All declared arterials are managed by the State Government.

CO-LOCATION

Adjoining land uses to enable complementary programs, activities and services and shared use of resources and facilities. For example, the co-location of schools and active open space.

COMMUNITY FACILITIES

Infrastructure provided by government or non-government organisations for accommodating a range of community support services, programs and activities. This includes facilities for education and learning (e.g. government and non-government schools, universities, adult learning centres); early years (e.g. preschool, maternal and child health, childcare); health and community services (eg. hospitals, aged care, doctors, dentists, family and youth services, specialist health services); community (e.g. civic centres, libraries, neighbourhood houses); arts and culture (e.g. galleries, museums, performance space); sport, recreation and leisure (e.g. swimming pools); justice (e.g. law courts); voluntary and faith (e.g. places of worship) and emergency services (e.g. police, fire and ambulance stations).

CONNECTOR STREET

A lower order street providing for low to moderate volumes and moderate speeds linking local streets to the arterial network Managed by the relevant local council. (See Table C1 in clause 56)

CONVENTIONAL DENSITY HOUSING

Housing with an average density of 1O to 15 dwellings per net developable hectare.

DEVELOPMENT CONTRIBUTIONS PLAN

Document that sets out the contributions expected from each individual landowner to fund infrastructure and services. Refer to Part 3B of the Planning and Environment Act l987.

ENCUMBERED LAND

Land that is constrained for development purposes. Includes easements for power/transmission lines, sewers, gas, waterways, drainage, retarding basins/wetlands, landfill, conservation and heritage areas. This land may be used for a range of activities (e.g. walking trails, sports fields).

FREEWAY

A high speed and high volume road with the highest level of access control and typically used for longer distance journeys across the metropolitan area and country Victoria. All freeways are managed by VicRoads.

GROWTH AREA

Areas on the fringe of metropolitan Melbourne around major regional transport corridors that are designated for large-scale change, over many years from rural to urban use. Melbourne has six growth areas called Casey-Cardinia; Hume; Melton-Caroline Springs; Whittlesea, Wyndham and Mitchell.

GROWTH AREA FRAMEWORK PLAN

Government document that sets long-term strategic planning direction to guide the creation of a more sustainable community in the growth areas.

HIGH DENSITY HOUSING

Housing with an average density of more than 30 dwellings per net developable hectare.

HOUSING DENSITY (NET)

The number of houses divided by net developable area

LINEAR OPEN SPACE NETWORK

Corridors of open space, mainly along waterways that link together, forming a network.

LAND BUDGET TABLE

A table setting out the total precinct area, net developable area and constituent land uses proposed within the precinct.

LOT

A part (consisting of one or more pieces) of any land (except a road, a reserve, or common property) shown on a plan, which can be disposed of separately and includes a unit or accessory unit on a registered plan of strata subdivision and a lot or accessory lot on a registered cluster plan.

LOWER DENSITY HOUSING

Housing with an average density of less than 10 dwellings per hectare.

MAJOR ACTIVITY CENTRE

Activity centres that have similar characteristics to Principal Activity Centres, but serve smaller catchment areas. For further information refer to Melbourne 2030.

MAJOR EMPLOYMENT AREA

Areas identified on the Growth Area Framework Plan for economic and employment growth.

MEDIUM DENSITY HOUSING

Housing with an average density of 16 to 30 dwellings per net developable hectare.

NATIVE VEGETATION

Plants that are indigenous to Victoria, including trees, shrubs, herbs, and grasses.

NET DEVELOPABLE AREA

Total amount of land within the precinct that is made available for development of housing and employment buildings, including lots, local and connector streets. Total precinct area minus community facilities, schools and educational facilities and open space, arterial roads and encumbered land. Small local parks defined at subdivision stage are included in net developable area.

NET RESIDENTIAL AREA

As per net developable area but excluding neighbourhood activity centres, non-government schools and golf course sites.

PASSIVE OPEN SPACE

Open space that is set aside for parks, gardens, linear corridors (including SBB Corridors), conservation bushlands, nature reserves, public squares and community gardens that are made available for passive recreation, play and unstructured physical activity including walking, cycling, hiking, revitalisation, contemplation and enjoying nature.

PRECINCT STRUCTURE PLAN

A statutory document that describes how a precinct or series of sites within a growth area will be developed over time. A precinct structure plan sets out the broad environmental, social and economic parameters for the use and development of land within the precinct.

PRINCIPAL PUBLIC TRANSPORT NETWORK

A high-quality public transport network that connects Principal and Major Activity Cent res, and comprises the existing radial fixed-rail network, extensions to this radial network and new cross-town bus routes.

PUBLIC OPEN SPACE

Land that is set aside in the precinct structure plan for public recreation or public resort, or as parklands, or for similar purposes. Incorporates active and passive open space.

TOWN CENTRE

Provide the focus for services, commercial and retail based employment and social interaction. They are where people shop, work, meet, relax and live. They are well-served by public transport, they range in size and intensity of use. In the growth areas, these are referred to as principal activity centres, major activity centres, neighbourhood activity centres and local centres. For further information refer to Melbourne 2030.

URBAN GROWTH BOUNDARY

A statutory planning management tool used to set clear limits to metropolitan Melbourne's urban development.

URBAN GROWTH ZONE

Statutory zone that applies to land that has been identified for future urban development. The UGZ has four purposes: (1) to manage transition of non-urban land into urban land; (2) to encourage development of well-planned and well-serviced new urban communities in accordance with an overall plan; (3) to reduce the number of development approvals needed in areas where an agreed plan is in place; and (4) to safeguard non-urban land from use and development that could prejudice its future urban development.

4.3 SUPPORTING INFORMATION

The following documents may assist in understanding the background to this DCP.

Transport Assessment for PSP 10: Botanic Ridge (November 2011), GTA Addendum Note (November 2011) and Memorandum - Additional Traffic Analysis (March 2012).

Community Infrastructure Assessment (November 2009) ASR Research and Addendum October 2010, ASR Research.

5.0 PROJECT CONSTRUCTION DETAILS

This section includes cost estimates for each DCP construction project accompanied by plans that show the extent of costed works.

All DCP projects are subject to detailed design and approval of the responsible authority.

5.1 TRANSPORT PROJECTS (refer plan 4 for locations)

The drawings and specifications for the Transport Projects included in this DCP are included in the following DCP Costing Sheets:

RD03 - Casey Fields Boulevard - Interim construction from South Gippsland Highway to intersection with Craig Road

RD04 - Craig Road upgrade to interim urban standard from Browns Road to Casey Fields Boulevard Junction

RD05 - Browns Road upgrade from Smiths Lane to Craig Road

RD06 - Ballarto Road connector upgrade – DCP Costing Sheet 4;

RD08 - Construction for retirement village constrained access street

IN01 - South Gippsland Highway / Casey Fields Boulevard Intersection

IN02 - South Gippsland Highway / Ballarto Road Intersection

IN03 - Craig Road / Casey Fields Boulevard Intersection

IN04 - Craig Road / Connector C1 Intersection – DCP Costing Sheet 10;

IN06 - Browns Road / Connector C4 Intersection

IN07 - Browns Road / Connector C3 Intersection – DCP Costing Sheet 12

5.2 COMMUNITY FACILITIES PROJECTS (refer Plan 5 for locations)

The drawings and specifications for the Community Facilities Projects included in this DCP are included in the following DCP Costing Sheet:

C2 - Community Centre 1 – Construction

5.3 RECREATION PROJECTS (refer Plan 5 for locations)

The drawings and specifications for the Active Recreation Projects included in this DCP are included in the following DCP Costing Sheets:

AR02 - Active Recreation West – Sportsfields

AR03 - Active Recreation West - Main Pavilion

AR04 - Active Recreation West - Tennis Pavilion

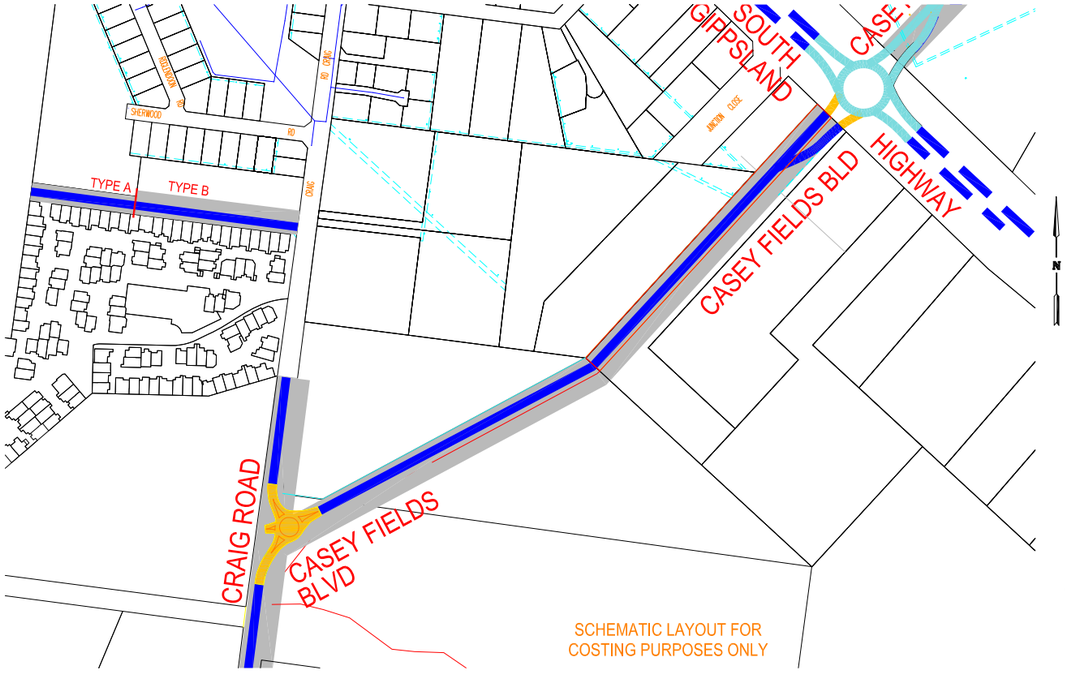
AR05 – Active Recreation West - Indoor Sports Facility

AR07 - Active Recreation East - Sportsfields AR08 - Active Recreation East - Pavilion

AR08 - Active Recreation East - Pavillion

BOTANIC RIDGE DCP

RD-03 ROAD COSTS
CASEY FIELDS BLD

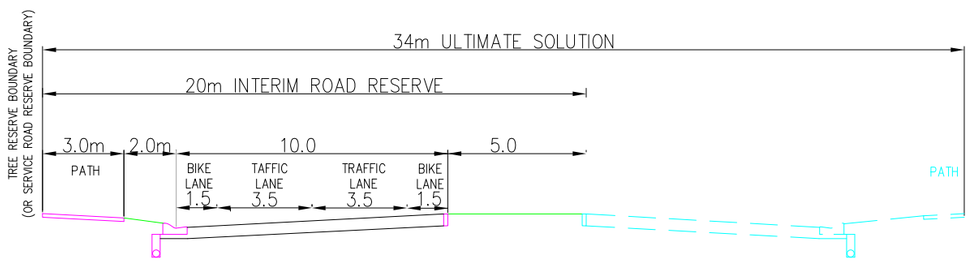


ULTIMATEROAD RESERVE

COSTED ROAD

NOTES AND ASSUMPTIONS FOR RD-03
1: Traffic Management considered to be low in complexity
2: Survey & Design is considered to be medium complexity due to green field site
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) green field site
b) no connection road
c) length of proposed works

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TYPICAL CASEY FIELDS CROSS SECTION INTERIM SOLUTION SUPPLIED BY GAA

Botanic Ridge DCP Roadworks Estimate

Road Name: RD-03 Casey Fields Boulevard south of Sth Gippsland Hwy
Limit of works: As shown on drawing road reserve and cross section vary
Length of Job: length 737Lm without intersections as shown
Notes: Costs based on City of Casey specification for road construction & GAA Interim Cross Section

Item	Unit	Rate	wide	units		cost/m		Unit cost	Quantity		subset	subtotal	Amount
			m			width			metres		cost		
Roadworks	subtotal							$ 1,850.75	737	m		$ 1,363,999.94	$ 1,364,000
Bulk Earthworks	m²	$ 40.00	5.3	metres	$ 40	$ 212.00	m					$ 156,244.00	$ -
Pavement (Urban)	m²	$ 91.50	9.475	metres	$ 92	$ 866.96	m					$ 638,951.36	$ -
Crushed Rock Shoulder	m²	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Pavement Removal	m²	$ 10.00	0	metres		$ -						$ -	
Concrete Edge Strip 225mm wide	m	$ 25.00	1	sides	$ 25	$ 25.00	m					$ 18,425.00	$ -
Kerb and Channel S 504		$ 45.00	1	sides	$ 45	$ 45.00	m					$ 33,165.00	
side entry pits std drw S305	unit	$ 2,000.00	1	interval metres	$ 90	$ 22.22	m					$ 16,377.78	
Bike path 2.0m wide conc	m²	$ 60.00	0	metres		$ -						$ -	
Shared pathway 2.5m wide gravel	m²	$ 30.00	0	metres		$ -						$ -	
Shared Pedestrian Footpath 3.0m wide	m²	$ 60.00	3	metres		$ 180.00						$ 132,660.00	
Pedestrian Footpath 1.5m gravel	m²	$ 30.00	0	metres		$ -						$ -	
Drainage , subgrade drain	m	$ 14.70	2		$ 15	$ 29.40	m					$ 21,667.80	$ -
Linemarking & Signage	m	$ 25.00	2	metres	$ 25	$ 50.00	m					$ 36,850.00	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Concrete Path	m²	$ 45.00	0		$ 45	$ -	m					$ -	$ -
level / trim nature strip as per standard drawings	m²	$ 4.00	7.5	metres	$ 4	$ 30.00	m					$ 22,110.00	$ -
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tree	2	rows of trees	$ 25	$ 50.00	m					$ 36,850.00	
Tube Stock Plantings as per draft tree strategy	unit	$5.27 -$6.78	0	metres wide	$ 6	$ -	m					$ -	$ -
root barriers	optional as per design in respect of appropriate offset criteria											$ -	
300mm dia conc drain Stormwater Cr BF	per metre	$182	168	metres		$ 41.49						$ 30,576	
375mm conc drain stormwater Cr Bk fill	per metre	$231	168	metres		$ 52.66						$ 38,808	
450mm conc drain stormwater Cr BF	per metre	$294	168	metres		$ 67.02						$ 49,392	
525mm conc drain stormwater Cr BF	per metre	$351	0	metres		$ -						$ -	
Service Road			1E-04	Lmetres									$ 0
Traffic Signals	Unit	$ 102,500.00	0	No of intesection								$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	0	metres								$ -	$ -
Street Lighting	m	$ 129.00	1	rows of lights		$ 129.00	m					$ 95,073.00	
lighting conduit	m	$ 50.00	1	No of runs/sides		$ 50.00	m					$ 36,850.00	
Subtotal						$ 1,850.75						$ 1,363,999.94	
estimated total													$ 1,364,000
Traffic Management										5.0%			$ 68,200
Contingency										20.0%			$ 272,800
Total + contingencies													
Services relocation Sec Poles, water fitting	Item												$ 50,000
Survey and Design										10%			$ 136,400
Overheads (supervision etc)										15%			$ 204,600
Total excluding land cost													**$ 2,096,000**
CPI adjustment to Nov 2012										2.2%			$ 46,112
Total Estimated Cost													$ 2,142,112
Adopted Cost													

Estimate Prepared by: CDCE Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.
PROPOSED KERB AND CHANNEL	POWER OVERHEAD	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN	POWER PIT	HOUSE DRAIN	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
SCALE NOT TO SCALE
COORD. SYS.: N/A
F.B. L.B.
JOB NO. TO 111
SURVEY N/A
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DESIGN SAF
REG. FILE NO. XXXXXXX
TRACED SAF
MELWAY REF.
CHECKED SAF
CONTRACT NO.

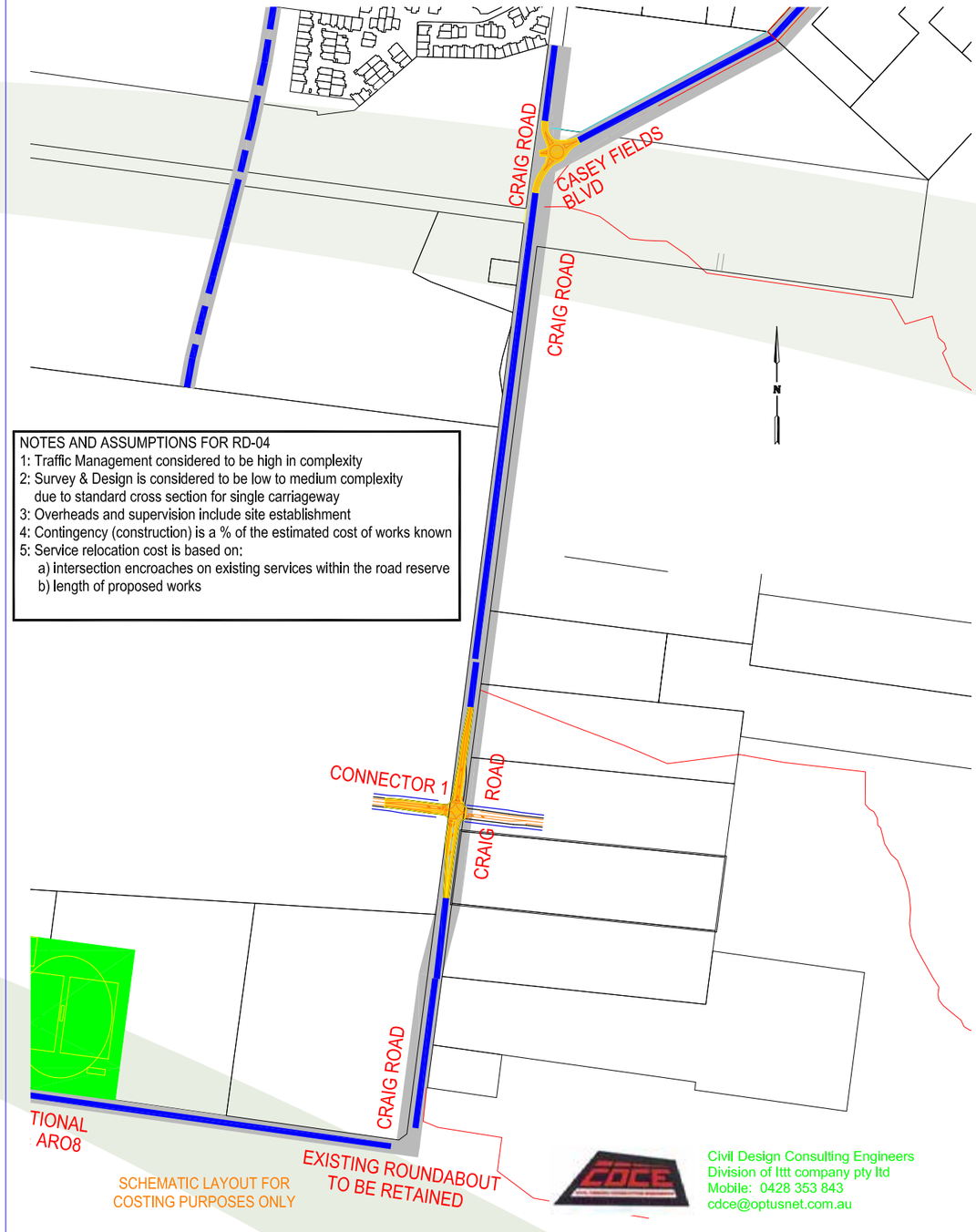
CIVIL DESIGN CONSULTING ENGINEERS

BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
CASEY FIELDS BLD RD-03
ROAD WORKS

DATE: November 2012
10 111
SHEET 1 OF 1

BOTANIC RIDGE DCP

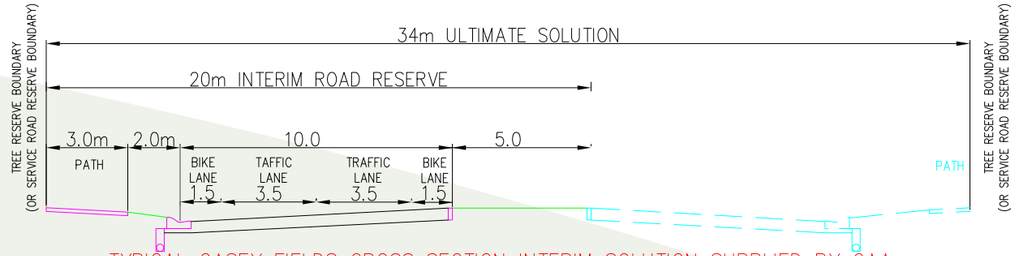
RD-04 ROAD COSTS
CRAIG RD



NOTES AND ASSUMPTIONS FOR RD-04
1: Traffic Management considered to be high in complexity
2: Survey & Design is considered to be low to medium complexity due to standard cross section for single carriageway
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) intersection encroaches on existing services within the road reserve
b) length of proposed works

ULTIMATEROAD RESERVE

COSTED ROAD



TYPICAL CASEY FIELDS CROSS SECTION INTERIM SOLUTION SUPPLIED BY GAA

Botanic Ridge DCP Roadworks Estimate

Road Name: RD-04 Craig Rd from Casey Fields Bld to Browns Rd intersection & 110m north of roundabout
Limit of works: As shown on drawing road reserve and cross section vary
Length of Job: length = 1189Lm without intersections as shown
Notes: Costs based on City of Casey specification for road construction & GAA Interim Cross Section

Item	Unit	Rate	wide	units		cost/m		Unit cost	Quantity		subset	subtotal	Amount
			m			width			metres		cost		
Roadworks	subtotal							$ 1,862.02	1189	m		$ 2,213,946.23	$ 2,213,946
Bulk Earthworks	m²	$ 40.00	5.3	metres	$ 40	$ 212.00	m					$ 252,068.00	$ -
Pavement (Urban)	m²	$ 91.50	9.48	metres	$ 92	$ 868.96	m					$ 1,030,818.41	$ -
Crushed Rock Shoulder	m²	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Pavement Removal	m²	$ 10.00	0	metres		$ -						$ -	
Drainage Culvert 6No. 1.2x0.9	m	$ 960.00	129.6	No.	2							$ 248,832.00	$ 248,832
Drainage Culvert 1No. 1.2x0.6	m	$ 660.00	21.6	No.	1							$ 14,256.00	$ 14,256
Concrete Edge Strip 225mm wide	m	$ 25.00	1	sides	$ 25	$ 25.00	m					$ 29,725.00	$ -
Kerb and Channel S 504	m	$ 45.00	1	sides	$ 45	$ 45.00	m					$ 53,505.00	
side entry pits std drw S305	unit	$ 2,000.00	1	interval m	90.00	$ 22.22	m					$ 26,422.22	
Bike path 2.0m wide conc	m²	$ 60.00	0	metres		$ -						$ -	
Shared pathway 2.5m wide gravel	m²	$ 30.00	0	metres		$ -						$ -	
Share Pedestrian Footpath 3.0m wide	m²	$ 60.00	3	metres		$ 180.00						$ 214,020.00	
Pedestrian Footpath 1.5m gravel	m²	$ 30.00	0	metres		$ -						$ -	
Drainage , subgrade drain	m	$ 14.70	2		$ 15	$ 29.40	m					$ 34,956.60	$ -
Linemarking & Signage	m	$ 25.00	2	metres	$ 25	$ 50.00	m					$ 59,450.00	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Concrete Path	m²	$ 45.00	0		$ 45	$ -	m					$ -	$ -
level / trim nature strip	m²	$ 4.00	7.5	metres	$ 4	$ 30.00	m					$ 35,670.00	$ -
as per standard drawings													
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tr	2	rows of tr	$ 25	$ 50.00	m					$ 59,450.00	
Tube Stock Plantings	unit	$5.27 -$6.78	0	metres wi	$ 6	$ -	m					$ -	$ -
as per draft tree strategy												$ -	
root barriers	optional as per design in respect of appropriate offset criteria												
300mm dia conc drain Stormwater Cr BF	per metre	$182	290	metres		$ 44.39						$ 52,780	
375mm conc drain stormwater Cr Bk fill	per metre	$231	290	metres		$ 56.34						$ 66,990	
450mm conc drain stormwater Cr BF	per metre	$294	290	metres		$ 71.71						$ 85,260	
525mm conc drain stormwater Cr BF	per metre	$351	0	metres		$ -						$ -	
Service Road				Lmetres									
Traffic Signals	Unit	$ 102,500.00	0	No of intesection								$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	0	metres								$ -	$ -
Street Lighting	m	$ 129.00	1	rows of lights		$ 129.00	m					$ 153,381.00	
lighting conduit	m	$ 50.00	1	No of runs/sides		$ 50.00	m					$ 59,450.00	
Subtotal						$ 1,862.02						$ 2,477,034.23	
estimated total													$ 2,477,034
Traffic Management										10.0%			$ 247,703
Contingency										20.0%			$ 495,407
Total + contingencies													
Services relocation Sec Poles, water fittin	item												$ 400,000
Survey and Design										10%			$ 247,703
Overheads (supervision etc)										15%			$ 371,555
Total excluding land cost													**$ 4,239,403**
CPI Adjustment to Nov 2012										2.2%			$ 93,267
Total Estimated Cost													**$ 4,332,670**
Adopted Cost													

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Estimate Prepared by: CDCE Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN	TREE STUMP	PIT NO.

AMENDMENTS

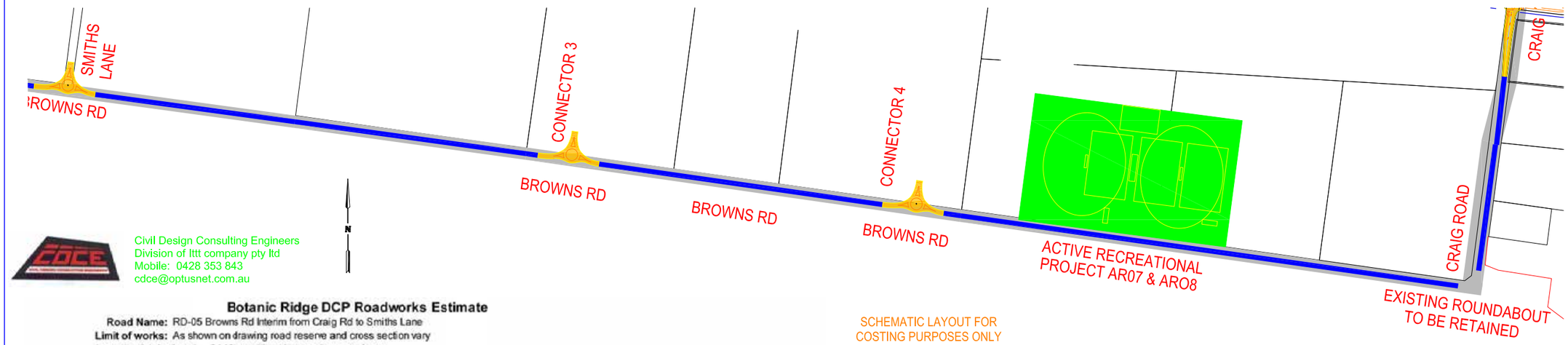
DATE	REMARKS	REV	APP

DATUM: N/A
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F.B. L.B.
JOB NO. 10 111
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REG. FILE NO. XXXXXXX
TRACED SAF
MELWAY REF.
CHECKED SAF
CONTRACT NO.

CIVIL DESIGN CONSULTING ENGINEERS

BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
CRAIG ROAD COST RD-04

DATE: November 2012
REVISION
10 111
SHEET 1 OF 1



SCHEMATIC LAYOUT FOR
COSTING PURPOSES ONLY

Civil Design Consulting Engineers
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Botanic Ridge DCP Roadworks Estimate

Road Name: RD-05 Browns Rd Interim from Craig Rd to Smiths Lane
Limit of works: As shown on drawing road reserve and cross section vary
Length of Job: length = 2140_m without intersections as shown
Notes: Costs based on City of Casey specification for road construction & GAA Interim Cross Section
resheeting existing asphalt surface & installation of kerb & channel with swale drains, allowance for minor piped drainage

Item	Unit	Rate	wide	units		cost/m		Unit cost	Quantity		subset	subtotal	Amount
			m			width			metres		cost		
Roadworks	subtotal							$ 703.59	2140	m		$ 1,505,685.00	$ 1,505,685
Bulk Earthworks	m³	$ 40.00	2.4	metres	$ 40	$ 96.00	m					$ 205,440.00	$ -
Pavement (profiling & resheet)	m²	$ 32.00	7.2	metres	$ 32	$ 230.40	m					$ 493,056.00	$ -
Crushed Rock Shoulder	m²	$ 20.00	9	metres	$ 20	$ -	m					$ -	$ -
Pavement Removal	m³	$ 10.00	9	metres		$ -						$ -	
Drainage Culvert 3No. 1.2x0.6	m	$ 660.00	67.5	No.	1							$ 44,550.00	$ 44,550
kerb and Channel S 503	m	$ 40.00	9	sides	$ 40	$ -	m					$ -	$ -
Kerb and Channel S 504		$ 45.00	2	sides	$ 45	$ 90.00	m					$ 192,600.00	
side entry pits std drw S305	unit	$ 2,000.00	2	interval metres	250	$ 16.00	unit					$ 34,240.00	
Bike path 2.0m wide conc	m²	$ 60.00	9	metres		$ -						$ -	
Shared pathway 2.5m wide gravel	m²	$ 30.00	9	metres		$ -						$ -	
Shared Pedestrian Footpath 2.5m wide	m²	$ 60.00	2.5	metres		$ 150.00						$ 321,000.00	
Pedestrian Footpath 1.5m gravel	m²	$ 30.00	9	metres		$ -						$ -	
Drainage , subgrade drain	m	$ 14.70	2		$ 15	$ 29.40	m					$ 62,916.00	$ -
Linemarking & Signage	m	$ 25.00	1	metres	$ 25	$ 25.00	m					$ 53,500.00	$ -
Landscaping refer plantings	m	$ 20.00	9	metres	$ 20	$ -	m					$ -	$ -
Concrete Path	m²	$ 45.00	9		$ 45	$ -	m					$ -	$ -
Wire rope barrier guard rail	m²	$ 115.00	329	metres	$ 115		m					$ 36,800.00	$ 36,800
level / trim nature strip	m²	$ 4.00	7	metres	$ 4	$ 28.00	m					$ 59,920.00	$
as per standard drawings													
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tree	9	rows of trees	$ 25	$ -	m					$ -	
Tube Stock Plantings	unit	$5.27 -$6.78	9	metres wide	$ 6	$ -	m					$ -	$ -
as per draft tree strategy												$ -	
root barriers	optional as per design in respect of appropriate offset criteria												
300mm dia conc drain Stormwater Cr BF	per metre	$182	20	metres		$ 17.09						$ 36,582	
375mm conc drain stormwater Cr Bk fill	per metre	$231	20	metres		$ 21.70						$ 46,431	
450mm conc drain stormwater Cr BF	per metre	$294	9	metres		$ -						$ -	
525mm conc drain stormwater Cr BF	per metre	$351	9	metres		$ -						$ -	
Service Road			1E-04	Lmetres									$ 0
Traffic Signals	Unit	$ 102,500.00	9	No of intesection								$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	9	metres								$ -	$ -
Street Lighting	m	$ 129.00	9	rows of lights		$ -	m					$ -	
lighting conduit	m	$ 50.00	9	No of runs/sides		$ -	m					$ -	
Subtotal						$ 703.59						$ 1,587,035.00	
estimated total													$ 1,587,035
Traffic Management										10.0%			$ 158,704
Contingency										20.0%			$ 317,407
Total + contingencies													
Services relocation Sec Poles, water fitting	Item												$ 250,000
Survey and Design										8%			$ 126,963
Overheads (supervision etc)										15%			$ 238,055
Total excluding land cost													**$ 2,678,164**
CPI Adjustment to Nov 2012										2.2%			$ 58,920
Total Estimated Cost													**$ 2,737,083**
Adopted Cost													

Estimate Prepared by: CDCE Nov-12

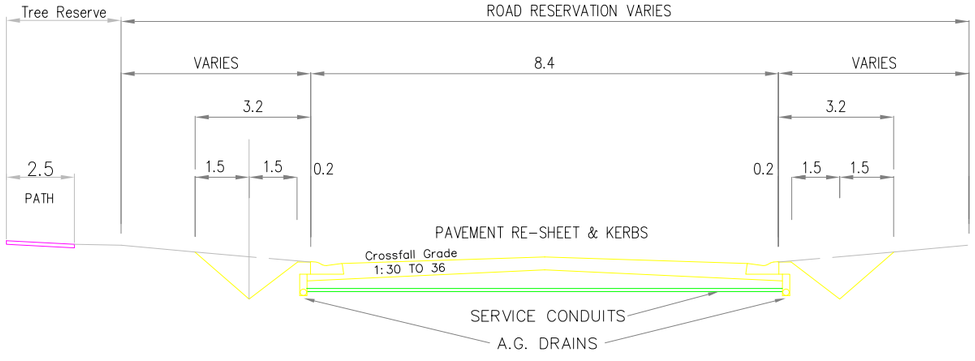
BOTANIC RIDGE DCP

RD-05 ROAD COSTS
BROWNS ROAD INTERIM UPGRADE

EXISTING ROAD RESERVE

COSTED ROAD

NOTES AND ASSUMPTIONS FOR RD-05
1: Traffic Management considered to be high in complexity
2: Survey & Design is considered to be medium complexity due to length of road and number of access points.
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) re sheet and new kerb works over existing alignment for the majority
b) length of proposed works



TYPICAL CROSS SECTION FOR INTERIM BROWNS RD

LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM	
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG	
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE	
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.	
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65	
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN	TREE STUMP	PIT NO.	

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
SCALE NOT TO SCALE
COORD. SYS.: N/A
F.B. L.B.
SURVEY N/A
DESIGN SAF
TRACED SAF
CHECKED SAF
JOB NO. 10 1110
DWG. FILE NAME XXXXXX.DWG
REG. FILE NO. XXXXXXX
MELWAY REF.
CONTRACT NO.

CIVIL DESIGN CONSULTING ENGINEERS

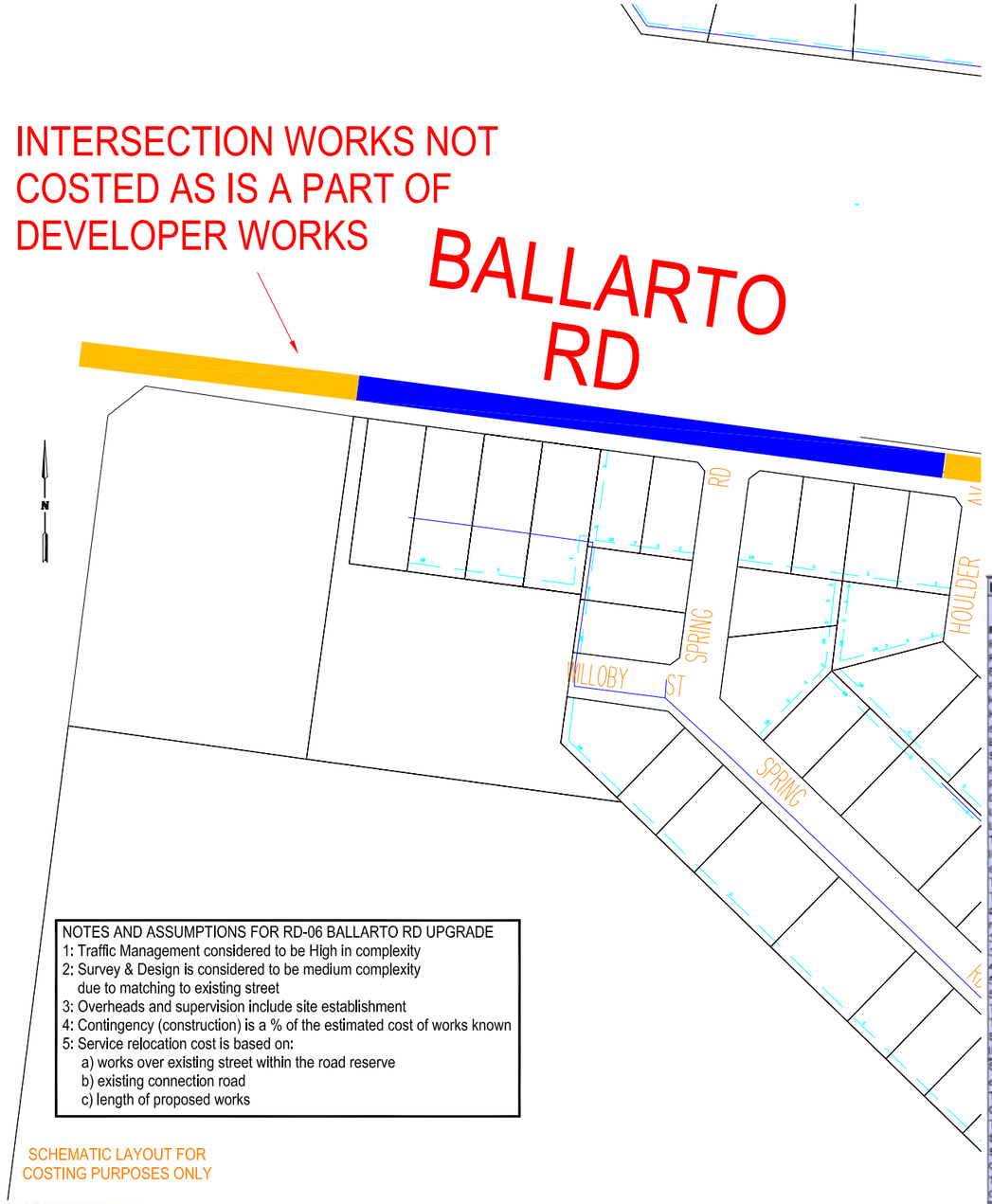
BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
BROWNS RD INTERIM UPGRADE RD-05

DATE: November 2012
REVISION
10 111
SHEET 1 OF 1

BOTANIC RIDGE DCP

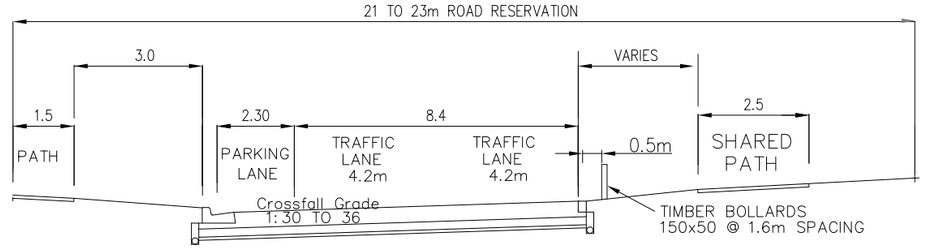
RD-06 ROAD COSTED
BALLARTO ROAD WEST CONNECTOR UPGRADE

INTERSECTION WORKS NOT COSTED AS IS A PART OF DEVELOPER WORKS



ROAD COSTED

INTERSECTION AREA



TYPICAL CONNECTOR CROSS SECTION SOLUTION SUPPLIED BY GAA

Botanic Ridge DCP Roadworks Estimate

Road Name: RD-06 Ballarto Rd. West extension
Limit of works: As shown on drawing
Length of Job: Based on road lengths combined 202Lm for the connection only
Notes: Costs based on road lengths typical section provided by GAA as shown

Item	Unit	Rate	wide	units		cost/m		Unit cost	Quantity		subset	subtotal	Amount
			m			width			metres		cost		
Roadworks	subtotal							$ 1,506.44	202	m		$ 304,301.78	$ 304,302
Bulk Earthworks	m³	$ 40.00	3	metres	$ 40	$ 120.00	m					$ 24,240.00	$ -
Pavement (Urban)	m²	$ 91.50	4.2	metres	$ 92	$ 384.30	m					$ 77,628.60	$ -
Ex Pavement Profiling and Resheeting	m²	$ 32.00	6.5	metres	$ 20	$ 130.00	m					$ 26,260.00	$ -
Pavement Removal	m³	$ 10.00	0	metres		$ -						$ -	
kerb and Channel S 503	m	$ 40.00	1	sides	$ 40	$ 40.00	m					$ 8,080.00	$ -
Kerb and Channel S 504		$ 45.00	1	sides	$ 45	$ 45.00	m					$ 9,090.00	
side entry pits std drw S305	unit	$ 2,000.00	2	interval metres	98.00	$ 44.44	m					$ 8,977.78	
Bike path 2.5m wide conc	m²	$ 60.00	0	metres		$ -						$ -	
Shared pathway 2.5m wide gravel	m²	$ 30.00	2.5	metres		$ 75.00						$ 15,150.00	
Pedestrian Footpath 1.5m wide	m²	$ 60.00	1.5	metres		$ 90.00						$ 18,180.00	
Pedestrian Footpath 1.5m gravel	m²	$ 30.00	0	metres		$ -						$ -	
Drainage , subgrade drain	m	$ 14.70	2		$ 15	$ 29.40	m					$ 5,938.80	$ -
Linemarking & Signage	m	$ 25.00	2	metres	$ 25	$ 50.00	m					$ 10,100.00	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Timber bollards	unit	$ 45.00	127		$ 45		m					$ 5,715.00	$ 5,715
level / trim nature strip	m²	$ 4.00	8.7	metres	$ 4	$ 34.80	m					$ 7,029.60	$
as per standard drawings													
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tree	2	rows of trees	$ 25	$ 50.00	m					$ 10,100.00	
Tube Stock Plantings	unit	$5.27 -$8.78	0	metres wide	$ 6	$ -	m					$ -	$ -
as per draft tree strategy												$ -	
root barriers	optional as per design in respect of appropriate offset criteria												
300mm dia conc drain Stormwater Cr BF	per metre	$182	67	metres		$ 60.37						$ 12,194	
375mm conc drain stormwater Cr Bk fill	per metre	$231	67	metres		$ 76.62						$ 15,477	
450mm conc drain stormwater Cr BF	per metre	$294	67	metres		$ 97.51						$ 19,698	
525mm conc drain stormwater Cr BF	per metre	$351	0	metres		$ -						$ -	
Service Road			0	Lmetres									$ -
Traffic Signals	Unit	$ 102,500.00	0	No of intesection								$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	0	metres								$ -	$ -
Street Lighting	m	$ 129.00	1	rows of lights		$ 129.00	m					$ 26,058.00	
lighting conduit	m	$ 50.00	1	No of runs/sides		$ 50.00	m					$ 10,100.00	
Subtotal						$ 1,506.44						$ 310,916.78	
estimated total													$ 310,017
Traffic Management									10.0%				$ 31,002
Contingency									20.0%				$ 62,003
Total + contingencies												$ 403,022	
Services relocation Sec Poles, water fitting	Item												$ 150,000
Survey and Design									10%				$ 31,002
Overheads (supervision etc)									15%				$ 46,503
Total excluding land cost													**$ 630,526**
CPI Adjustment to Nov 2012									2.2%				$ 13,872
Total Estimated Cost													**$ 644,398**
Adopted Cost													

NOTES AND ASSUMPTIONS FOR RD-06 BALLARTO RD UPGRADE
1: Traffic Management considered to be High in complexity
2: Survey & Design is considered to be medium complexity due to matching to existing street
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) works over existing street within the road reserve
b) existing connection road
c) length of proposed works

SCHEMATIC LAYOUT FOR COSTING PURPOSES ONLY

Civil Design Consulting Engineers
Division of Ittt company pty ltd
Mobile: 0428 353 843
cdce@optusnet.com.au

Estimate Prepared by: CDCE Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO. (25)
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN H	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

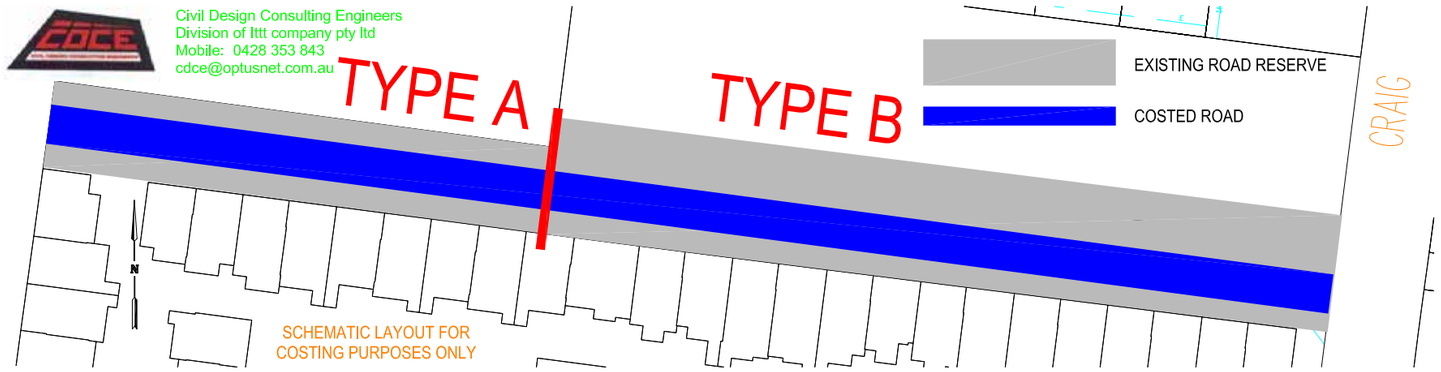
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DESIGN SAF REG. FILE NO. XXXXXXX
TRACED SAF MELWAY REF.
CHECKED SAF CONTRACT NO.

CIVIL DESIGN CONSULTING ENGINEERS

BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
RD–06 ROAD BALLARTO RD WEST
CONNECTOR UPGRADE

DATE: November 2012
REVISION
10 111
SHEET 1 OF 1

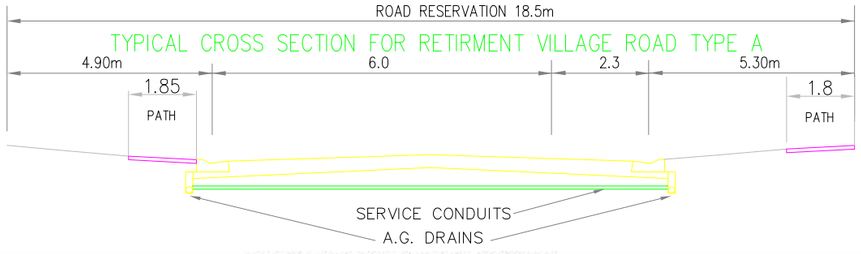
Civil Design Consulting Engineers
Division of Ittt company pty ltd
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BOTANIC RIDGE DCP

RD-08 ROAD COSTS
ACCESS LEVEL 2 ROAD 2 PARTS

NOTES AND ASSUMPTIONS FOR RD-08
1: Traffic Management considered to be low & medium in complexity
2: Survey & Design is considered to be medium complexity due to length of road and number of access points.
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) green field site
b) connection ot Craig Rd

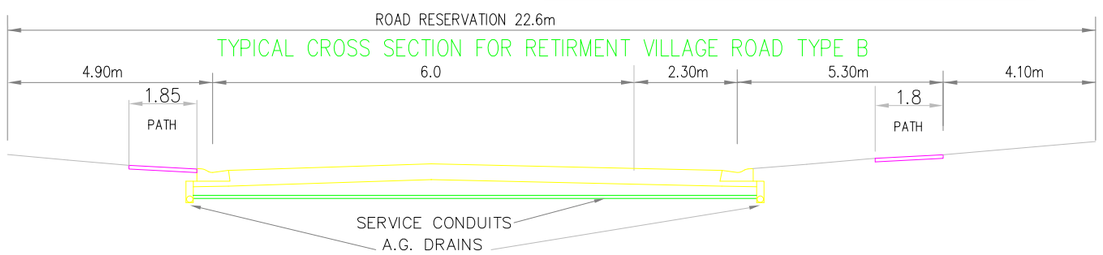


Botanic Ridge DCP Roadworks Estimate

Road Name: RD-08 18.5m wide road reserve Type A
Limit of works: As shown on drawing road reserve and cross section vary
Length of Job: length = 107Lm without intersections as shown
Notes: Costs based on City of Casey specification for road construction & GAA Cross Section 18.5m wide road reserve

Item	Unit	Rate	wide m	units		cost/m width		Unit cost	Quantity metres		subset cost	subtotal	Amount
Roadworks	subtotal							$ 1,833.47	107	m		$ 196,181.74	$ 196,182
Bulk Earthworks	m²	$ 40.00	4.45	metres	$ 40	$ 178.00	m					$ 19,046.00	$ -
Pavement (Urban)	m²	$ 91.40	7.1	metres	$ 91	$ 648.94	m					$ 69,436.58	$ -
Crushed Rock Shoulder	m²	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Pavement Removal	m²	$ 10.00	0	metres		$ -						$ -	
kerb and Channel S 503	m	$ 40.00	0	sides	$ 40	$ -	m					$ -	$ -
Kerb and Channel S 504		$ 45.00	2	sides	$ 45	$ 90.00	m					$ 9,630.00	
side entry pits std drw S305	unit	$ 2,000.00	2	interval metres	90	$ 44.44	unit					$ 4,755.56	
Bike path 2.0m wide conc	m²	$ 80.00	0	metres		$ -						$ -	
Shared pathway 2.5m wide gravel	m²	$ 30.00	0	metres		$ -						$ -	
Pedestrian Footpath 1.8m wide	m²	$ 60.00	1.8	metres		$ 108.00						$ 11,556.00	
Pedestrian Footpath 1.85m concrete	m²	$ 60.00	1.85	metres		$ 111.00						$ 11,877.00	
Drainage , subgrade drain	m	$ 14.70	2		$ 15	$ 29.40	m					$ 3,145.80	$ -
Linemarking & Signage	m	$ 25.00	1	metres	$ 25	$ 25.00	m					$ 2,675.00	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Double sided paling (acoustic) fence	m²	$ 67.00	2		$ 67	$ 134.00	m					$ 14,338.00	$ -
level / trim nature strip	m²	$ 4.00	5.85	metres	$ 4	$ 23.40	m					$ 2,503.80	$
as per standard drawings													
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tree	2	rows of trees	$ 25	$ 50.00	m					$ 5,350.00	
Tube Stock Plantings	unit	$5.27 -$6.78	0	metres wide	$ 6	$ -	m					$ -	$ -
as per draft tree strategy												$ -	
root barriers	optional as per design in respect of appropriate offset criteria												
300mm dia conc drain Stormwater Cr BF	per metre	$182	55	metres		$ 93.55						$ 10,010	
375mm conc drain stormwater Cr Bk fill	per metre	$231	55	metres		$ 118.74						$ 12,705	
450mm conc drain stormwater Cr BF	per metre	$294	0	metres		$ -						$ -	
525mm conc drain stormwater Cr BF	per metre	$361	0	metres		$ -						$ -	
Service Road			1E-04	Lmetres									$ 0
Traffic Signals	Unit	$ 192,500.00	0	No of intesection								$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	0	metres								$ -	$ -
Street Lighting	m	$ 129.00	1	rows of lights		$ 129.00	m					$ 13,803.00	
lighting conduit	m	$ 50.00	1	No of runs/sides		$ 50.00	m					$ 5,350.00	
Subtotal						$ 1,833.47						$ 196,181.74	
estimated total													$ 196,182
Traffic Management										5.0%			$ 9,809
Contingency										20.0%			$ 39,236
Total + contingencies													
Services relocation Sec Poles, water fitting	item												$ -
Survey and Design										8%			$ 15,695
Overheads (supervision etc)										15%			$ 29,427
Total excluding land cost													**$ 290,349**
CPI Adjustment to Nov 2012										2.2%			$ 6,388
Total Estimated Cost													**$ 296,737**
Adopted Cost													

Estimate Prepared by: CDCE Nov-12



Botanic Ridge DCP Roadworks Estimate

Road Name: RD-08 22.6m wide road reserve Type B
Limit of works: As shown on drawing road reserve and cross section vary
Length of Job: length = 168Lm without intersections as shown
Notes: Costs based on City of Casey specification for road construction & GAA Cross Section 25.0m wide road reserve

Item	Unit	Rate	wide m	units		cost/m width		Unit cost	Quantity metres		subset cost	subtotal	Amount
Roadworks	subtotal							$ 1,802.28	168	m		$ 302,783.79	$ 302,784
Bulk Earthworks	m²	$ 40.00	4.45	metres	$ 40	$ 178.00	m					$ 29,904.00	$ -
Pavement (Urban)	m²	$ 91.40	7.1	metres	$ 91	$ 648.94	m					$ 109,021.92	$ -
Crushed Rock Shoulder	m²	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Pavement Removal	m²	$ 10.00	0	metres		$ -						$ -	
kerb and Channel S 503	m	$ 40.00	0	sides	$ 40	$ -	m					$ -	$ -
Kerb and Channel S 504		$ 45.00	2	sides	$ 45	$ 90.00	m					$ 15,120.00	
side entry pits std drw S305	unit	$ 2,000.00	2	interval metres	90	$ 44.44	unit					$ 7,466.67	
Bike path 2.0m wide conc	m²	$ 80.00	0	metres		$ -						$ -	
Shared pathway 2.5m wide gravel	m²	$ 30.00	0	metres		$ -						$ -	
Pedestrian Footpath 1.8m wide	m²	$ 60.00	1.8	metres		$ 108.00						$ 18,144.00	
Pedestrian Footpath 1.85m concrete	m²	$ 60.00	1.85	metres		$ 111.00						$ 18,648.00	
Drainage , subgrade drain	m	$ 14.70	2		$ 15	$ 29.40	m					$ 4,939.20	$ -
Linemarking & Signage	m	$ 25.00	2	metres	$ 25	$ 50.00	m					$ 8,400.00	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Double sided paling (acoustic) fence	m²	$ 67.00	1		$ 67	$ 67.00	m					$ 11,266.00	$ -
level / trim nature strip	m²	$ 4.00	10	metres	$ 4	$ 40.00	m					$ 6,720.00	$ -
as per standard drawings													
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tree	2	rows of trees	$ 25	$ 50.00	m					$ 8,400.00	
Tube Stock Plantings	unit	$5.27 -$6.78	0	metres wide	$ 6	$ -	m					$ -	$ -
as per draft tree strategy												$ -	
root barriers	optional as per design in respect of appropriate offset criteria												
300mm dia conc drain Stormwater Cr BF	per metre	$182	84	metres		$ 91.00						$ 15,288	
375mm conc drain stormwater Cr Bk fill	per metre	$231	84	metres		$ 115.50						$ 19,404	
450mm conc drain stormwater Cr BF	per metre	$294	0	metres		$ -						$ -	
525mm conc drain stormwater Cr BF	per metre	$351	0	metres		$ -						$ -	
Service Road			1E-04	Lmetres									$ 0
Traffic Signals	Unit	$ 192,500.00	0	No of intesection								$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	0	metres								$ -	$ -
Street Lighting	m	$ 129.00	1	rows of lights		$ 129.00	m					$ 21,672.00	
lighting conduit	m	$ 50.00	1	No of runs/sides		$ 50.00	m					$ 8,400.00	
Subtotal						$ 1,802.28						$ 302,783.79	
estimated total													$ 302,784
Traffic Management										7.0%			$ 21,195
Contingency										20.0%			$ 60,557
Total + contingencies													
Services relocation Sec Poles, water fitting	item												$ 50,000
Survey and Design										8%			$ 24,223
Overheads (supervision etc)										15%			$ 45,418
Total excluding land cost													**$ 504,176**
CPI Adjustment to Nov 2012										2.2%			$ 11,092
Total Estimated Cost													**$ 515,268**
Adopted Cost													

Estimate Prepared by: CDCE Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
SCALE NOT TO SCALE
COORD. SYS.: N/A
F.B. L.B.
SURVEY N/A
DESIGN SAF
TRACED SAF
CHECKED SAF
JOB NO. 10 1110
DWG. FILE NAME XXXXXX.DWG
REG. FILE NO. XXXXXXX
MELWAY REF.
CONTRACT NO.

CIVIL DESIGN CONSULTING ENGINEERS

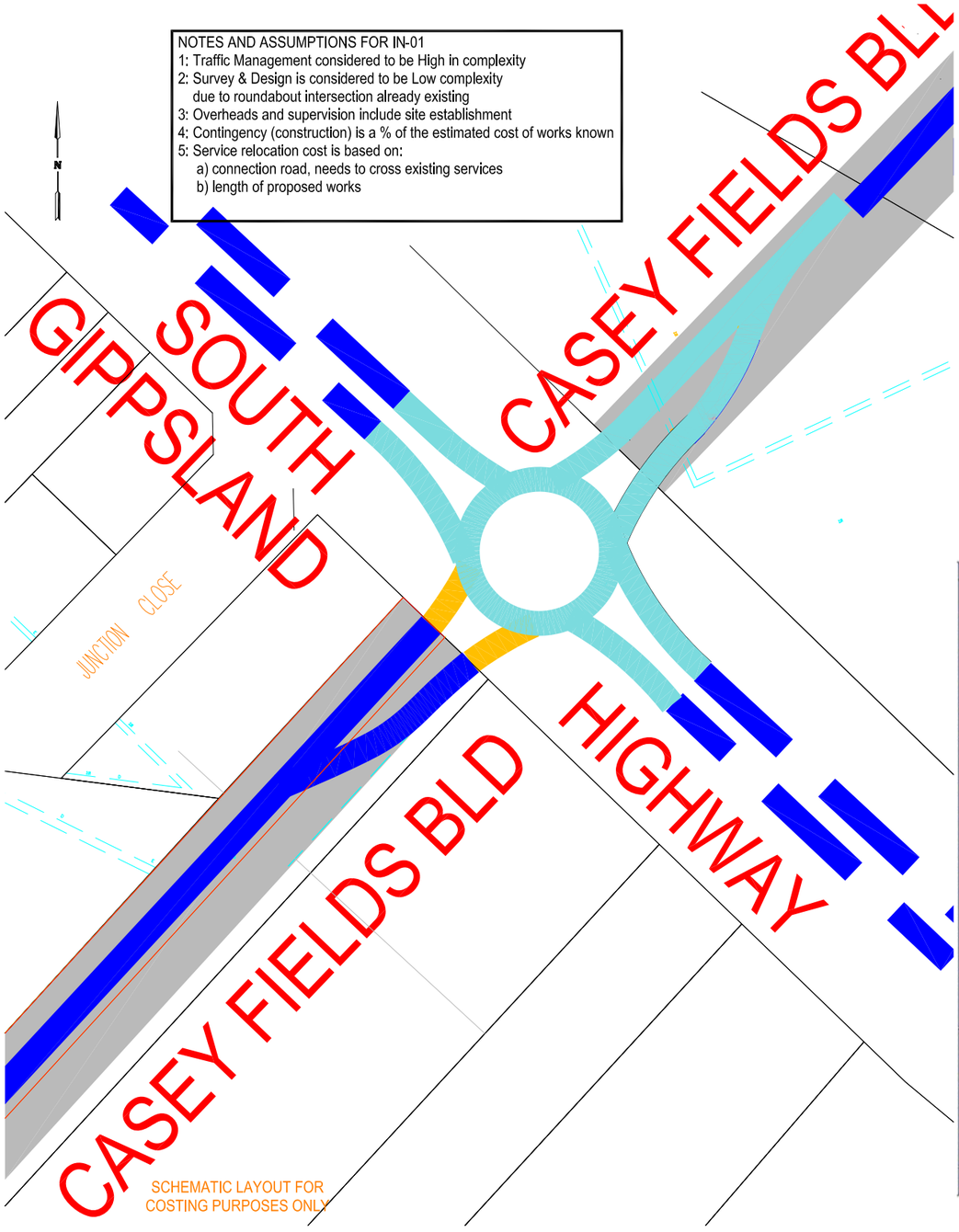
BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
CONSTRAINED ACCESS LEVEL 2 RD RD08

DATE: November 2012
REVISION
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SHEET 1 OF 1

BOTANIC RIDGE DCP

IN-01 INTERSECTION COSTS CONNECTION TO SOUTH GIPPSLAND HWY & CASEY FIELDS BLD ROUNDABOUT

NOTES AND ASSUMPTIONS FOR IN-01
1: Traffic Management considered to be High in complexity
2: Survey & Design is considered to be Low complexity due to roundabout intersection already existing
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) connection road, needs to cross existing services
b) length of proposed works



ULTIMATE ROAD RESERVE

ROAD

COSTED INTERSECTION AREA

Civil Design Consulting Engineers
Division of Ittt company pty ltd
Mobile: 0428 353 843
cdce@optusnet.com.au

Botanic Ridge DCP Roadworks Estimate

Road Name: IN-01 connection to roundabout SGH & Casey Fields Bld.
Limit of works: As shown on drawing
Length of Job: Based on road lengths combined 41Lm for the connection only
Notes: Costs based on road lengths confined with in the connection to roundabout as shown

Item	Unit	Rate	wide m	units		cost/m width		Unit cost	Quantity metres		subset cost	subtotal	Amount
Roadworks	subtotal							$ 1,835.31	41	m		$ 75,247.75	$ 75,248
Bulk Earthworks	m³	$ 40.00	4.75	metres	$ 40	$ 190.00	m					$ 7,790.00	$ -
Pavement (Urban)	m³	$ 91.50	8.7	metres	$ 92	$ 796.05	m					$ 32,638.05	$ -
Crushed Rock Shoulder	m³	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Pavement Removal	m³	$ 10.00	0	metres		$ -						$ -	
kerb and Channel S 503	m	$ 40.00	1	sides	$ 40	$ 40.00	m					$ 1,640.00	$ -
Kerb and Channel S 504		$ 45.00	1	sides	$ 45	$ 45.00	m					$ 1,845.00	
side entry pits std drw S305	unit	$ 2,000.00	2	interval metres	50.00	$ 97.56	m					$ 4,900.00	
Bike path 2.0m wide conc	m²	$ 60.00	0	metres		$ -						$ -	
Shared pathway 2.5m wide gravel	m²	$ 30.00	0	metres		$ -						$ -	
Pedestrian Footpath 1.5m wide	m²	$ 60.00	1.5	metres		$ 90.00						$ 3,690.00	
Pedestrian Footpath 1.5m gravel	m²	$ 30.00	0	metres		$ -						$ -	
Drainage , subgrade drain	m	$ 14.70	1		$ 15	$ 14.70	m					$ 602.70	$ -
Linemarking & Signage	m	$ 25.00	2	metres	$ 25	$ 50.00	m					$ 2,050.00	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Concrete Path	m²	$ 45.00	0		$ 45	$ -	m					$ -	$ -
level / trim nature strip	m²	$ 4.00	13	metres	$ 4	$ 52.00	m					$ 2,132.00	$ -
as per standard drawings													
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tree	2	rows of trees	$ 25	$ 50.00	m					$ 2,050.00	
Tube Stock Plantings	unit	$5.27 -$6.78	0	metres wide	$ 6	$ -	m					$ -	$ -
as per draft trees strategy												$ -	
root barriers	optional as per design in respect of appropriate offset criteria												
300mm dia conc drain Stormwater Cr BF	per metre	$182	0	metres		$ -						$ -	
375mm conc drain stormwater Cr Bk fill	per metre	$231	41	metres		$ 231.00						$ 9,471	
450mm conc drain stormwater Cr BF	per metre	$294	0	metres		$ -						$ -	
525mm conc drain stormwater Cr BF	per metre	$351	0	metres		$ -						$ -	
Service Road			0	Lmetres									$ -
Traffic Signals	Unit	$ 102,500.00	0	No of intesection								$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	0	metres								$ -	$ -
Street Lighting	m	$ 129.00	1	rows of lights		$ 129.00	m					$ 5,289.00	
lighting conduit	m	$ 50.00	1	No of runs/sides		$ 50.00	m					$ 2,050.00	
Subtotal						$ 1,835.31						$ 75,247.75	
estimated total													$ 75,248
Traffic Management									5.0%				$ 3,762
Contingency									20.0%				$ 15,050
Total + contingencies													
Services relocation Sec Poles, water fitting	item												$ 100,000
Survey and Design									10%				$ 7,525
Oxheads (supervision etc)									15%				$ 11,287
Total excluding land cost													**$ 212,872**
CPI Adjusted to Nov. 2012									2.2%				$ 4,683
Total Estimated Cost													**$ 217,555**
Adopted Cost													

Estimate Prepared by: CDCE Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO. (25)
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN H	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A SCALE NOT TO SCALE
COORD. SYS.: N/A
F.B. L.B. JOB NO. 10 111
SURVEY N/A DWG. FILE NAME XXXXXX.DWG
DESIGN SAF REG. FILE NO. XXXXXXX
TRACED SAF MELWAY REF.
CHECKED SAF CONTRACT NO.

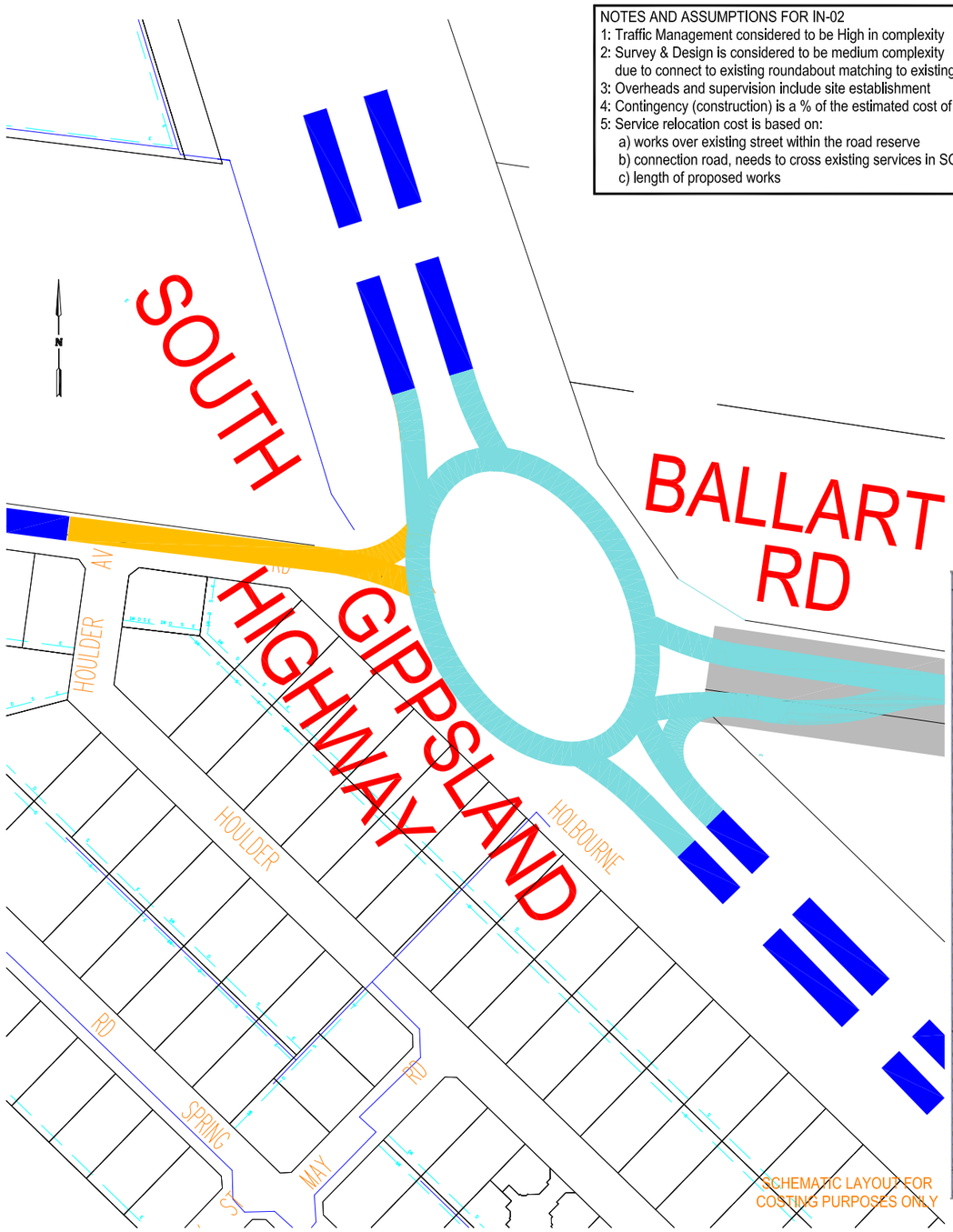
CIVIL DESIGN CONSULTING ENGINEERS

BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
IN-01 INTERSECTION COST NEW CONNECTION TO
STH GIPPSLAND HWY & CASEY FIELD BLD ROUNDABOUT

DATE: November 2012
REVISION
10 111
SHEET 1 OF 1

NOTES AND ASSUMPTIONS FOR IN-02
1: Traffic Management considered to be High in complexity
2: Survey & Design is considered to be medium complexity due to connect to existing roundabout matching to existing street
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) works over existing street within the road reserve
b) connection road, needs to cross existing services in SGH
c) length of proposed works

BOTANIC RIDGE DCP
IN-02 INTERSECTION COSTS
SOUTH GIPPSLAND HWY & BALLARTO ROAD ROUNDABOUT



ULTIMATE ROAD RESERVE
ROAD
COSTED INTERSECTION AREA

Civil Design Consulting Engineers
Division of Ittt company pty ltd
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Botanic Ridge DCP Roadworks Estimate

Road Name: IN-02 connection to roundabout SGH & Ballarto Rd.
Limit of works: As shown on drawing
Length of Job: Based on road lengths combined 157Lm for the connection only
Notes: Costs based on road lengths confined with in the connection to roundabout as shown

Item	Unit	Rate	wide m	units		cost/m width		Unit cost	Quantity metres		subset cost	subtotal	Amount
Roadworks	subtotal							$ 1,748.23	157	m		$ 274,471.75	$ 274,472
Bulk Earthworks	m²	$ 40.00	4.75	metres	$ 40	$ 190.00	m					$ 29,830.00	$ -
Pavement (Urban)	m²	$ 91.50	8.7	metres	$ 92	$ 796.05	m					$ 124,979.85	$ -
Crushed Rock Shoulder	m²	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Pavement Removal	m²	$ 10.00	0	metres		$ -						$ -	
kerb and Channel S 503	m	$ 40.00	1	sides	$ 40	$ 40.00	m					$ 6,280.00	$ -
Kerb and Channel S 504		$ 45.00	1	sides	$ 45	$ 45.00	m					$ 7,065.00	
side entry pits std drw S305	unit	$ 2,000.00	2	interval metres	90.00	$ 25.48	m					$ 4,000.00	
Bike path 2.0m wide conc	m²	$ 60.00	0	metres		$ -						$ -	
Shared pathway 2.5m wide gravel	m²	$ 30.00	2.5	metres		$ 75.00						$ 11,775.00	
Pedestrian Footpath 1.5m wide	m²	$ 60.00	0	metres		$ -						$ -	
Pedestrian Footpath 1.5m gravel	m²	$ 30.00	0	metres		$ -						$ -	
Drainage , subgrade drain	m	$ 14.70	1		$ 15	$ 14.70	m					$ 2,307.90	$ -
Linemarking & Signage	m	$ 25.00	2	metres	$ 25	$ 50.00	m					$ 7,850.00	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Concrete Path	m²	$ 45.00	0		$ 45	$ -	m					$ -	$ -
level / trim nature strip	m²	$ 4.00	13	metres	$ 4	$ 52.00	m					$ 8,164.00	$ -
as per standard drawings													
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tree	2	rows of trees	$ 25	$ 50.00	m					$ 7,850.00	
Tube Stock Plantings	unit	$5.27 -$6.78	0	metres wide	$ 6	$ -	m					$ -	$ -
as per draft tree strategy												$ -	
root barriers	optional as per design in respect of appropriate offset criteria												
300mm dia conc drain Stormwater Cr BF	per metre	$182	0	metres		$ -						$ -	
375mm conc drain stormwater Cr Bk fill	per metre	$231	157	metres		$ 231.00						$ 36,267	
450mm conc drain stormwater Cr BF	per metre	$294	0	metres		$ -						$ -	
525mm conc drain stormwater Cr BF	per metre	$351	0	metres		$ -						$ -	
Service Road			0	Lmetres									$ -
Traffic Signals	Unit	$ 102,500.00	0	No of intesection								$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	0	metres								$ -	$ -
Street Lighting	m	$ 129.00	1	rows of lights		$ 129.00	m					$ 20,253.00	
lighting conduit	m	$ 50.00	1	No of runs/sides		$ 50.00	m					$ 7,850.00	
Subtotal						$ 1,748.23						$ 274,471.75	
estimated total													$ 274,472
Traffic Management										10.0%			$ 27,447
Contingency										20.0%			$ 54,894
Total + contingencies													
Services relocation Sec Poles, water fitting	Item												$ 100,000
Survey and Design										7%			$ 19,213
Overheads (supervision etc)										15%			$ 41,171
Total excluding land cost													**$ 517,197**
CPI Adjustment to Nov. 2012										2.2%			$ 11,378
Total Estimated Cost													**$ 528,575**
Adopted Cost													

Estimate Prepared by: CDCE Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN	TREE STUMP	PIT NO.

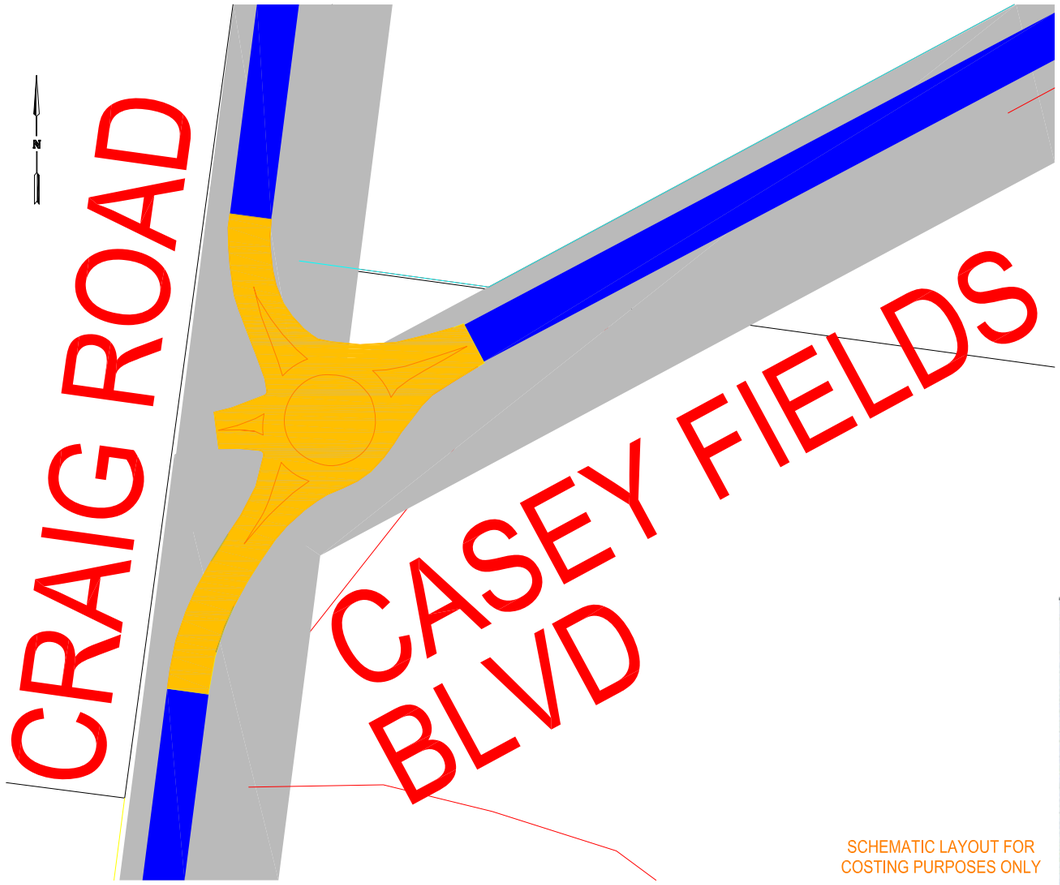
AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
COORD. SYS.: N/A
SCALE NOT TO SCALE
F.B. L.B.
JOB NO. 10 111
SURVEY N/A
DWG. FILE NAME XXXXXX.DWG
DESIGN SAF
REG. FILE NO. XXXXXXX
TRACED SAF
MELWAY REF.
CHECKED SAF
CONTRACT NO.

CIVIL DESIGN CONSULTING ENGINEERS
BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
IN-02 INTERSECTION COST NEW CONNECTION TO STH GIPPSLAND HWY & BALLARTO RD ROUNDABOUT

DATE: November 2012
REVISION
10 111
SHEET 1 OF 1



SCHEMATIC LAYOUT FOR COSTING PURPOSES ONLY

BOTANIC RIDGE DCP

IN-03 INTERSECTION COSTS
CRAIG RD & CASEY FIELDS BLVD

ULTIMATE ROAD RESERVE

COSTED INTERSECTION AREA

NOTES AND ASSUMPTIONS FOR IN-03 BROWNS & CASEY FIELDS BLD
1: Traffic Management considered to be medium in complexity
2: Survey & Design is considered to be medium complexity due to roundabout intersection proposal
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) intersection has one sided encroachment on existing services
b) connection road is greenfield
c) length of proposed works

Botanic Ridge DCP Roadworks Estimate

Road Name: IN-03 Craigs Rd & Casey Fields Bld
Limit of works: As shown on drawing road reserve and cross section vary
Length of Job: Intersection works only 20m dia central island
Notes: Costs based on City of Casey specification for road construction & GAA schematic intersection

Item	Unit	Rate	wide m	units		cost/m width		Unit cost	Quantity metres		subset cost	subtotal	Amount
Roadworks	subtotal							$ 339,111.60	1	m		$ 339,111.60	$ 339,112
Bulk Earthworks	m²	$ 40.00	846	metres	$ 40	$ 33,840.00	m					$ 33,840.00	$ -
Pavement (urban)	m²	$ 91.50	1489	metres	$ 92	$ 136,243.50	m					$ 136,243.50	$ -
Crushed Rock Shoulder	m²	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Pavement Removal	m²	$ 10.00	0	metres		$ -						$ -	
Drainage Culvert 3No. 1.2x0.6	m	$ 660.00	0	No.	1							$ -	$ -
kerb and Channel S 503	m	$ 40.00	224	sides	$ 40	$ 8,960.00	m					$ 8,960.00	$ -
Kerb and Channel S 504		$ 45.00	279	sides	$ 45	$ 12,555.00	m					$ 12,555.00	
side entry pits std drw S305	unit	$ 2,000.00	9	interval metres	90	$ 18,000.00	unit					$ 18,000.00	
Bike path 2.5m wide conc	m²	$ 60.00	429	metres		$ 25,740.00						$ 25,740.00	
Shared pathway 2.5m wide gravel	m²	$ 30.00	0	metres		$ -						$ -	
Pedestrian Footpath 1.5m wide	m²	$ 60.00	0	metres		$ -						$ -	
Pedestrian Footpath 1.5m gravel	m²	$ 30.00	0	metres		$ -						$ -	
Drainage , subgrade drain	m	$ 14.70	503		$ 15	$ 7,394.10	m					$ 7,394.10	$ -
Linemarking & Signage	m	$ 25.00	503	metres	$ 25	$ 12,575.00	m					$ 12,575.00	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Concrete Island infill	m²	$ 45.00	153		$ 45	$ 6,885.00	m					$ 6,885.00	$ -
level / trim nature strip	m²	$ 4.00	2045	metres	$ 4	$ 8,180.00	m					$ 8,180.00	$ -
as per standard drawings													
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tree	16	rows of trees	$ 25	$ 400.00	m					$ 400.00	
Tube Stock Plantings	unit	$5.27 -$6.78	0	metres wide	$ 6	$ -	m					$ -	$ -
as per draft tree strategy												$ -	
root barriers	optional as per design in respect of appropriate offset criteria												
300mm dia conc drain Stormwater Cr BF	per metre	$182	90	metres		$ 16,380.00						$ 16,380	
375mm conc drain stormwater Cr Bk fill	per metre	$231	87	metres		$ 20,097.00						$ 20,097	
450mm conc drain stormwater Cr BF	per metre	$294	0	metres		$ -						$ -	
525mm conc drain stormwater Cr BF	per metre	$351	0	metres		$ -						$ -	
Service Road			0	Lmetres									$ -
Traffic Signals	Unit	$ 102,500.00	0	No of intesection								$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	0	metres								$ -	$ -
Street Lighting	m	$ 129.00	178	rows of lights		$ 22,962.00	m					$ 22,962.00	
lighting conduit	m	$ 50.00	178	No of runs/sides		$ 8,900.00	m					$ 8,900.00	
Subtotal						$ 339,111.60						$ 339,111.60	
estimated total													$ 339,112
Traffic Management										7.0%			$ 23,738
Contingency										20.0%			$ 67,822
Total + contingencies													
Services relocation Sec Poles, water fitting	item												$ 100,000
Survey and Design										10%			$ 33,911
Overheads (supervision etc)										15%			$ 50,867
Total excluding land cost													**$ 615,450**
CPI Adjustment to Nov 2012										2.2%			$ 13,540
Total Estimated Cost													**$ 628,990**
Adopted Cost													

Estimate Prepared by: CDCE Nov-12

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LEGEND

PROPOSED DRAINAGE	GAS MAIN	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.
PROPOSED KERB AND CHANNEL	POWER OVERHEAD	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN	POWER PIT	HOUSE DRAIN	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
COORD. SYS.: N/A
SCALE NOT TO SCALE
F.B. L.B.
JOB NO. 10 111
SURVEY N/A
DWG. FILE NAME XXXXXX.DWG
DESIGN SAF
REG. FILE NO. XXXXXXX
TRACED SAF
W.E.WAY REF.
CHECKED SAF
CONTRACT NO.

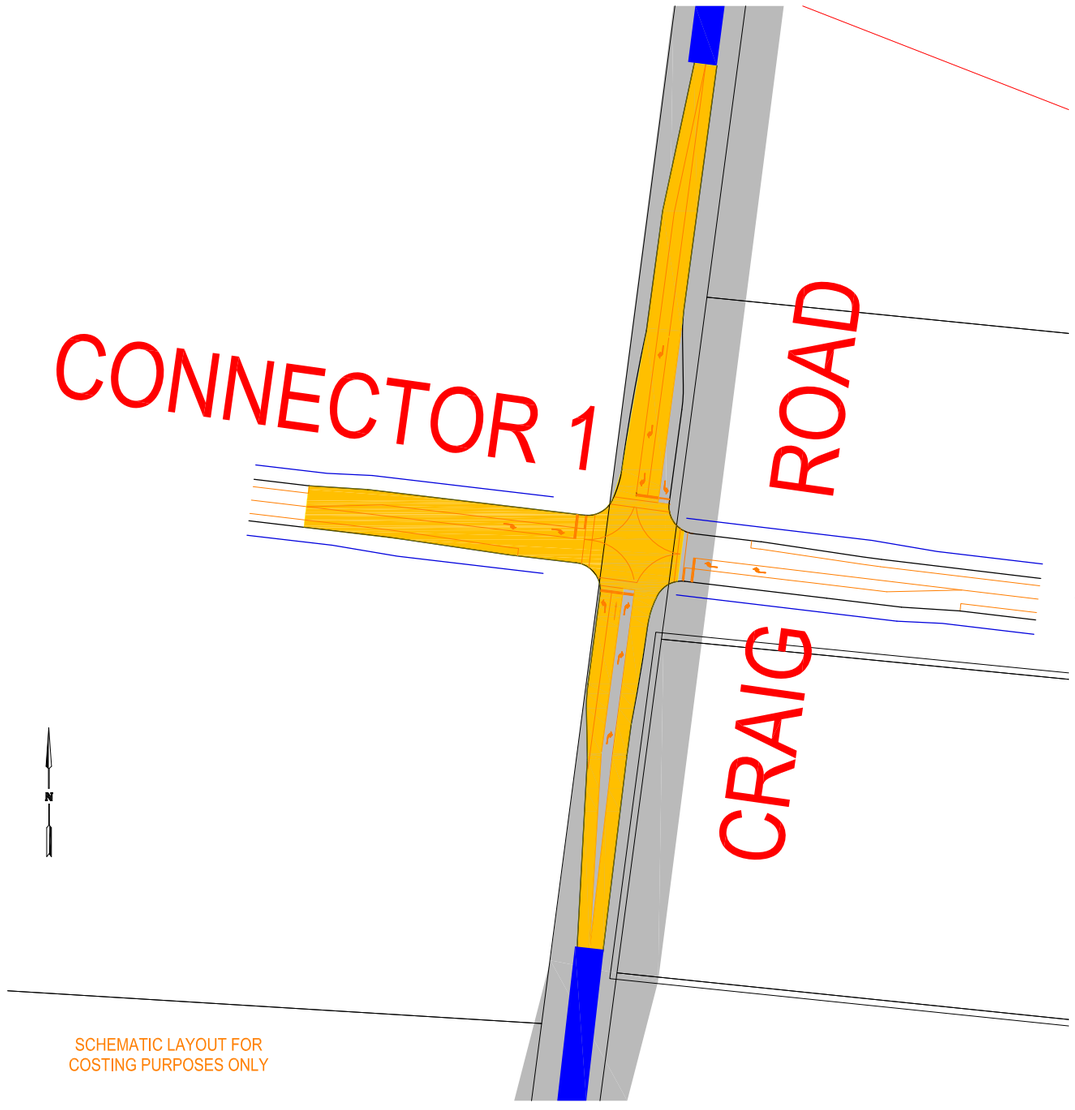
CIVIL DESIGN CONSULTING ENGINEERS

BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
CRAIG RD & CASEY FIELDS BLVD
IN-03 INTERSECTION

DATE: November 2012
REVISION
10 111
SHEET 1 OF 1

BOTANIC RIDGE DCP

IN-04 INTERSECTION COSTS
CRAIG RD & CONNECTOR 1 EAST



ULTIMATE ROAD RESERVE

COSTED INTERSECTION AREA

Botanic Ridge DCP Roadworks Estimate

Road Name: IN-04 Craigs Rd & connector 1 east
Limit of works: As shown on drawing road reserve and cross section vary
Length of Job: Intersection works only
Notes: Costs based on City of Casey specification for road construction & GAA schematic intersection

Item	Unit	Rate	wide	units		cost/m		Unit cost	Quantity		subset	subtotal	Amount
			m			width			metres		cost		
Roadworks	subtotal							$ 868,367.10	1	m		$ 868,367.10	$ 868,367
Bulk Earthworks	m²	$ 40.00	2575	metres	$ 40	$ 103,000.00	m					$ 103,000.00	$ -
Pavement (urban)	m²	$ 91.50	4549	metres	$ 92	$ 415,233.50	m					$ 416,233.50	$ -
Crushed Rock Shoulder	m²	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Pavement Removal	m²	$ 10.00	0	metres		$ -						$ -	
Drainage Culvert 3No. 1.2x0.6	m	$ 660.00	0	No.	1							$ -	$ -
kerb and Channel S 503	m	$ 40.00	0	sides	$ 40	$ -	m					$ -	$ -
Kerb and Channel S 504		$ 45.00	668	sides	$ 45	$ 30,060.00	m					$ 30,060.00	
side entry pits std drw S305	unit	$ 2,000.00	12	interval metres	90	$ 24,000.00	unit					$ 24,000.00	
Bike path 2.5m wide conc	m²	$ 60.00	535	metres		$ 32,100.00						$ 32,100.00	
Shared pathway 3.0m wide concrete	m²	$ 60.00	834	metres		$ 50,040.00						$ 50,040.00	
Pedestrian Footpath 1.5m wide	m²	$ 60.00	0	metres		$ -						$ -	
Pedestrian Footpath 1.5m gravel	m²	$ 30.00	0	metres		$ -						$ -	
Drainage , subgrade drain	m	$ 14.70	668		$ 15	$ 9,819.60	m					$ 9,819.60	$ -
Linemarking & Signage	m	$ 25.00	790	metres	$ 25	$ 19,750.00	m					$ 19,750.00	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Concrete Island infill	m²	$ 45.00	0		$ 45	$ -	m					$ -	$ -
level / trim nature strip	m²	$ 4.00	4740	metres	$ 4	$ 18,960.00	m					$ 18,960.00	$ -
as per standard drawings													
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tree	15	rows of trees	$ 25	$ 375.00	m					$ 375.00	
Tube Stock Plantings	unit	$5.27 -$6.78	0	metres wide	$ 6	$ -	m					$ -	$ -
as per draft tree strategy												$ -	
root barriers	optional as per design in respect of appropriate offset criteria												
300mm dia conc drain Stormwater Cr BF	per metre	$182	132	metres		$ 24,024.00						$ 24,024	
375mm conc drain stormwater Cr Bk fill	per metre	$231	132	metres		$ 30,492.00						$ 30,492	
450mm conc drain stormwater Cr BF	per metre	$294	132	metres		$ 38,808.00						$ 38,808	
525mm conc drain stormwater Cr BF	per metre	$351	0	metres		$ -						$ -	
Service Road			0	Lmetres									$ -
Traffic Signals	Unit	$ 102,500.00	4	No of intersection								$ 410,000	$ 410,000
Traffic Signal Conduit subset	m	$ 35.00	258	metres								$ 9,030.00	$ 9,030
Street Lighting	m	$ 129.00	395	rows of lights		$ 50,955.00	m					$ 50,955.00	
lighting conduit	m	$ 50.00	395	No of runs/sides		$ 19,750.00	m					$ 19,750.00	
Subtotal						$ 868,367.10						$ 1,287,397.10	
estimated total													$ 1,287,397
Traffic Management									10.0%				$ 128,740
Contingency									20.0%				$ 257,479
Total + contingencies													
Services relocation Sec Poles, water fitting	Item												$ 250,000
Survey and Design									10%				$ 128,740
Overheads (supervision etc)									15%				$ 193,110
Total excluding land cost													**$ 2,245,466**
CPI Adjustment to Nov. 2012									2.2%				$ 49,400
Total Estimated Cost													**$ 2,294,866**
Adopted Cost													

Estimate Prepared by: CDCE Nov-12

NOTES AND ASSUMPTIONS FOR IN-04 CRIAG RD & C1 west
1: Traffic Management considered to be high in complexity
2: Survey & Design is considered to be medium complexity due to signalised intersection proposal
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) intersection has one sided encroachment on existing services
b) connection road green field
c) length of proposed works

Civil Design Consulting Engineers
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LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN H	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
COORD. SYS.: N/A
F.B. L.B.
SURVEY N/A
DESIGN SAF
TRACED SAF
CHECKED SAF

SCALE NOT TO SCALE
JOB NO. 10 111
DWG. FILE NAME XXXXXX.DWG
REG. FILE NO. XXXXXXX
MELWAY REF.
CONTRACT NO.

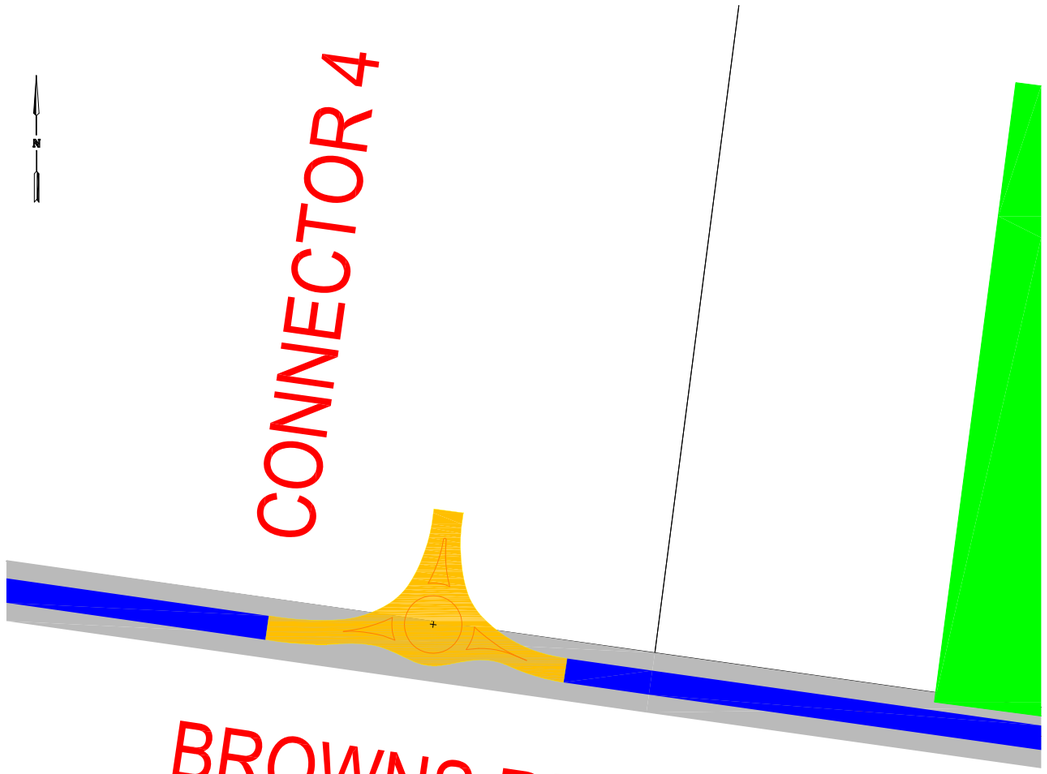
CIVIL DESIGN CONSULTING ENGINEERS

BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
CRAIG RD & CONNECTOR 1
IN–04 INTERSECTION

DATE: November 2012
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10 111
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BOTANIC RIDGE DCP

IN-06 INTERSECTION COSTS
BROWNS RD & CONNECTOR 4



SCHEMATIC LAYOUT FOR
COSTING PURPOSES ONLY

EXISTING ROAD RESERVE

COSTED INTERSECTION AREA

NOTES AND ASSUMPTIONS FOR IN-06 BROWNS & C4
1: Traffic Management considered to be high in complexity
2: Survey & Design is considered to be medium complexity
due to roundabout intersection proposal
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) intersection has one sided encroachment on existing services
b) connection road is greenfield
c) length of proposed works

Botanic Ridge DCP Roadworks Estimate

Road Name: IN-06 Browns Rd & Connector C4
Limit of works: As shown on drawing road reserve and cross section vary
Length of Job: Intersection works only 20m dia central island
Notes: Costs based on City of Casey specification for road construction & GAA schematic intersection

Item	Unit	Rate	wide	units		cost/m		Unit cost	Quantity		subset	subtotal	Amount
			m			width			metres		cost		
Roadworks	subtotal							$ 293,790.50	1	m		$ 293,790.50	$ 293,791
Bulk Earthworks	m³	$ 40.00	885	metres	$ 40	$ 35,400.00	m					$ 35,400.00	$ -
Pavement (urban)	m²	$ 91.50	1355	metres	$ 92	$ 123,982.50	m					$ 123,982.50	$ -
Crushed Rock Shoulder	m²	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Pavement Removal	m²	$ 10.00	0	metres		$ -						$ -	
Drainage Culvert 3No. 1.2x0.6	m	$ 660.00	0	No.	1							$ -	$ -
kerb and Channel S 503	m	$ 40.00	201	sides	$ 40	$ 8,040.00	m					$ 8,040.00	$ -
Kerb and Channel S 504		$ 45.00	259	sides	$ 45	$ 11,655.00	m					$ 11,655.00	
side entry pits std drw S305	unit	$ 2,000.00	7	interval metres	90	$ 14,000.00	unit					$ 14,000.00	
Bike path 2.5m wide conc	m²	$ 60.00	362	metres		$ 21,720.00						$ 21,720.00	
Shared pathway 2.5m wide gravel	m²	$ 30.00	0	metres		$ -						$ -	
Pedestrian Footpath 1.5m wide	m²	$ 60.00	0	metres		$ -						$ -	
Pedestrian Footpath 1.5m gravel	m²	$ 30.00	0	metres		$ -						$ -	
Drainage , subgrade drain	m	$ 14.70	460		$ 15	$ 6,762.00	m					$ 6,762.00	$ -
Linemarking & Signage	m	$ 25.00	46	metres	$ 25	$ 1,150.00	m					$ 1,150.00	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Concrete Island infill	m²	$ 45.00	140		$ 45	$ 6,300.00	m					$ 6,300.00	$ -
level / trim nature strip	m²	$ 4.00	1869	metres	$ 4	$ 7,476.00	m					$ 7,476.00	$
as per standard drawings													
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tree	15	rows of trees	$ 25	$ 375.00	m					$ 375.00	
Tube Stock Plantings	unit	$5.27 -$8.78	0	metres wide	$ 6	$ -	m					$ -	$ -
as per draft tree strategy												$ -	
root barriers	optional as per design in respect of appropriate offset criteria												
300mm dia conc drain Stormwater Cr BF	per metre	$182	75	metres		$ 13,650.00						$ 13,650	
375mm conc drain stormwater Cr Bk fill	per metre	$231	75	metres		$ 17,325.00						$ 17,325	
450mm conc drain stormwater Cr BF	per metre	$294	0	metres		$ -						$ -	
525mm conc drain stormwater Cr BF	per metre	$351	0	metres		$ -						$ -	
Service Road			0	Lmetres									$ -
Traffic Signals	Unit	$ 102,500.00	0	No of intesection								$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	0	metres								$ -	$ -
Street Lighting	m	$ 129.00	145	rows of lights		$ 18,705.00	m					$ 18,705.00	
lighting conduit	m	$ 50.00	145	No of runs/sides		$ 7,250.00	m					$ 7,250.00	
Subtotal						$ 293,790.50						$ 293,790.50	
estimated total													$ 293,791
Traffic Management									10.0%				$ 29,379
Contingency									20.0%				$ 58,758
Total + contingencies												$ 381,928	
Services relocation Sec Poles, water fitting	Item												$ 250,000
Survey and Design									10%				$ 29,379
Overheads (supervision etc)									15%				$ 44,069
Total excluding land cost													**$ 705,375**
CPI Adjustment to Nov. 2012									2.2%				$ 15,518
Total Estimated Cost													**$ 720,894**
Adopted Cost													

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Estimate Prepared by: CDCE Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.
PROPOSED KERB AND CHANNEL	POWER OVERHEAD	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN	POWER PIT	HOUSE DRAIN	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
COORD. SYS.: N/A
SCALE NOT TO SCALE
F.B. L.B.
JOB NO. 10 111
SURVEY N/A
DWG. FILE NAME XXXXXX.DWG
DESIGN SAF
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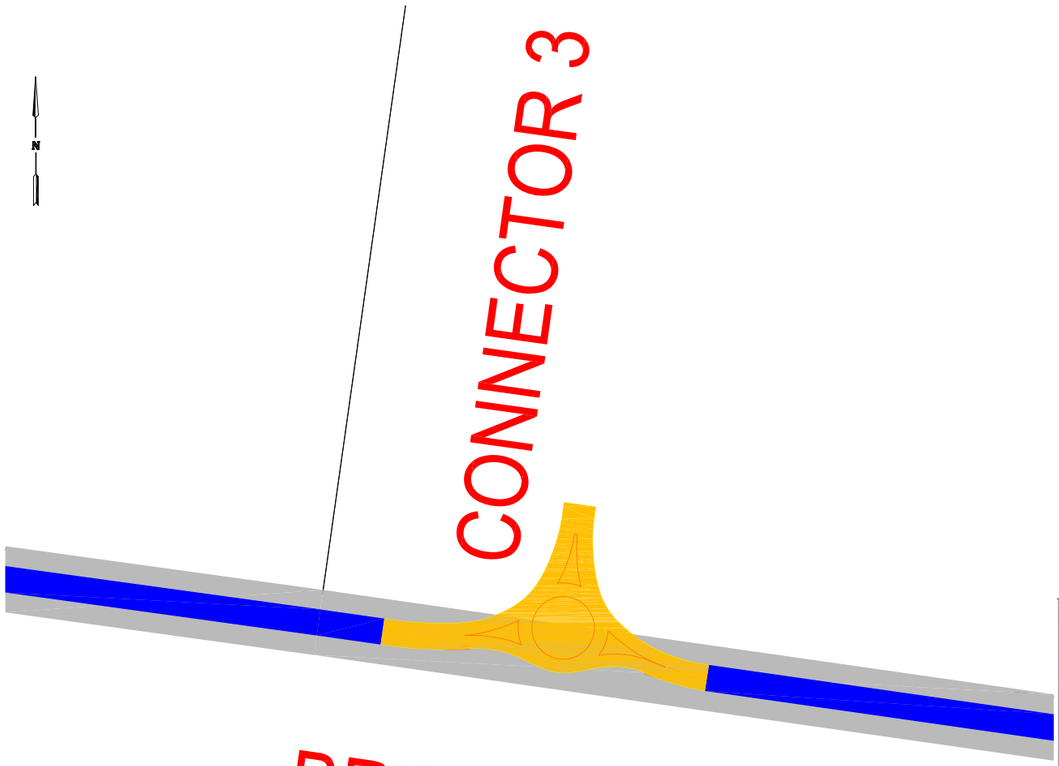
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BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
BROWNS RD & CONNECTOR 4
IN–06 INTERSECTION

DATE: November 2012
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BOTANIC RIDGE DCP

IN-07 INTERSECTION COSTS
BROWNS RD & CONNECTOR 3



SCHEMATIC LAYOUT FOR COSTING PURPOSES ONLY

EXISTING ROAD RESERVE

COSTED INTERSECTION AREA

NOTES AND ASSUMPTIONS FOR IN-07 BROWNS & C3
1: Traffic Management considered to be high in complexity
2: Survey & Design is considered to be medium complexity due to roundabout intersection proposal
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) intersection has one sided encroachment on existing services
b) connection road is greenfield
c) length of proposed works

Botanic Ridge DCP Roadworks Estimate

Road Name: IN-07 Browns Rd & Connector C3
Limit of works: As shown on drawing road reserve and cross section vary
Length of Job: Intersection works only 20m dia central island
Notes: Costs based on City of Casey specification for road construction & GAA schematic intersection

Item	Unit	Rate	wide m	units		cost/m width		Unit cost	Quantity metres		subset cost	subtotal	Amount
Roadworks	subtotal							$ 293,790.50	1	m		$ 293,790.50	$ 293,791
Bulk Earthworks	m²	$ 40.00	885	metres	$ 40	$ 35,400.00	m					$ 35,400.00	$ -
Pavement (urban)	m²	$ 91.50	1355	metres	$ 92	$ 123,982.50	m					$ 123,982.50	$ -
Crushed Rock Shoulder	m²	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Pavement Removal	m²	$ 10.00	0	metres		$ -						$ -	
Drainage Culvert 3No. 1.2x0.6	m	$ 660.00	0	No.	1							$ -	$ -
kerb and Channel S 503	m	$ 40.00	201	sides	$ 40	$ 8,040.00	m					$ 8,040.00	$ -
Kerb and Channel S 504		$ 45.00	259	sides	$ 45	$ 11,655.00	m					$ 11,655.00	
side entry pits std drw S305	unit	$ 2,000.00	7	interval metres	90	$ 14,000.00	unit					$ 14,000.00	
Bike path 2.5m wide conc	m²	$ 60.00	362	metres		$ 21,720.00						$ 21,720.00	
Shared pathway 2.5m wide gravel	m²	$ 30.00	0	metres		$ -						$ -	
Pedestrian Footpath 1.5m wide	m²	$ 60.00	0	metres		$ -						$ -	
Pedestrian Footpath 1.5m gravel	m²	$ 30.00	0	metres		$ -						$ -	
Drainage , subgrade drain	m	$ 14.70	460		$ 15	$ 6,762.00	m					$ 6,762.00	$ -
Linemarking & Signage	m	$ 25.00	46	metres	$ 25	$ 1,150.00	m					$ 1,150.00	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Concrete Island infill	m²	$ 45.00	140		$ 45	$ 6,300.00	m					$ 6,300.00	$ -
level / trim nature strip	m²	$ 4.00	1869	metres	$ 4	$ 7,476.00	m					$ 7,476.00	$ -
as per standard drawings													
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tree	15	rows of trees	$ 25	$ 375.00	m					$ 375.00	
Tube Stock Plantings	unit	$5.27 -$6.78	0	metres wide	$ 6	$ -	m					$ -	$ -
as per draft tree strategy												$ -	
root barriers	optional as per design in respect of appropriate offset criteria												
300mm dia conc drain Stormwater Cr BF	per metre	$182	75	metres		$ 13,650.00						$ 13,650	
375mm conc drain stormwater Cr Bk fill	per metre	$231	75	metres		$ 17,325.00						$ 17,325	
450mm conc drain stormwater Cr BF	per metre	$294	0	metres		$ -						$ -	
525mm conc drain stormwater Cr BF	per metre	$351	0	metres		$ -						$ -	
Service Road			0	Lmetres									$ -
Traffic Signals	Unit	$ 102,500.00	0	No of intesection								$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	0	metres								$ -	$ -
Street Lighting	m	$ 129.00	145	rows of lights		$ 18,705.00	m					$ 18,705.00	
lighting conduit	m	$ 50.00	145	No of runs/sides		$ 7,250.00	m					$ 7,250.00	
Subtotal						$ 293,790.50						$ 293,790.50	
estimated total													$ 293,791
Traffic Management										10.0%			$ 29,379
Contingency										20.0%			$ 58,758
Total + contingencies												$ 381,928	
Services relocation Sec Poles, water fitting	item												$ 250,000
Survey and Design										10%			$ 29,379
Overheads (supervision etc)										15%			$ 44,069
Total excluding land cost													**$ 705,375**
CPI Adjustment to Nov. 2012										2.2%			$ 15,518
Total Estimated Cost													**$ 720,894**
Adopted Cost													

Civil Design Consulting Engineers
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Estimate Prepared by: CDCE Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN H	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

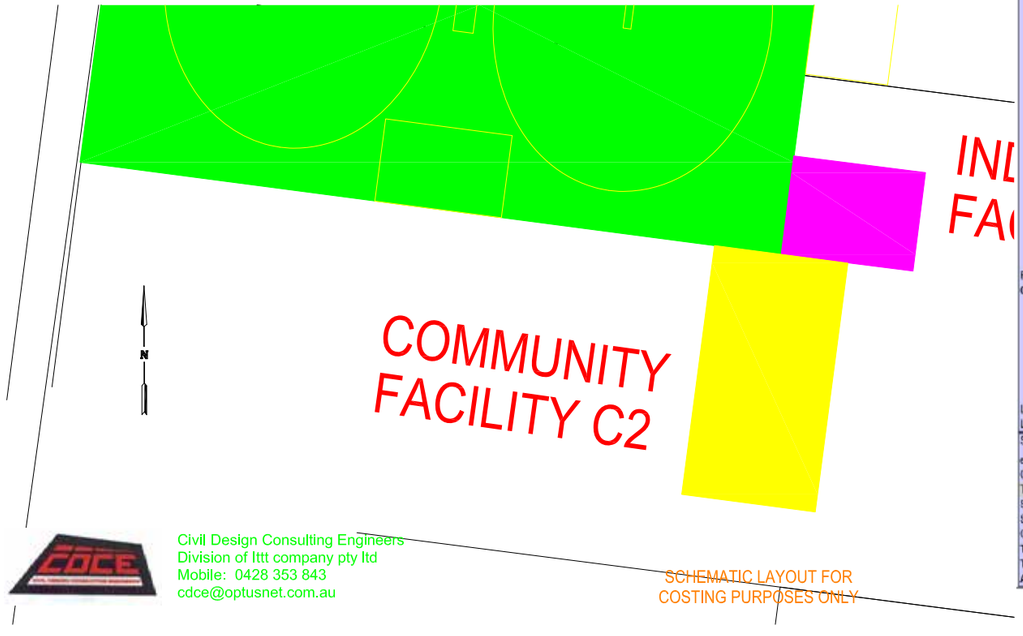
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JOB NO. 10 111
SURVEY N/A | DWG. FILE NAME XXXXXX.DWG
DESIGN SAF | REG. FILE NO. XXXXXXX
TRACED SAF | MELWAY REF.
CHECKED SAF | CONTRACT NO.

CIVIL DESIGN CONSULTING ENGINEERS

BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
BROWNS RD & CONNECTOR 3
IN-07 INTERSECTION

DATE: November 2012
REVISION
10 111
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BOTANIC RIDGE DCP
INTEGRATED COMMUNITY CENTRE
C2



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SCHEMATIC LAYOUT FOR
COSTING PURPOSES ONLY

Community Centre 2

Name: C2 Community Centre West Site 0.8ha
Scope of works: Integrated Community Centre
Detail Quad Kindergarten, Maternal and child health care, consulting suites, community meeting facilities
Notes: Costs based on estimate from previous community centres
Estimate based on normal earthworks on fairly level site

Item	Unit	Rate	area	comments	No.	cost	Unit cost	Quantity		subset	subtotal	Amount
								Number		cost		
Playing Fields												
Football/Cricket:												
Soccer:												
Netball:												
Tennis:												
Community Centre												
Building	subtotal		1597.4	square metres			$ 3,753,369.80	1				$ 3,753,370
Kindergarten	m²	$ 2,377.00	290.4	4 rooms	1	$ 690,280.80					$ 690,280.80	
Main Office Admin. Room	m²	$ 2,280.00	45		1	$ 102,600.00					$ 102,600.00	
Staff rm, amenties	m²	$ 2,280.00	45		1	$ 102,600.00					$ 102,600.00	
Storage & Laundry	m²	$ 2,000.00	8		1	$ 16,000.00					$ 16,000.00	
Kitchens	m²	$ 2,377.00	24	2 No.	1	$ 57,048.00					$ 57,048.00	
Childrens amenities	m²	$ 2,377.00	48	4 No.	1	$ 114,096.00					$ 114,096.00	
Internal Storage rooms	m²	$ 2,000.00	64	4 No.	1	$ 128,000.00					$ 128,000.00	
Foyer & Circulation space	m²	$ 2,280.00	150		1	$ 342,000.00					$ 342,000.00	
External Storage Space	m²	$ 500.00	80	4 No.	1	$ 40,000.00					$ 40,000.00	
Playground	m²	$ 400.00	616	2 No.	1	$ 246,400.00					$ 246,400.00	
MCH & consulting rooms	m²	$ 2,280.00	40	2 No.	1	$ 91,200.00					$ 91,200.00	
Waiting & circulation	m²	$ 2,280.00	45		1	$ 102,600.00					$ 102,600.00	
Consulting rooms, offices	m²	$ 2,280.00	48		1	$ 109,440.00					$ 109,440.00	
Community & Learning	m²				1							
Foyer & Circulation space	m²	$ 2,280.00	150	4 rooms	1	$ 342,000.00					$ 342,000.00	
Main Office	m²	$ 2,280.00	35		1	$ 79,800.00					$ 79,800.00	
Large meeting space	m²	$ 2,280.00	250		1	$ 570,000.00					$ 570,000.00	
Store rooms	m²	$ 2,000.00	50		1	$ 100,000.00					$ 100,000.00	
Large kitchen	m²	$ 2,377.00	35		1	$ 83,195.00					$ 83,195.00	
Amenities	m²	$ 2,377.00	30		1	$ 71,310.00					$ 71,310.00	
Class room 4No.	m²	$ 2,280.00	160		1	$ 364,800.00					$ 364,800.00	
Bushfire premium for BAL 19			4.8%			$ 168,334.55					$ 168,334.55	$ 168,335
Carpark works	subtotal		per	100	car spaces		$ 157,775.00	0.7	70	Car Spaces		$ 110,443
Bulk Earthworks	m³	$ 35.00	700	metres		$24,500					$ 24,500.00	
Crushed Rock (Gravel)	m²	$ 32.00	2000	metres		$64,000					$ 64,000.00	
Timber edging	Lm	$ 15.00	500			$7,500					$ 7,500.00	
Side entry pits std drw S305	unit	$ 1,200.00	2	internal metres		$2,400					$ 2,400.00	
Drainage , subgrade drain	m	$ 14.70	250			$3,675					$ 3,675.00	
Granitic path	m²	$ 30.00	1250		1	$37,500					$ 37,500.00	
300mm dia conc drain Cr BF	per metre	$ 182.00	100	metres		$18,200					$ 18,200.00	
Landscaping Level B	m²	$ 30.00	4500	Level B								$ 135,000
Landscaping Level A	m²	$ 55.00	270	Level A								$ 14,850
Subtotal						$ 4,079,479					$ 4,079,479.35	
estimated total												$ 4,181,997
Contingence									20.0%			$ 836,399
Total + contingencies											$ 5,018,396	
Services for buildings	Item											$ 150,000
Survey and Design									5%			$ 209,100
Overheads (supervision etc)									10%			$ 418,200
Total excluding land cost												**$ 5,795,696**
Total Estimatec Cost												**$ 5,795,696**
Adopted Cost												

Estimate Prepared by: CDCE Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/P NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
COORD. SYS.: N/A
SURVEY N/A
DESIGN SAF
TRACED SAF
CHECKED SAF

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JOB NO. 10 111
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REG. FILE NO. XXXXXXX
MGA94 REF.
CONTRACT NO.

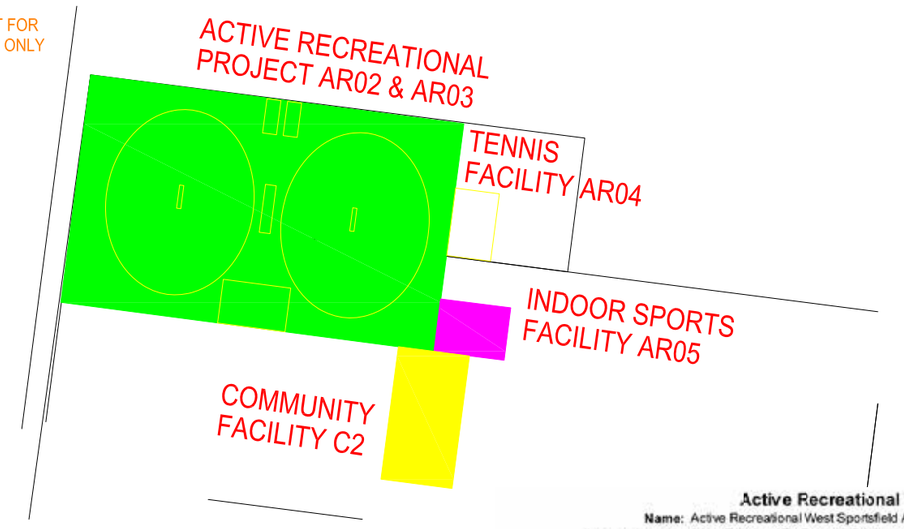
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BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
INTEGRATED COMMUNITY CENTRE C2

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SCHEMATIC LAYOUT FOR COSTING PURPOSES ONLY



BOTANIC RIDGE DCP

ACTIVE RECREATIONAL PROJECT AR02 & AR03

Active Recreation West Pavilion AR03 Costing

Name: Pavilion AR03 for Active Recreation West Sports Field
Scope of works: Pavilion (combined) 720m2, water tanks
Detail Pavilion for Football & Cricket ovals and netball
Notes: Costs based on estimate from previous playing fields
Estimate based on normal earthworks on fairly level site after earthworks surplus

Item	Unit	Rate	area	comments	No.	cost	Unit cost	Quantity		subset	subtotal	Amount
								Number		cost		
Playing Fields												
Football/Cricket:												
Pavilion			**720**	**square meters**			**$ 1,419,318.00**	1				**$ 1,419,318**
Function Room	m²	$ 2,377.00	120	multi function room	1	$ 285,240.00					$ 285,240.00	
Change Rooms	m²	$ 2,377.00	204	57m home 45m away	1	$ 484,908.00					$ 484,908.00	
Toilets & showers	m²	$ 2,377.00	92	1 male + 1 female	1	$ 218,684.00					$ 218,684.00	
Public Toilets	m²	$ 2,377.00	25	10m M + 10m F +5m	1	$ 59,425.00					$ 59,425.00	
Kitchen/Kiosk/bar	m²	$ 2,377.00	28		1	$ 66,556.00					$ 66,556.00	
External covered viewing area	m²	$ 1,500.00	100		1	$ 150,000.00					$ 150,000.00	
Storage	m²	$ 2,377.00	30		1	$ 71,310.00					$ 71,310.00	
Ancillary rooms: Referees, scorers, Office, Cleaners	m²	$ 2,377.00	35		1	$ 83,195.00					$ 83,195.00	
Soccer:												
Netball:												
Courts:	**subtotal**						**$ 204,422.00**	1				**$ 204,422**
Pavilion	unit											
Change Rooms	m²	$ 2,377.00	40	2 rooms	1	$ 95,080.00					$ 95,080.00	
Toilets & showers	m²	$ 2,377.00	36	2 rooms	1	$ 85,572.00					$ 85,572.00	
Storage	m²	$ 2,377.00	10		1	$ 23,770.00					$ 23,770.00	
Tennis:												
Bushfire premium for BAL 19			4.8%			$ 77,939.52					$ 77,939.52	$ 77,940
Earthworks surplus for unlevel site	m3	$ 30.00	360	deep or cut to fill	3	$ 32,400.00					$ 32,400.00	$ 32,400
Water tanks for pavilion	unit	$ 20,000.00			2							**$ 40,000**
Subtotal						$ 1,734,080					$ 1,734,079.52	
estimated total												$ 1,774,080
Traffic Management									0.5%			$ 8,870
Contingency									20.0%			$ 354,816
Total + contingencies											$ 2,137,766	
Services to pavilion	Item											$ 75,000
Survey and Design									10%			$ 177,408
Overheads (supervision etc)									10%			$ 177,408
Total excluding land cost												**$ 2,567,582**
Total Estimated Cost												**$ 2,567,582**
Adopted Cost												

Estimate Prepared by: CDCE Nov-12

Active Recreational West Sportsfield AR02 Costing

Name: Active Recreational West Sportsfield AR02 - 10.1ha site
Scope of works: 2 No. Full size AFL Football/Cricket ovals, lighting: 1No. for night play & 1No. practise, 3 practice cricket nets, 2 netball courts, lighting 1No. night play, 4 tennis courts, Gravel carpark (100 cars), & paths etc
Detail Playing Fields only with associated equipment and carpark
Notes: Costs based on estimate from previous playing fields site
Estimate allowed for 50% playing fields surface to have 3m cut to fill and then new material ontop of this level.

Item	Unit	Rate	area	comments	No.	cost	Unit cost	Quantity		subset	subtotal	Amount
								Number		cost		
Playing Fields												
Football/Cricket:												
Oval	**subtotal**			**Senior District Level**			**$ 972,500.00**	2				**$ 1,945,000**
Bulk Earthworks	m³	$ 30.00	14250	cut to fill	1	$ 427,500.00					$ 427,500.00	
Sandy loam	m³	$ 20.00	7600	0.4 metres deep	1	$ 152,000.00					$ 152,000.00	
Top soil & seeding	m²	$ 40.00	3800	0.2 metres deep	1	$ 152,000.00					$ 152,000.00	
irrigation	unit	$ 35,000.00			1	$ 35,000.00					$ 35,000.00	
fences oval perimeter	unit	$ 40.00		1.2m highfence	300	$ 12,000.00					$ 12,000.00	
fences perimeter	unit	$ 40.00		1m timber post Gal pipe	900	$ 36,000.00					$ 36,000.00	
goals	unit	$ 2,000.00		per post	8	$ 16,000.00					$ 16,000.00	
Cricket pitch	unit	$ 25,000.00			1	$ 25,000.00					$ 25,000.00	
Practise Cricket pitches & nets	unit	$ 28,000.00			1.5	$ 42,000.00					$ 42,000.00	
Lighting night play	unit	$ 25,000.00		4 pole play 2 practise	3	$ 75,000.00					$ 75,000.00	
Soccer:												
Netball:												
Courts:	**subtotal**						**$ 67,500.00**	2				**$ 135,290**
Bulk Earthworks	m³	$ 40.00	80	metres	1	$ 3,200.00					$ 3,200.00	
Sub-Surface	m³	$ 32.00	200	4 layers 100mm thick	1	$ 6,400.00					$ 6,400.00	
Surface	m²	$ 60.00	200	metres	1	$ 12,000.00					$ 12,000.00	
Drainage	unit	$ 25,000.00			1	$ 25,000.00					$ 25,000.00	
Rings, No fencing, lines etc	unit	$ 1,000.00			1	$ 1,000.00					$ 1,000.00	
Lighting night play 1 crt only	unit	$ 10,000.00		4 poles all 1 court	2	$ 20,000.00					$ 20,000.00	
Pavilion	unit											
Tennis:												
Courts:	**subtotal**						**$ 55,000.00**	4				**$ 220,000**
Carpark works	**subtotal**			**100 cars Crushed Rock**			**$247,025.60**	1				**$ 247,025**
Bulk Earthworks	m³	$ 35.00	1750	metres		$61,250					$ 61,250.00	
Crushed Rock (Gravel)	m²	$ 32.00	2000	metres		$64,000					$ 64,000.00	
Timber edging	Lm	$ 15.00	500			$7,500					$ 7,500.00	
Side entry pits std drw S305	unit	$ 1,200.00	2	interval metres		$2,400					$ 2,400.00	
Drainage , subgrade drain	m	$ 14.70	250			$3,675					$ 3,675.00	
Granitic path	m²	$ 30.00	3000	1.22km of 2.5m wide		$90,000					$ 90,000.00	
300mm dia conc drain Cr BF	per metre	$ 182.00	100	metres		$18,200					$ 18,200.00	
interchange shelters	unit	$ 5,000.00			4	$20,000					$ 20,000.00	$ 20,000
Landscaping Level B	m²	$ 35.00	21930	Level B								**$ 767,050**
Landscaping Level A	m²	$ 55.00	5980	Level A								**$ 328,980**
Subtotal											$ 1,307,125.00	
estimated total												$ 3,653,176
Traffic Management									0.5%			$ 18,266
Contingency									20.0%			$ 730,635
Total + contingencies											$ 4,402,076	
Services lighting, irrigation	Item											$ 100,000
Survey and Design									5%			$ 182,659
Overheads (supervision etc)									10%			$ 365,318
Total excluding land cost												**$ 5,050,052**
Total Estimated Cost												**$ 5,050,052**
Adopted Cost												

Estimate Prepared by: CDCE Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO. (25)
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
SCALE NOT TO SCALE
COORD. SYS.: N/A
F.B. L.B.
JOB NO. 10 111
SURVEY N/A
DWG. FILE NAME XXXXXX.DWG
DESIGN SAF
REG. FILE NO. XXXXXX
TRACED SAF
MELWAY REF.
CHECKED SAF
CONTRACT NO.

CIVIL DESIGN CONSULTING ENGINEERS

BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
ACTIVE RECREATIONAL PROJECT AR02, AR03

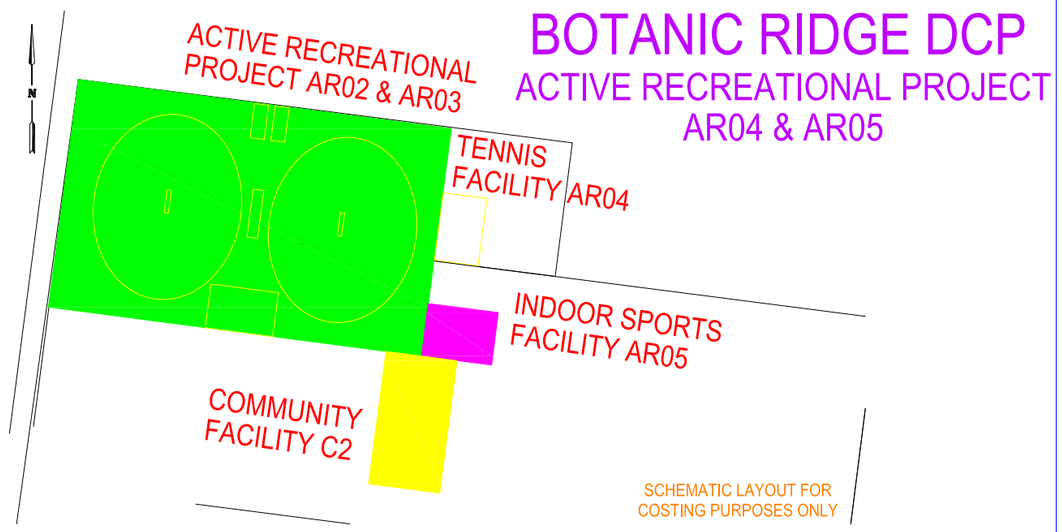
DATE: November 2012
REVISION
10 111
SHEET 1 OF 1

Indoor Sports Facility AR05 Costing

Name: AR05 Indoor Sports Facility
Scope of works: Indoor Sports Facility for 2 lower order basket ball courts
Detail Indoor Sports Facility
Notes: Costs based on estimate for Separate Indoor Sports Facility.
Estimate based on normal earthworks on fairly level site after earthworks surplus

Item	Unit	Rate	area	comments	No.	cost	Unit cost	Quantity		subset	subtotal	Amount
								Number		cost		
Playing Fields												
Football/Cricket:												
Soccer:												
Netball:												
Pavilion	unit											
Tennis:					0	$ -						
Indoor sport Facility	m²		1295	square metres			$ 1,752,307.06	1				$ 1,752,307
Basket Ball Courts	m²	$ 1,200.00	640		2	$ 1,536,000					$ 1,536,000.00	
Change Rooms	m²	$ 2,377.00	20	2 rooms	2	$ 95,080.00					$ 95,080.00	
Toilets & showers	m²	$ 2,377.00	18	2 rooms	2	$ 85,572.00					$ 85,572.00	
Ancillary rooms: Referees, scorers, Office, Cleaners	m²	$ 2,377.00	15		1	$ 35,655					$ 35,655.00	
Bushfire premium for BAL 19			4.8%			$ 84,110.74					$ 84,110.74	$ 84,111
Earthworks surplus for unlevel site	m3	$ 30.00	647.5	deep or cut to fill	3	$ 58,275.00					$ 58,275.00	$ 58,275
Water tanks for pavilion	unit	$ 20,000.00			2							$ 40,000
Subtotal						$ 1,894,693					$ 1,894,692.74	
estimated total												$ 1,934,693
Traffic Management									2.0%			$ 38,694
Contingency									20.0%			$ 386,939
Total + contingencies											$ 2,360,326	
Services to pavilion	Item											$ 75,000
Survey and Design									9%			$ 174,122
Overheads (supervision etc)									10%			$ 193,469
Total excluding land cost												$ 2,802,917
Total Estimated Cost												$ 2,802,917
Adopted Cost												

Estimate Prepared by: CDCE Nov-12



Active Recreation West Tennis Pavilion AR04 Costing

Name: Tennis Pavilion AR04 for Active Recreation West Sports Field
Scope of works: Pavilion (combined) 204m2, water tanks
Detail Pavilion for tennis only
Notes: Costs based on estimate from previous playing fields
Estimate based on normal earthworks on fairly level site after earthworks surplus

Item	Unit	Rate	area	comments	No.	cost	Unit cost	Quantity		subset	subtotal	Amount
								Number		cost		
Playing Fields												
Football/Cricket:												
Soccer:												
Netball:												
Pavilion	unit											
Tennis:												
Pavilion			196	square meters			$ 430,812.00	1				$ 430,812
Change Rooms	m²	$ 2,377.00	20	2 rooms	1	$ 47,540.00					$ 47,540.00	
Toilets & showers	m²	$ 2,377.00	16	2 rooms	1	$ 38,032.00					$ 38,032.00	
Public Toilets	m²	$ 2,377.00	8	accessible toilet & shc	1	$ 19,016.00					$ 19,016.00	
Community Space	m²	$ 2,377.00	60		1	$ 142,620.00					$ 142,620.00	
Kitchen/Kiosk	m²	$ 2,377.00	20		1	$ 47,540.00					$ 47,540.00	
External covered viewing area	m²	$ 1,500.00	40		1	$ 60,000.00					$ 60,000.00	
Storage	m²	$ 2,377.00	15		1	$ 35,655.00					$ 35,655.00	
Ancillary rooms: Referees, scorers, Office, Cleaners	m²	$ 2,377.00	17		1	$ 40,409.00					$ 40,409.00	
Bushfire premium for BAL 19			4.8%			$ 20,678.98					$ 20,678.98	$ 20,679
Earthworks surplus for unlevel site	m3	$ 30.00	204	deep or cut to fill	3	$ 18,360.00					$ 18,360.00	$ 18,360
Water tanks for pavilion	unit	$ 20,000.00			1							$ 20,000
Subtotal						$ 469,851					$ 469,850.98	
estimated total												$ 489,851
Traffic Management									0.5%			$ 2,449
Contingency									20.0%			$ 97,970
Total + contingencies											$ 590,270	
Services to pavilion	Item											$ 50,000
Survey and Design									10%			$ 48,985
Overheads (supervision etc)									10%			$ 48,985
Total excluding land cost												$ 738,241
Total Estimated Cost												$ 738,241
Adopted Cost												

Estimate Prepared by: CDCE Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN H	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
COORD. SYS.: N/A
F.B. L.B.
SURVEY N/A
DESIGN SAF
TRACED SAF
CHECKED SAF

SCALE NOT TO SCALE
JOB NO. 10 111
DWG. FILE NAME XXXXXX.DWG
REG. FILE NO. XXXXXX
MGA/AHD REF.
CONTRACT NO.

CIVIL DESIGN CONSULTING ENGINEERS

BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
ACTIVE RECREATIONAL PROJECT AR04 & AR05

DATE: November 2012
REVISION
10 111
SHEET 1 OF 1

BOTANIC RIDGE DCP

ACTIVE RECREATIONAL PROJECT AR07 & AR08

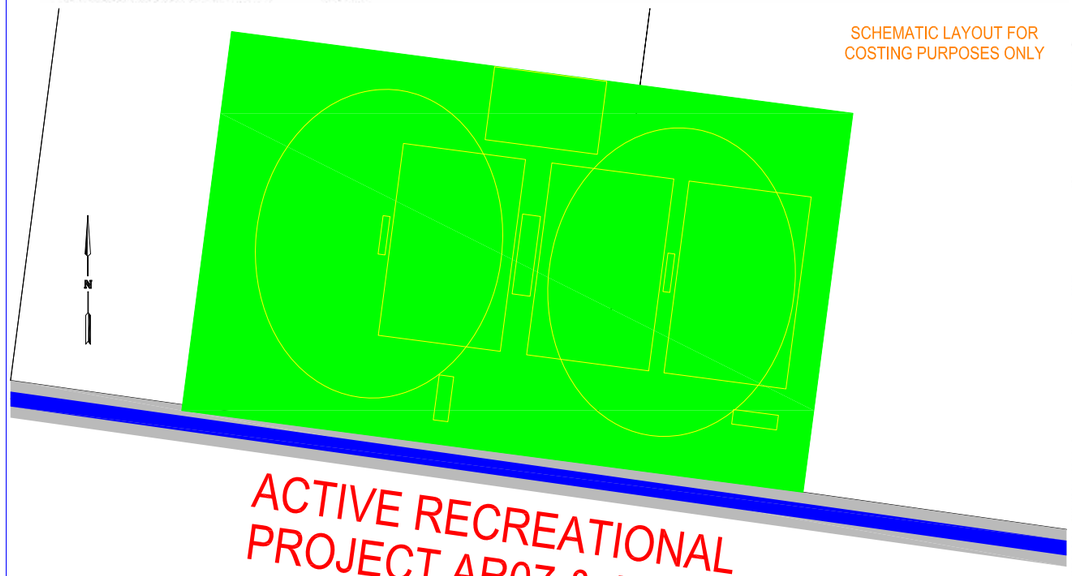
Active Recreation East Pavilion AR08 Costing

Name: Pavilion AR08 for Active Recreation East Sports Field
Scope of works: Pavilion (combined) 654m2, water tanks
Detail Pavilion for, 2 no. Cricket & 3no. Soccer
Notes: Costs based on estimate from previous playing fields
Estimate based on normal earthworks on fairly level site

Item	Unit	Rate	area	comments	No.	cost	Unit cost	Quantity		subset	subtotal	Amount
								Number		cost		
Playing Fields												
Football/Cricket:												
Pavilion			**654**	**square meters**			**$ 1,486,938.00**	**1**				**$ 1,486,938**
Function Room	m²	$ 2,377.00	100	multi function room	1	$ 237,700.00					$ 237,700.00	
Change Rooms	m²	$ 2,377.00	204	57m home 45m away	1	$ 484,908.00					$ 484,908.00	
Toilets & showes	m²	$ 2,377.00	110	1 male + 1 female	1	$ 261,470.00					$ 261,470.00	
Public Toilets	m²	$ 2,377.00	45	10m M + 10m F +5m	1	$ 106,965.00					$ 106,965.00	
Kitchen/Kiosk/bar	m²	$ 2,377.00	28		1	$ 66,556.00					$ 66,556.00	
External covered vewing area	m²	$ 1,500.00	50		1	$ 75,000.00					$ 75,000.00	
Storage	m²	$ 2,377.00	60		1	$ 142,620.00					$ 142,620.00	
Ancillary rooms: Referees, scorers, Office, Cleanes	m²	$ 2,377.00	47		1	$ 111,719.00					$ 111,719.00	
Soccer:												
Netball:												
Pavilion	unit											
Tennis:												
Water tanks for pavilion	unit	$ 20,000.00			2							**$ 40,000**
Subtotal						$ 1,486,938					$ 1,486,938.00	
estimated total												$ 1,526,938
Traffic Management									0.5%			$ 7,635
Contingency									20.0%			$ 305,388
Total + contingencies											$ 1,839,960	
Services to pavilion	item											$ 75,000
Survey and Design									10%			$ 152,694
Overheads (supervision etc)									10%			$ 152,694
Total excluding land cost												**$ 2,220,348**
Total Estimated Cost												**$ 2,220,348**
Adopted Cost												

Estimate Prepared by: **CDCE** Nov-12

SCHEMATIC LAYOUT FOR COSTING PURPOSES ONLY



ACTIVE RECREATIONAL PROJECT AR07 & AR08

Civil Design Consulting Engineers
Division of Lttt company pty ltd
Mobile: 0428 353 843
cdce@optusnet.com.au

Active Recreational East Sportsfield AR07 Costing

Name: Active Recreational East Sportsfield AR07 - 7.7ha site
Scope of works: 2 No. Full size Cricket ovals/3 soccer Pitches, lighting: 1No. for night play & 1No. practise, 3 practice cricket nets.
Gravel carpark (75 cars), & paths etc
Detail Playing Fields only withassociated equipment and carpark
Notes: Costs based on estimate from previous playing fields site
Estimate based on using balanced cut to fill earthworks

Item	Unit	Rate	area	comments	No.	cost	Unit cost	Quantity		subset	subtotal	Amount
								Number		cost		
Playing Fields												
Football/Cricket:												
Oval:	**subtotal**			**Senior District Level**			**$ 115,000.00**	**2**				**$ 230,000**
fences oval perimeter	unit	$ 40.00		1.2m highfence	300	$ 12,000.00					$ 12,000.00	
fences perimeter	unit	$ 40.00		1m timber post Gal pipe	900	$ 36,000.00					$ 36,000.00	
Cricket pitch	unit	$ 25,000.00			1	$ 25,000.00					$ 25,000.00	
Practise Cricket pitches & nets	unit	$ 28,000.00			1.5	$ 42,000.00					$ 42,000.00	
Soccer:												
Pitchs	**subtotal**						**$ 895,000.00**	**3**				**$ 2,685,000**
Bulk Earthworks	m³	$ 40.00	11400	0.6 metres deep	1	$ 456,000.00					$ 456,000.00	
Sandy loam	m³	$ 20.00	7600	0.4 metres deep	1	$ 152,000.00					$ 152,000.00	
Top soil & seeding	m²	$ 40.00	3800	0.2 metres deep	1	$ 152,000.00					$ 152,000.00	
irrigation	unit	$ 75,000.00			1	$ 75,000.00					$ 75,000.00	
fences goals etc	unit	$ 35,000.00			1	$ 35,000.00					$ 35,000.00	
Lighting night play	unit	$ 50,000.00		rows of lights	0.5	$ 25,000.00					$ 25,000.00	
Netball:												
Pavilion	unit											
Tennis:												
Carpark works	**subtotal**			**100 cars Crushed Rock**			**$219,275.00**	**0.75**				**$ 157,706**
Bulk Earthworks	m³	$ 35.00	700	metres		$24,500					$ 24,500.00	
Crushed Rock (Gravel)	m³	$ 32.00	2000	metres		$64,000					$ 64,000.00	
Timber edging	Lm	$ 15.00	500			$7,500					$ 7,500.00	
Side entry pits std drw S305	unit	$ 1,200.00	2	interval metres		$2,400					$ 2,400.00	
Drainage , subgrade drain	m	$ 14.70	250			$3,675					$ 3,675.00	
Granitic path	m²	$ 30.00	3000	1.20km of 2.5m wide		$90,000					$ 90,000.00	
300mm dia conc drain Cr BF	per metre	$ 182.00	100	metres		$18,200					$ 18,200.00	
Landscaping Level B	m²	$ 35.00	19850	Level B								**$ 694,750**
Landscaping Level A	m²	$ 55.00	3240	Level A								**$ 178,200**
Subtotal						$ 1,240,275					$ 1,240,275.00	
estimated total												$ 3,965,656
Traffic Management									0.5%			$ 19,828
Contingency									20.0%			$ 793,131
Total + contingencies											$ 4,778,616	
Services lighting, irrigation	item											$ 100,000
Survey and Design									5%			$ 198,283
Overheads (supervisior etc)									10%			$ 396,566
Total excluding land cost												**$ 5,473,464**
Total Estimated Cost												**$ 5,473,464**
Adopted Cost												

Estimate Prepared by: **CDCE** Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN —G—	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN —W—	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G —T—	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G —E—	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.
PROPOSED KERB AND CHANNEL	POWER OVERHEAD —E—	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN —S—	POWER PIT	HOUSE DRAIN —H—	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
COORD. SYS.: N/A
F.B. L.B.
SURVEY N/A
DESIGN SAF
TRACED SAF
CHECKED SAF

SCALE NOT TO SCALE
JOB NO. 10 111
DWG. FILE NAME XXXXXX.DWG
REG. FILE NO. XXXXXXX
MELWAY REF.
CONTRACT NO.

CIVIL DESIGN CONSULTING ENGINEERS

BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
ACTIVE RECREATIONAL PROJECT AR07 & AR08

DATE: November 2012
REVISION
10 111
SHEET 1 OF 1

6.0 APPENDIX A: CONTINGENCY PROJECTS

6.1 ADMINISTRATION OF CONTINGENCY PROJECTS

If the Commonwealth Government Minister administering the EPBC Act makes a determination requiring the setting aside of land within the Botanic Ridge precinct for SBB habitat, the following contingency projects as described and costed in Table 9 and shown on Plan 7 will be required to be delivered via the Botanic Ridge DCP. This will be in addition to projects already outlined in the body of this DCP.

If the Commonwealth Government Minister administering the EPBC Act makes a determination that **does not** require the setting aside of land within the Botanic Ridge precinct for SBB habitat, the following contingency projects described and costed in Table 9, and shown on Plan 7, will be deemed to be redundant and no longer applicable to the Botanic Ridge DCP.

6.2 CONTINGENCY PROJECT CONSTRUCTION COST SHEETS

This section includes detailed cost estimate sheets for the construction of following Contingency Projects (if required)

CP01 - Connector Road C5 (portion that would cross SBB Corridor)

CP02 - Connector Road C1 (portion that would cross SBB Corridor)

CP03 – East SBB Corridor – Shared Path Crossing

CP04a & 4b – South SBB Corridor – Shared Path Crossings

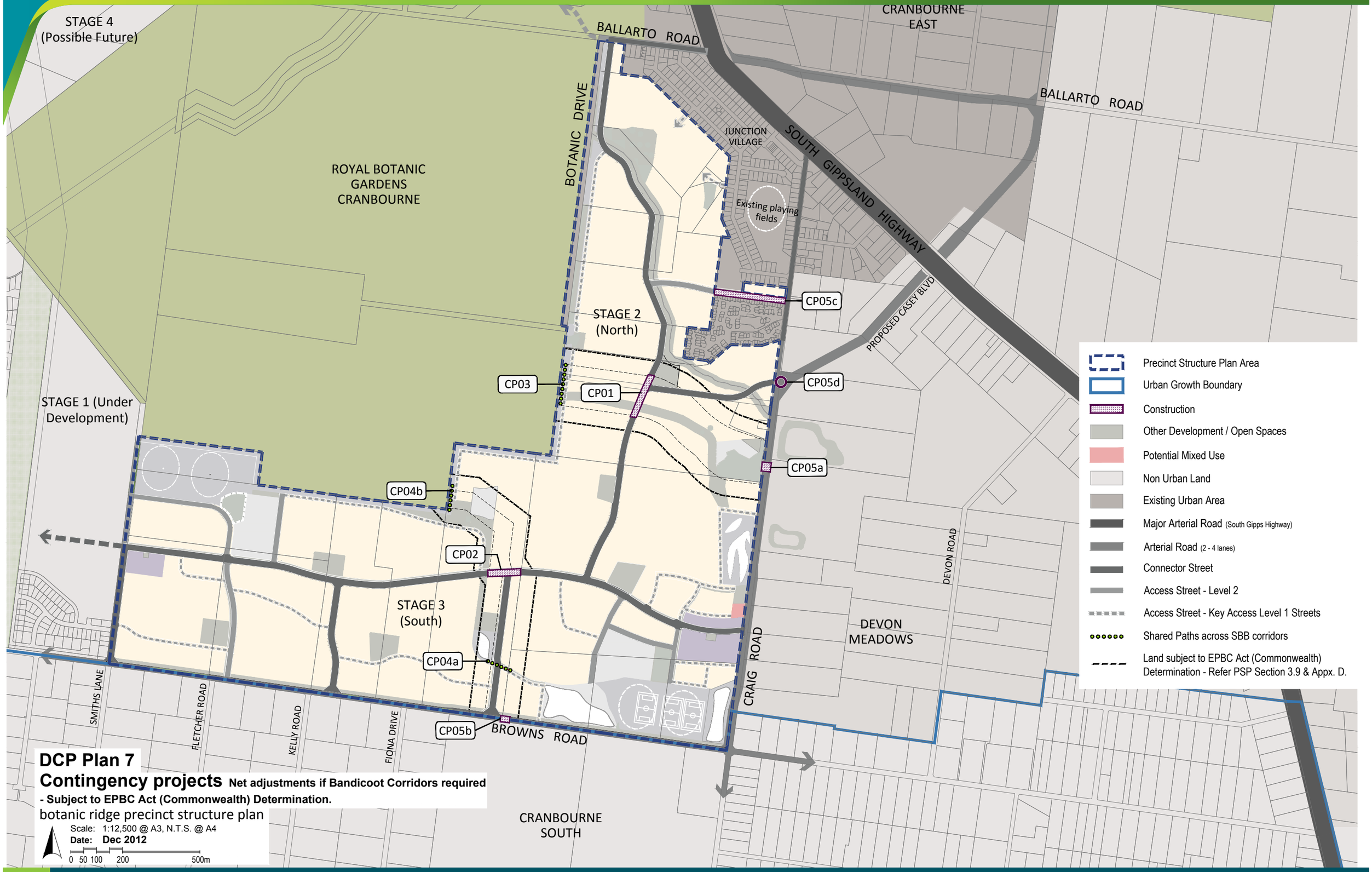
CP05a – Addition of Culverts for SBB Corridor Crossing to RD04

CP05b – Addition of Culverts for SBB Corridor Crossing to RD05

CP05c – Upgrading of RD08 to Constrained Connector Road Status

CP05d – Deletion of West Leg of IN03 Intersection

All Contingency construction projects are subject to detailed design and approval of the responsible authority.



DCP Plan 7

Contingency projects **Net adjustments if Bandicoot Corridors required - Subject to EPBC Act (Commonwealth) Determination.**

botanic ridge precinct structure plan

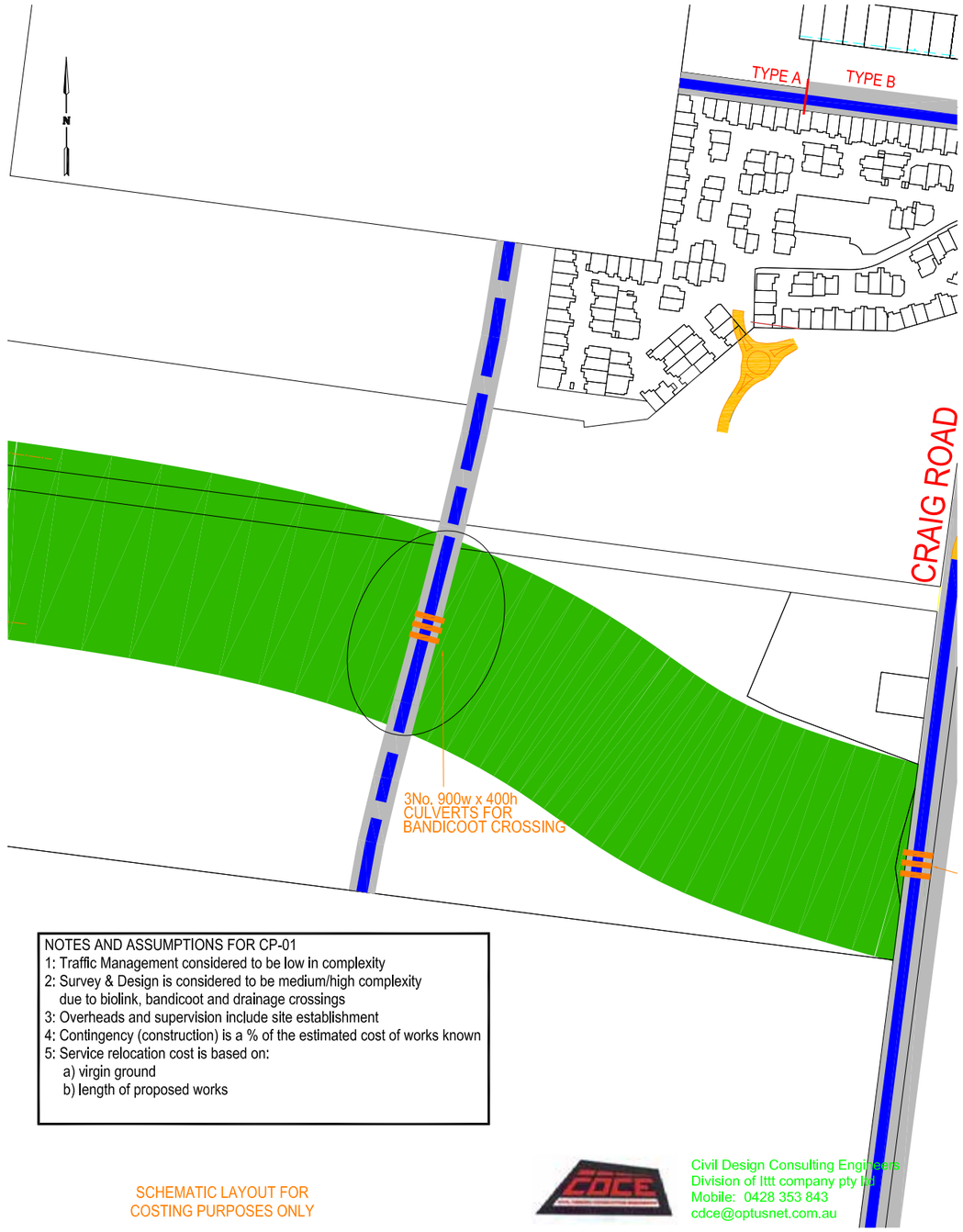
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Date: Dec 2012

0 50 100 200 500m

Table 9: CONTINGENCY PROJECTS

NEW PROJECT NO	PROJECT	INFRASTRUCTURE CATEGORY	WORKS DESCRIPTION	LAND AREA (HA)	PROJECT COST: LAND	PROJECT COST: CONSTRUCTION	TOTAL ESTIMATED PROJECT COST: LAND & CONSTRUCTION	PROPORTION OF COST ATTRIBUTABLE TO MCA	DEVELOPMENT TYPES MAKING CONTRIBUTION	TOTAL COST ATTRIBUTABLE TO MCA	TOTAL COST PER NDHA	TOTAL FUNDED VIA BOTANIC RIDGE DCP
CP01	Construction - Connector Road C5 - portion that would cross eastern SBB Corridor.	Development	Construction of 22.5m wide and 172m long connector road and reservation, including pedestrian and shared path and culverts for bandicoot movement.	0.00	$0	$659,462	$659,462	100%	Residential and town centres	$659,462	$2,906	$659,462
CP02	Construction - Connector Road C1 - portion that would cross southern SBB Corridor.	Development	Construction of 22.5m wide and 82m long connector road and reservation, including pedestrian and shared path and culverts for bandicoot movement.	0.00	$0	$293,889	$293,889	100%	Residential and town centres	$293,889	$1,295	$293,889
CP03	Construction - Shared Path Crossing of Eastern SBB Corridor.	Development	Construction of 11.5m wide and 172m long shared path, associated landscaping and culverts for bandicoot movenment. Includes separate culverts for waterway.	0.00	$0	$124,831	$124,831	100%	Residential and town centres	$124,831	$550	$124,831
CP04a / 04b	Construction - Shared Path Crossings of Southern SBB Corridor.	Development	Construction of 2no. 11.5m wide and 82m long shared path crossings, associated landscaping and culverts for bandicoot movement.	0.00	$0	$98,063	$98,063	100%	Residential and town centres	$98,063	$432	$98,063
CP05a	Construction - Addition of Culverts for SBB corridor crossing to DCP project RD04	Development	Addition of 3 no. 0.9mx0.4m culverts for bandicoot movement	0.00	$0	$42,086	$42,086	100%	Residential and town centres	$42,086	$185	$42,086
CP05b	Construction - Addition of Culverts for SBB corridor crossing to DCP project RD05	Development	Addition of 3 no. 0.9mx0.4m culverts for bandicoot movement	0.00	$0	$43,274	$43,274	100%	Residential and town centres	$43,274	$191	$43,274
CP05c	Construction - Addition of road pavement to upgrade road from constrained access level 2 status to Constrained connector status	Development	Construction of additional 2.4m width of road pavement to deliver 8.4m wide carriageway 275m long within a constrained reservation to facilitate public transport access.	0.00	$0	$73,642	$73,642	100%	Residential and town centres	$73,642	$325	$73,642
CP05d	Construction - deletion of west leg of IN03 intersection.	Development	Deletion of western (4th) leg of intersection due to adjoining east / west connector being removed to deliver SBB corridor.	0.00	$0	-$37,217	-$37,217	100%	Residential and town centres	-37,217	-164	-37,217
CP06	Land - Modified opportionment of Community Centre 1 attributed to MCA	Development	Apportionment applied to MCA reduced by 1% due to reduced residential yield caused by SBB corridors.	0.80	$440,550	$0	$440,550	-1%	Residential and town centres	-4,406	-19	-4,406
CP07	Construction - Modified opportionment of Community Centre 1 attributed to MCA	Development	Apportionment applied to MCA reduced by 1% due to reduced residential yield caused by SBB corridors.	0.00	$0	$5,795,696	$5,795,696	-1%	Residential and town centres	-57,957	-255	-57,957
CO08	Net Developable Area Adjustment	Development	Adjustment to NDHA Development Levy due to reduction in Net Developable Area by inclusion of SBB Corridors						Residential and town centres		$10,634	$2,412,980
										$27,764,804	**$16,080**	**$3,648,648**



NOTES AND ASSUMPTIONS FOR CP-01
1: Traffic Management considered to be low in complexity
2: Survey & Design is considered to be medium/high complexity due to biolink, bandicoot and drainage crossings
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) virgin ground
b) length of proposed works

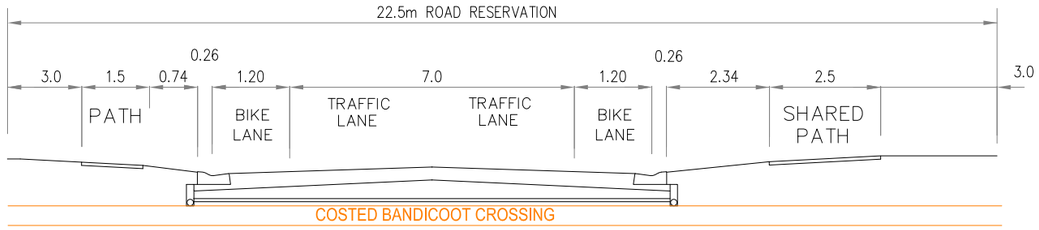
SCHEMATIC LAYOUT FOR
COSTING PURPOSES ONLY

Civil Design Consulting Engineers
Division of Ittt company pty ltd
Mobile: 0428 353 843
cdce@optusnet.com.au

BOTANIC RIDGE DCP

CP-01 ROAD COSTS
EAST SBB CORRIDOR CONNECTOR RD
WITH BANDICOOT CROSSING

ULTIMATEROAD RESERVE
COSTED ROAD
COSTED BANDICOOT CROSSINGS



TYPICAL BIOLINK COLLECTOR ROAD WITH A SHARED PATH

Botanic Ridge DCP Roadworks Estimate

Road Name: CP-01 SBB Corridor Connector Street with Shared Path & Bandicoot crossings
Limit of works: As shown on drawing road reserve and cross section
Length of Job: length = 172Lm Biolink crossing only as shown
Notes: Costs based on City of Casey specification for road construction & GAA Cross Section

Item	Unit	Rate	wide	units		cost/m		Unit cost	Quantity	subset	subtotal	Amount
			m			width			metres	cost		
Roadworks	subtotal							$ 2,081.02	172 m		$ 357,936.20	$ 357,936
Bulk Earthworks	m²	$ 40.00	5.41	metres	$ 40	$ 216.40	m				$ 37,220.80	$ -
Pavement (Urban)	m²	$ 91.50	8.72	metres	$ 92	$ 797.88	m				$ 137,235.36	$ -
Crushed Rock Shoulder	m²	$ 20.00	0	metres	$ 20	$ -	m				$ -	$ -
Pavement Removal	m²	$ 10.00	0	metres		$ -					$ -	
Bio Link Bandicoot Crossing 3No. 0.9x0.4	m	$ 410.00	67.5	No.	1						$ 27,675.00	$ 27,675
Drainage Culvert 2No. 1.2x1.8	m	$ 1,320.00	45.0	No.	1						$ 59,400.00	$ 59,400
Kerb and Channel S 504		$ 45.00	2	sides	$ 45	$ 90.00	m				$ 15,480.00	
side entry pits std drw S305	unit	$ 2,000.00	2	internal ma	90.00	$ 44.44	m				$ 7,644.44	
Shared path 2.5m wide conc	m²	$ 60.00	2.5	metres		$ 150.00					$ 25,800.00	
Shared pathway 2.5m wide gravel	m²	$ 30.00	0	metres		$ -					$ -	
Pedestrian Footpath 1.5m wide	m²	$ 60.00	1.5	metres		$ 90.00					$ 15,480.00	
Pedestrian Footpath 1.5m gravel	m²	$ 30.00	0	metres		$ -					$ -	
Drainage , subgrade drain	m	$ 14.70	2		$ 15	$ 29.40	m				$ 5,056.80	$ -
Linemarking & Signage	m	$ 25.00	3	metres	$ 25	$ 75.00	m				$ 12,900.00	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m				$ -	$ -
Concrete Path	m²	$ 45.00	0		$ 45	$ -	m				$ -	$ -
level / trim nature strip	m²	$ 4.00	9.1	metres	$ 4	$ 36.40	m				$ 6,260.80	$ -
as per standard drawings												
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tr	2	rows of tre	$ 25	$ 50.00	m				$ 8,600.00	
Tube Stock Plantings	unit	$5.27 -$6.78	0	metres wi	$ 6	$ -	m				$ -	$ -
as per draft tree strategy											$ -	
root barriers	optional as per design in respect of appropriate offset criteria											
300mm dia conc drain Stormwater Cr BF	per metre	$182	0	metres		$ -					$ -	
375mm conc drain stormwater Cr Bk fill	per metre	$231	0	metres		$ -					$ -	
450mm conc drain stormwater Cr BF	per metre	$294	86	metres		$ 147.00					$ 25,284	
525mm conc drain stormwater Cr BF	per metre	$351	86	metres		$ 175.50					$ 30,186	
Traffic Signals	Unit	$ 102,500.00	0	No of intersection							$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	0	metres							$ -	$ -
Street Lighting	m	$ 129.00	1	rows of lights		$ 129.00	m				$ 22,188.00	
lighting conduit	m	$ 50.00	1	No of runs/sides		$ 50.00	m				$ 8,600.00	
Subtotal						$ 2,081.02					$ 445,011.20	
estimated total												$ 445,011
Traffic Management									0.0%			$ -
Contingency									20.0%			$ 89,002
Total + contingencies												
Services relocation Sec Poles, water fittings	Item											$ -
Survey and Design									10%			$ 44,501
Overheads (supervision etc)									15%			$ 66,752
Total excluding land cost												**$ 645,266**
CPI Adjustment to Nov. 2012									2.2%			$ 14,196
Total Estimated Cost												**$ 659,462**
Adopted Cost												

cost per lineal metre $ 3,834

Estimate Prepared by: CDCE Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM		
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG		
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE		
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.		
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO.	65	
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN	TREE STUMP	PIT NO.		

AMENDMENTS

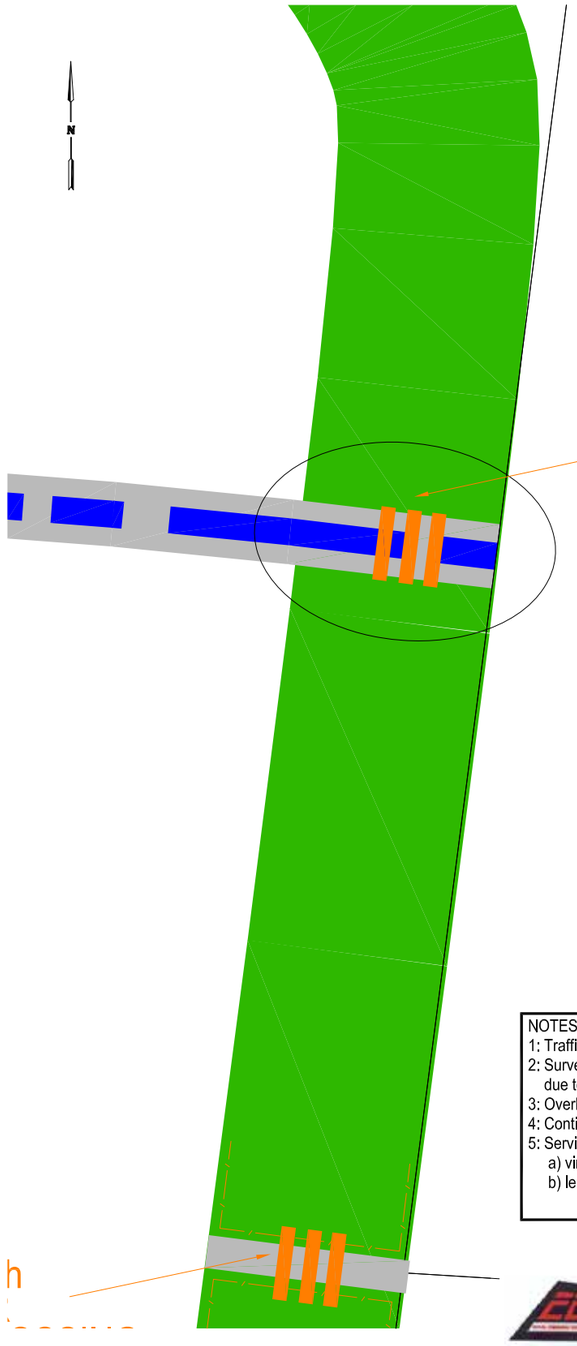
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SURVEY N/A
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TRACED SAF
CHECKED SAF
JOB NO. 10 111
DWG. FILE NAME XXXXXX.DWG
REG. FILE NO. XXXXXXX
MELWAY REF.
CONTRACT NO.

CIVIL DESIGN CONSULTING ENGINEERS

BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
CONNECTOR RD EAST SBB CORRIDOR

DATE: November 2012
REVISION
10 111 -
SHEET 1 OF 1



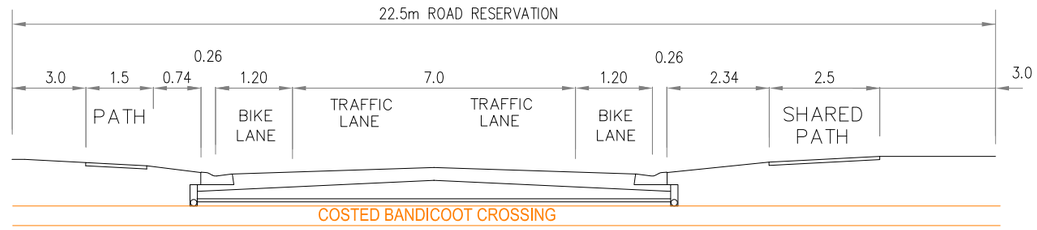
BOTANIC RIDGE DCP

CP-02 ROAD COSTS
SOUTH SBB CORRIDOR CONNECTOR RD
WITH BANDICOOT CROSSING

ULTIMATEROAD RESERVE

COSTED ROAD

COSTED BANDICOOT CROSSINGS



TYPICAL BIOLINK COLLECTOR ROAD WITH A SHARED PATH

Botanic Ridge DCP Roadworks Estimate

Road Name: CP-02 SBB Corridor Connector Street with Shared Path & Bandicoot crossings
Limit of works: As shown on drawing road reserve and cross section
Length of Job: length = 82Lm Biolink crossing only as shown
Notes: Costs based on City of Casey specification for road construction & GAA Cross Section

Item	Unit	Rate	wide m	units		cost/m width		Unit cost	Quantity metres		subset cost	subtotal	Amount
Roadworks	subtotal							$ 2,061.02	82	m		$ 170,644.00	$ 170,644
Bulk Earthworks	m²	$ 40.00	5.41	metres	$ 40	$ 216.40	m					$ 17,744.80	$ -
Pavement (Urban)	m²	$ 91.50	8.72	metres	$ 92	$ 797.88	m					$ 65,425.16	$ -
Crushed Rock Shoulder	m²	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Pavement Removal	m²	$ 10.00	0	metres		$ -						$ -	
Bio Link Bandicoot Crossing 3No. 0.9x0.4	m	$ 410.00	67.5	No.	1							$ 27,675.00	$ 27,675
Kerb and Channel S 504		$ 45.00	2	sides	$ 45	$ 90.00	m					$ 7,380.00	
side entry pits std drw S305	unit	$ 2,000.00	2	interval m	90.00	$ 44.44	m					$ 3,644.44	
Shared path 2.5m wide conc	m²	$ 60.00	2.5	metres		$ 150.00						$ 12,300.00	
Shared pathway 2.5m wide gravel	m²	$ 30.00	0	metres		$ -						$ -	
Pedestrian Footpath 1.5m wide	m²	$ 60.00	1.5	metres		$ 90.00						$ 7,380.00	
Pedestrian Footpath 1.5m gravel	m²	$ 30.00	0	metres		$ -						$ -	
Drainage , subgrade drain	m	$ 14.70	2		$ 15	$ 29.40	m					$ 2,410.80	$ -
Linemarking & Signage	m	$ 25.00	3	metres	$ 25	$ 75.00	m					$ 6,150.00	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Concrete Path	m²	$ 45.00	0		$ 45	$ -	m					$ -	$ -
level / trim nature strip as per standard drawings	m²	$ 4.00	9.1	metres	$ 4	$ 36.40	m					$ 2,984.80	$ -
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tr	2	rows of tr	$ 25	$ 50.00	m					$ 4,100.00	
Tube Stock Plantings as per draft tree strategy	unit	$5.27 -$6.78	0	metres w	$ 6	$ -	m					$ -	$ -
root barriers	optional as per design in respect of appropriate offset criteria											$ -	
300mm dia conc drain Stormwater Cr BF	per metre	$182	0	metres		$ -						$ -	
375mm conc drain stormwater Cr Bk fill	per metre	$231	0	metres		$ -						$ -	
450mm conc drain stormwater Cr BF	per metre	$294	41	metres		$ 147.00						$ 12,054	
525mm conc drain stormwater Cr BF	per metre	$351	41	metres		$ 175.50						$ 14,391	
Traffic Signals	Unit	$ 102,500.00	0	No of intesection								$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	0	metres								$ -	$ -
Street Lighting	m	$ 129.00	1	rows of lights		$ 129.00	m					$ 10,578.00	
lighting conduit	m	$ 50.00	1	No of runs/sides		$ 50.00	m					$ 4,100.00	
Subtotal								$ 2,061.02				$ 198,319.00	
estimated total													$ 198,319
Traffic Management										0.0%			$ -
Contingency										20.0%			$ 39,664
Total + contingencies													
Services relocation Sec Poles, water fittings	item												$ -
Survey and Design										10%			$ 19,832
Overheads (supervision etc)										15%			$ 29,748
Total excluding land cost													**$ 287,563**
CPI Adjustment to Nov 2012										2.2%			$ 6,326
Total Estimated Cost													**$ 293,889**
Adopted Cost													

cost per lineal metre $ 3,584

NOTES AND ASSUMPTIONS FOR CP-02
1: Traffic Management considered to be low in complexity
2: Survey & Design is considered to be medium/high complexity due to biolink and bandicoot crossings
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) virgin ground
b) length of proposed works

Civil Design Consulting Engineers
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Estimate Prepared by: CDCE Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
SCALE NOT TO SCALE
COORD. SYS.: N/A
F.B. L.B.
JOB NO. 10 111
SURVEY N/A
DWG. FILE NAME XXXXXX.DWG
DESIGN SAF
REG. FILE NO. XXXXXXX
TRACED SAF
MELWAY REF.
CHECKED SAF
CONTRACT NO.

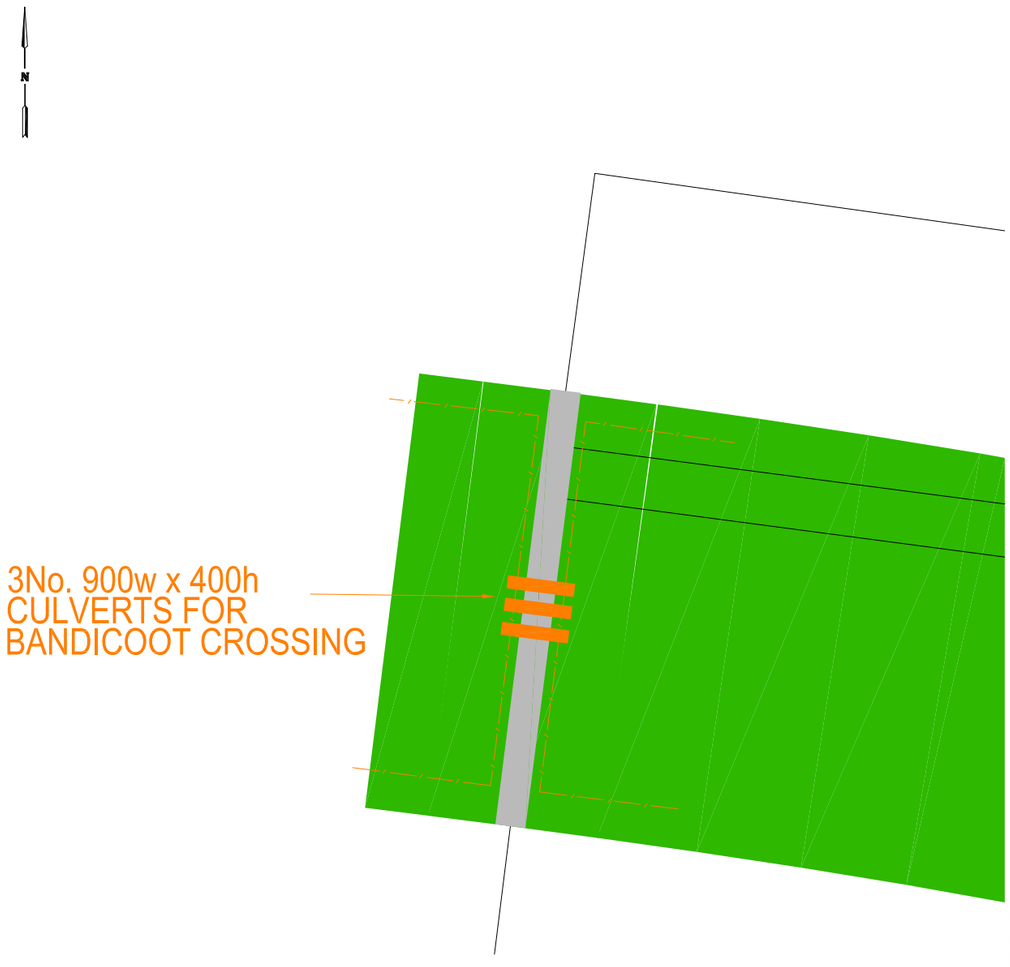
CIVIL DESIGN CONSULTING ENGINEERS

BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
CONNECTOR RD SOUTH SBB CORRIDOR

DATE: November 2012
REVISION
10 111
SHEET 1 OF 1

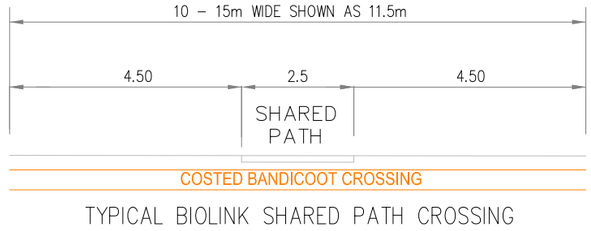
BOTANIC RIDGE DCP

CP-03 SHARED PATH COSTS SBB CORRIDOR WITH TYPICAL BANDICOOT CROSSING



COSTED PATH

COSTED BANDICOOT CROSSINGS



TYPICAL BIOLINK SHARED PATH CROSSING

Botanic Ridge DCP Roadworks Estimate

Road Name: CP-03 SBB Corridor Shared Path & Bandicoot Crossing
Limit of works: As shown on drawing road reserve and cross section
Length of Job: length = 172Lm Biolink crossing only as shown
Notes: Costs based on City of Casey specification for road construction & GAA Cross Section

Item	Unit	Rate	wide m	units		cost/m width		Unit cost	Quantity metres		subset cost	subtotal	Amount
Roadworks	subtotal							$ 231.00	172	m		$ 39,732.00	$ 39,732
Bulk Earthworks	m²	$ 40.00	0.5	metres	$ 40	$ 20.00	m					$ 3,440.00	$ -
Pavement (Urban)	m²	$ 91.50	0	metres	$ 92	$ -	m					$ -	$ -
Crushed Rock Shoulder	m²	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Pavement Removal	m²	$ 10.00	0	metres		$ -						$ -	
Drainage Culvert 2No. 1.2x1.8	m	$ 1,320.00	23.0	No.	1							$ 30,360.00	$ 30,360
Bio Link Bandicoot Crossing 3No. 0.9x0.4	m	$ 410.00	34.5	No.	1							$ 14,145.00	$ 14,145
Kerb and Channel S 504		$ 45.00	0	sides	$ 45	$ -	m					$ -	
side entry pits std drw S305	unit	$ 2,600.00	0	interval metres	90.00	$ -	m					$ -	
Shared path 2.5m wide conc	m²	$ 60.00	2.5	metres		$ 150.00						$ 25,800.00	
Shared pathway 2.5m wide gravel	m²	$ 30.00	0	metres		$ -						$ -	
Pedestrian Footpath 1.5m wide	m²	$ 60.00	0	metres		$ -						$ -	
Pedestrian Footpath 1.5m gravel	m²	$ 30.00	0	metres		$ -						$ -	
Drainage , subgrade drain	m	$ 14.70	0		$ 15	$ -	m					$ -	$ -
Linemarking & Signage	m	$ 25.00	0	metres	$ 25	$ -	m					$ -	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m					$ -	$ -
Concrete Path	m²	$ 45.00	0		$ 45	$ -	m					$ -	$ -
level / trim nature strip	m²	$ 4.00	9	metres	$ 4	$ 36.00	m					$ 6,192.00	$ -
as per standard drawings													
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tree	1	rows of trees	$ 25	$ 25.00	m					$ 4,300.00	
Tube Stock Plantings	unit	$5.27 -$6.78	0	metres wide	$ 6	$ -	m					$ -	$ -
as per draft tree strategy												$ -	
root barriers	optional as per design in respect of appropriate offset criteria												
300mm dia conc drain Stormwater Cr BF	per metre	$182	0	metres		$ -						$ -	
375mm conc drain stormwater Cr Bk fill	per metre	$231	0	metres		$ -						$ -	
450mm conc drain stormwater Cr BF	per metre	$294	0	metres		$ -						$ -	
525mm conc drain stormwater Cr BF	per metre	$351	0	metres		$ -						$ -	
Traffic Signals	Unit	$ 102,500.00	0	No of intesection								$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	0	metres								$ -	$ -
Street Lighting	m	$ 129.00	0	rows of lights		$ -	m					$ -	
lighting conduit	m	$ 50.00	0	No of runs/sides		$ -	m					$ -	
Subtotal						$ 231.00						$ 84,237.00	
estimated total													$ 84,237
Traffic Management										0.0%			$ -
Contingency										20.0%			$ 16,847
Total + contingencies													
Services relocation Sec Poles, water fittings	Item												$ -
Survey and Design										10%			$ 8,424
Overheads (supervision etc)										15%			$ 12,636
Total excluding land cost													**$ 122,144**
CPI Adjustment to Nov. 2012										2.2%			$ 2,687
Total Estimated Cost													**$ 124,831**
Adopted Cost													

cost per lineal metre $ 726

NOTES AND ASSUMPTIONS FOR CP-03
1: Traffic Management considered to be nil in complexity
2: Survey & Design is considered to be low to medium complexity due to biolink, bandicoot and drainage crossings
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) virgin ground
b) length of proposed works

SCHEMATIC LAYOUT FOR
COSTING PURPOSES ONLY

Civil Design Consulting Engineers
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Estimate Prepared by: CDCE Nov-12

LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO. (23)
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
COORD. SYS.: N/A
F.B. L.B.
SURVEY N/A
DESIGN SAF
TRACED SAF
CHECKED SAF
SCALE NOT TO SCALE
JOB NO. 10 111
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CONTRACT NO.

CIVIL DESIGN CONSULTING ENGINEERS

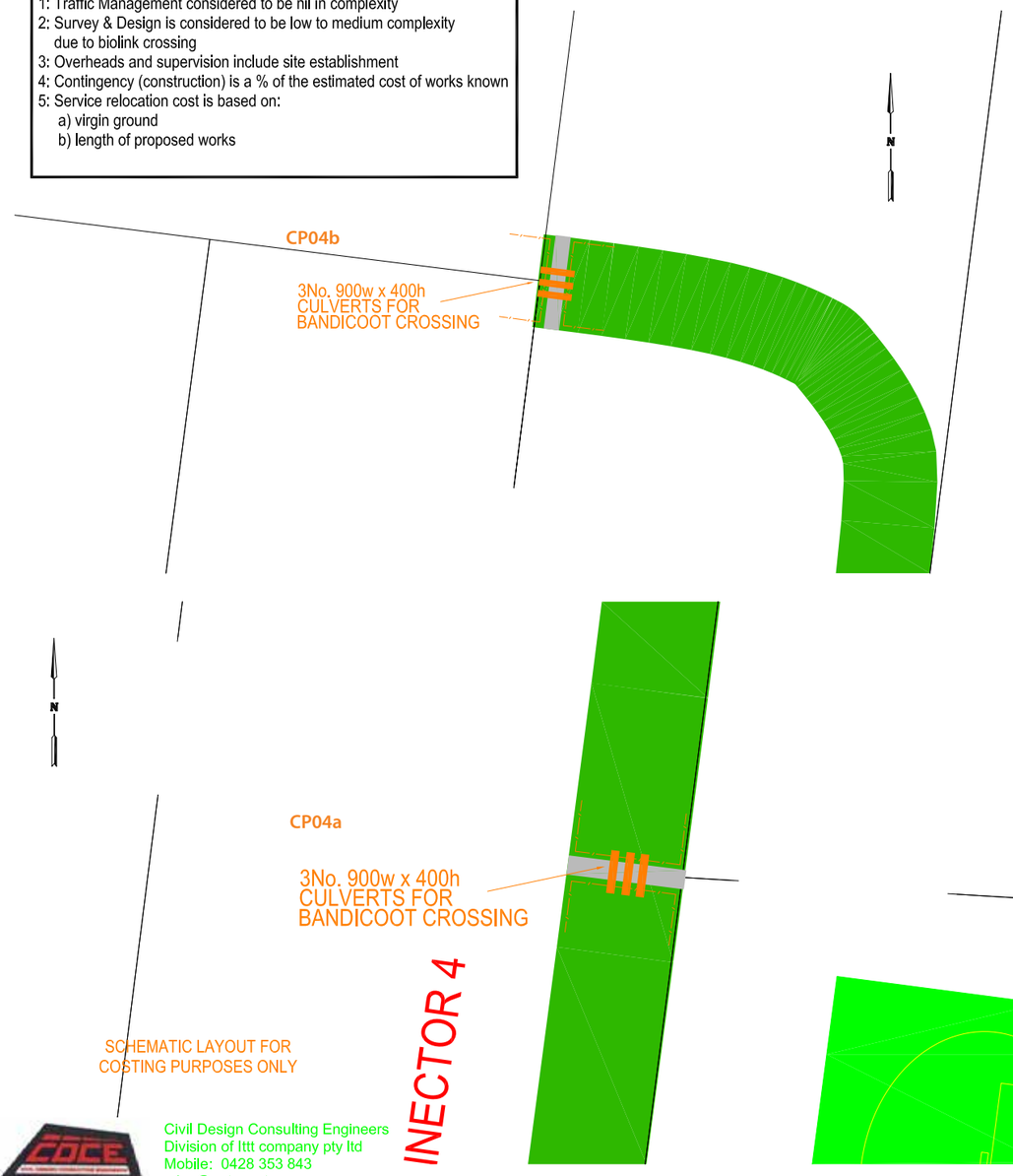
BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
SHARED PATH WITH EAST SBB CORRIDOR CROSSING

DATE: November 2012
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BOTANIC RIDGE DCP

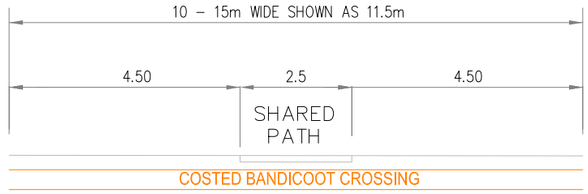
CP-04a & 4b SHARED PATH COSTS SBB CORRIDOR SOUTH WITH TYPICAL BANDICOOT CROSSING

NOTES AND ASSUMPTIONS FOR CP-04a & 4b
1: Traffic Management considered to be nil in complexity
2: Survey & Design is considered to be low to medium complexity due to biolink crossing
3: Overheads and supervision include site establishment
4: Contingency (construction) is a % of the estimated cost of works known
5: Service relocation cost is based on:
a) virgin ground
b) length of proposed works



COSTED PATH

COSTED BANDICOOT CROSSINGS



TYPICAL BIOLINK SHARED PATH CROSSING

Botanic Ridge DCP Roadworks Estimate

Road Name: CP-04a & 4b SBB Corridor Shared Path & South Bandicoot crossings
Limit of works: As shown on drawing road reserve and cross section
Length of Job: length = 82Lm for each Biolink crossing only as shown
Notes: Costs based on City of Casey specification for road construction & GAA Cross Section

Item	Unit	Rate	wide	units		cost/m		Unit cost	Quantity	subset	subtotal	Amount
			m			width			metres	cost		
Roadworks	subtotal							$ 231.00	164 m		$ 37,884.00	$ 37,884
Bulk Earthworks	m³	$ 40.00	0.5	metres	$ 40	$ 20.00	m				$ 3,280.00	$ -
Pavement (Urban)	m²	$ 91.50	0	metres	$ 52	$ -	m				$ -	$ -
Crushed Rock Shoulder	m²	$ 20.00	0	metres	$ 20	$ -	m				$ -	$ -
Pavement Removal	m²	$ 10.00	0	metres		$ -					$ -	
Bio Link Bandicoot Crossing 3No. 0.9x0.4	m	$ 410.00	34.5	No.	2						$ 28,290.00	$ 28,290
Kerb and Channel S 504		$ 45.00	0	sides	$ 45	$ -	m				$ -	
side entry pits std drw S305	unit	$ 2,000.00	0	interval m	90.00	$ -	m				$ -	
Shared path 2.5m wide conc	m²	$ 60.00	2.5	metres		$ 150.00					$ 24,600.00	
Shared pathway 2.5m wide gravel	m²	$ 30.00	0	metres		$ -					$ -	
Pedestrian Footpath 1.5m wide	m²	$ 60.00	0	metres		$ -					$ -	
Pedestrian Footpath 1.5m gravel	m²	$ 30.00	0	metres		$ -					$ -	
Drainage , subgrade drain	m	$ 14.70	0		$ 15	$ -	m				$ -	$ -
Linemarking & Signage	m	$ 25.00	0	metres	$ 25	$ -	m				$ -	$ -
Landscaping refer plantings	m	$ 20.00	0	metres	$ 20	$ -	m				$ -	$ -
Concrete Path	m²	$ 45.00	0		$ 45	$ -	m				$ -	$ -
level / trim nature strip	m²	$ 4.00	9	metres	$ 4	$ 36.00	m				$ 5,904.00	$ -
as per standard drawings												
Tree Planting 2 - 2.5m tall	unit	$25/m-$150/tr	1	rows of tre	$ 25	$ 25.00	m				$ 4,100.00	
Tube Stock Plantings	unit	$5.27 -$8.78	0	metres we	$ 6	$ -	m				$ -	$ -
as per draft tree strategy											$ -	
root barriers	optional as per design in respect of appropriate offset criteria											
300mm dia conc drain Stormwater Cr BF	per metre	$182	0	metres		$ -					$ -	
375mm conc drain stormwater Cr Bk fill	per metre	$231	0	metres		$ -					$ -	
450mm conc drain stormwater Cr BF	per metre	$294	0	metres		$ -					$ -	
525mm conc drain stormwater Cr BF	per metre	$351	0	metres		$ -					$ -	
Traffic Signals	Unit	$ 102,500.00	0	No of intesection							$ -	$ -
Traffic Signal Conduit subset	m	$ 35.00	0	metres							$ -	$ -
Street Lighting	m	$ 129.00	0	rows of lights		$ -	m				$ -	
lighting conduit	m	$ 50.00	0	No of runs/sides		$ -	m				$ -	
Subtotal						$ 231.00					$ 66,174.00	
estimated total												$ 66,174
Traffic Management									0.0%			$ -
Contingency									20.0%			$ 13,235
Total + contingencies												
Services relocation Sec Poles, water fittings	Item											$ -
Survey and Design									10%			$ 6,617
Overheads (supervision etc)									15%			$ 9,926
Total excluding land cost												**$ 95,952**
CPI Adjustment to Nov. 2012									2.2%			$ 2,111
Total Estimated Cost												**$ 98,063**
Adopted Cost												

cost per lineal metre $ 598

Estimate Prepared by: CDCE Nov-12

Civil Design Consulting Engineers
Division of Ittt company pty ltd
Mobile: 0428 353 843
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LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO. (25)
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN	TREE STUMP	PIT NO. <S2>

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
COORD. SYS.: N/A
SCALE NOT TO SCALE
F.B. L.B.
JOB NO. 10 111
SURVEY N/A
DWG. FILE NAME XXXXXX.DWG
DESIGN SAF
REG. FILE NO. XXXXXXX
TRACED SAF
MELWAY REF.
CHECKED SAF
CONTRACT NO.

CIVIL DESIGN CONSULTING ENGINEERS

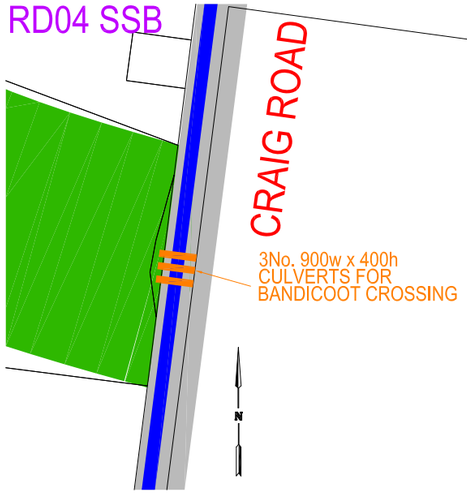
BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
SHARED PATH WITH SOUTH SBB CORRIDOR CROSSING

DATE: November 2012
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BOTANIC RIDGE DCP

CP-05 (a, b ,c, d)
NET ADJUSTMENTS & CONTINGENCIES
IF BANDICOOT CORRIDORS REQUIRED
FOR (RD-04, RD-05, RD-08 & IN-03)

RD04 SSB



a Botanic Ridge DCP Roadworks Estimate

Road Name: RD04 SSB culvert crossing addition
Limit of works: As shown on drawing
Length of Job: Based on additon of culverts
Notes: Costs based on culvert lengths to cross road reserve

Item		Amount
without SSB crossing		$ 2,477,034
with SSB crossing		$ 2,503,802
Subtotal		$ 26,568
estimated total		$ 26,568
Traffic Management	10.0%	$ 2,657
Contingency	20.0%	$ 5,314
Total + contingencies		
Services relocation Sec Poles, water fittings		$ -
Survey and Design	10%	$ 2,657
Overheads (supervision etc)	15%	$ 3,985
Total excluding land cost		**$ 41,180**
CPI Adjusted to Nov. 2012	2.2%	$ 906
Total Estimated Cost		**$ 42,086**
Adopted Cost		

Estimate Prepared by: **CDCE** Nov-12

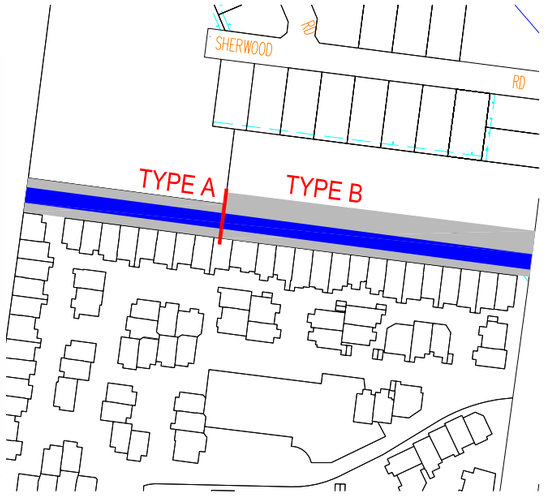
RD-08 ROAD CHANGES

c Botanic Ridge DCP Roadworks Estimate

Road Name: RD08 Road Changes
Limit of works: As shown on drawing
Length of Job: Based on increase in road pavement and road reserve widths
Notes: Costs based on connector/level road widths provided by GAA

Item		Amount
with reduced road pavement & road reserve		$ 498,986
without reduced road pavement & road reserve		$ 547,004
Subtotal		$ 48,038
estimated total		$ 48,038
Traffic Management	6.0%	$ 2,882
Contingency	20.0%	$ 9,608
Total + contingencies		
Services relocation Sec Poles, water fittings		$ -
Survey and Design	9%	$ 4,323
Overheads (supervision etc)	15%	$ 7,206
Total excluding land cost		**$ 72,057**
CPI Adjusted to Nov. 2012	2.2%	$ 1,585
Total Estimated Cost		**$ 73,642**
Adopted Cost		

Estimate Prepared by: **CDCE** Nov-12



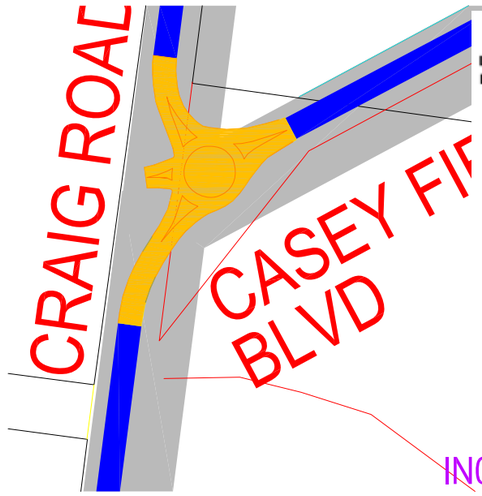


b Botanic Ridge DCP Roadworks Estimate

Road Name: RD05 SSB culvert crossing addition
Limit of works: As shown on drawing
Length of Job: Based on additon of culverts
Notes: Costs based on culvert lengths to cross road reserve

Item		Amount
without SSB crossing		$ 1,587,035
with SSB crossing		$ 1,614,710
Subtotal		$ 27,675
estimated total		$ 27,675
Traffic Management	10.0%	$ 2,768
Contingency	20.0%	$ 5,535
Total + contingencies		
Services relocation Sec Poles, water fittings		$ -
Survey and Design	8%	$ 2,214
Overheads (supervision etc)	15%	$ 4,151
Total excluding land cost		**$ 42,343**
CPI Adjusted to Nov. 2012	2.2%	$ 932
Total Estimated Cost		**$ 43,274**
Adopted Cost		

Estimate Prepared by: **CDCE** Nov-12



d Botanic Ridge DCP Roadworks Estimate

Road Name: IN-03 Fourth leg removed
Limit of works: As shown on drawing
Length of Job: Based on removal of fourth leg connection only
Notes: Costs based on road lengths confined with in the connection to roundabout as shown

Item		Amount
Three leg roundabout		$ 315,154
Four leg roundabout		$ 339,112
Subtotal		$ 23,958
estimated total		$ 23,958
Traffic Management	7.0%	$ 1,677
Contingency	20.0%	$ 4,792
Total + contingencies		
Services relocation Sec Poles, water fittings		$ -
Survey and Design	10%	$ 2,396
Overheads (supervision etc)	15%	$ 3,594
Total excluding land cost		**$ 36,416**
CPI Adjusted to Nov. 2012	2.2%	$ 801
Total Estimated Cost		**$ 37,217**
Adopted Cost		

Estimate Prepared by: **CDCE** Nov-12

IN03 ADDITONAL 4TH LEG REMOVAL

Civil Design Consulting Engineers
Division of Ittt company pty ltd
Mobile: 0428 353 843
cdce@optusnet.com.au

SCHEMATIC LAYOUT FOR
COSTING PURPOSES ONLY

LEGEND

PROPOSED DRAINAGE	GAS MAIN G	FIRE PLUG	SEWER MANHOLE	GRAVEL DRIVE	PSM
PROPOSED DRAINAGE PITS	WATER MAIN W	STOP VALVE	TELEPHONE PIT	CONCRETE DRIVE	TITLE PEG
EXISTING DRAINAGE	TELEPHONE U/G T	FIRE HYDRANT	TELEPHONE POLE	THICKENED PATH	NATURAL SURFACE
EXISTING DRAINAGE PITS	POWER U/G E	POWER POLE	TELEPHONE BOX	TREE	HOUSE NO.
PROPOSED KERB AND CHANNEL	POWER OVERHEAD E	LIGHT POLE	GAS VALVE	TREE REMOVAL	LOT/LP NO. 65
EXISTING KERB AND CHANNEL	SEWER MAIN S	POWER PIT	HOUSE DRAIN	TREE STUMP	PIT NO.

AMENDMENTS

DATE	REMARKS	REV	APP

DATUM: N/A
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SCALE NOT TO SCALE
F.B. L.B.
JOB NO. 10 111
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CIVIL DESIGN CONSULTING ENGINEERS

BOTANIC RIDGE DCP
DEVELOPMENT CONTRIBUTIONS PLAN (DCP)
COST TO ADD OR REMOVE COMPONENTS

DATE: November 2012
REVISION
10 111
SHEET 1 OF 1

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Botanic Ridge Development Contributions Plan - December 2012 (Amended April 2015)

METROPOLITAN
PLANNING
AUTHORITY

State
Government
Victoria